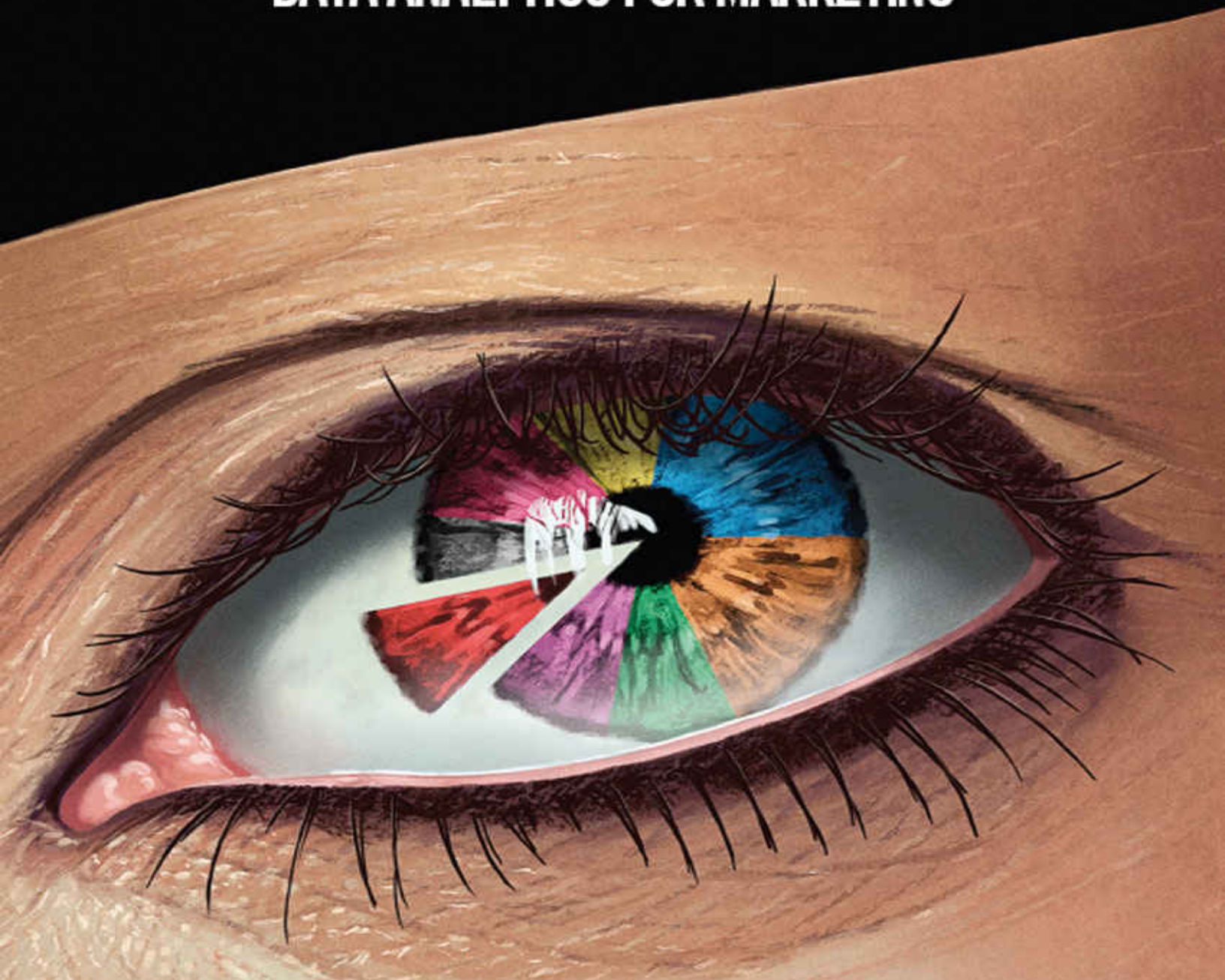
CULT OF ANALYTICS

DATA ANALYTICS FOR MARKETING



SECOND EDITION

STEVE JACKSON

ROUTLEDGE

Cult of Analytics

Cult of Analytics, Second Edition, enables professionals to build an analytics driven culture into their business or organisation. Marketers will learn how to turn tried and tested tactics into an actionable plan to change their culture to one that uses web analytics on a day-to-day basis.

Through use of the fictitious ACME plc case, Steve Jackson provides working examples based on real life situations from the various companies he has worked with, such as Nokia, KONE, Rovio, Amazon, Expert, IKEA, Vodafone and EMC. These examples will give the reader practical techniques for their own business regardless of size or situation making *Cult of Analytics* a must-have for any would-be digital marketer.

This new edition has been thoroughly updated, and now includes examples of how to get the best from Google analytics, as well as ways to use social media data, big data, tag management and advanced persona segmentation to drive real value in your organisation. It's also been expanded to include exercises and new cases for students and tutors using the book as a text.

Steve Jackson is a well-known analyst and entrepreneur. He currently serves as CEO of Quru Oy, a digital analytics agency, Finland. He is the author of *Cult of Analytics* and *The Sucking Manifesto*.

Praise for *Cult Of Analytics*

"Steve has spent years down in the weeds, talking to executives and meditating on a mountain. That unique perspective is the main reason *Cult of Analytics* is such a unique book. Easy to read, easy to follow, easier still to find relevant ideas you can implement in your company."

Avinash Kaushik, author of *Web Analytics: An Hour A Day*

"Web analysis is central to the success of most modern organisations. However, operationalising the insights gleaned and the next actions to take are what hold back many from reaping the benefits of web analytics. *Cult of Analytics* is a must read from someone who is not just sharing hope and theory but someone who has actually done it for others and is willing to show you the way."

Bryan Eisenberg, *NewYork Times* bestselling author of *Waiting ForYour Cat to Bark*

"A great read! I wish I had read this three years ago – it would have saved me a lot of time (and pain) at Google! I must admit I was initially not enthused to read another potentially dry analytics book, but this is really engaging right from page 1."

Brian Clifton, author of *AdvancedWeb Metrics*, senior strategist at Omega Digital Media, former head of web analytics at Google EMEA

"When I first wrote *Web Analytics Demystified* and *The Big Book of Key Performance Indicators* it was difficult to tell how deeply engrained in business culture web analytics would eventually become. Now, nearly five years later the question has been answered and the best companies around the world are actively working to better understand the online opportunity through better measurement. Steve Jackson in *Cult of Analytics* has provided a necessary update to much of my work providing one of most broad views of analytics in the Enterprise available in print. Steve covers the topic of analytics at a variety of levels – from corporate adoption of analytics down into the depths of key performance indicators – providing clear explanations throughout. I have always known that Steve is one of the

best and the brightest in our industry, *Cult of Analytics* will clarify that for the rest of the world."

Eric T. Peterson, author of *Web Analytics Demystified*, *Web Site Measurement Hacks* and *The Big Book of Key Performance Indicators*, founder of Web Analytics Wednesday and The Web Analytics Forum, and principal at Web Analytics Demystified, Inc.

"Creating a competitive advantage based on web analytics includes a shift towards the true data driven organisation. This shift demands that you master your web analytics technology, but more importantly that you understand how to derive insight from all this new data and that you have a setup and an ability to execute on the insights. For the above nirvana to happen, you need two books, one on your technology, the other, without any doubts, *Cult of Analytics*!"

Dennis R. Mortensen, author of *Yahoo! Web Analytics*, and CEO and founder of X.ai

"Great job, useful insights and smart pieces of business thinking tied to practical tips and tricks. Chapter 9 is the masterpiece of the work. The quality of Chapter 9 was a pleasant and unexpected surprise at the end of the book. You have a very personal style with a strong focus on marketing, especially with the REAN. The examples in Chapter 4 – Quick wins – shows the importance of customer process optimisation and how the online channel is much more than media and marketing. Very good!"

Seppo Roponen, Finnish Country Manager, comScore

"The long-awaited classic. Much needed and wanted. I have been waiting for this book a long time and now I was finally given the opportunity to read Steve's first published book. *Cult of Analytics* is by definition a classic as it will become a reference book for people working with web analytics and online marketing optimisation. It is required reading for those wanting to avoid the pain they would face if they didn't read *Cult of Analytics*.

"The book follows its own underlying theory, the REAN model. First the book cover, name and reviews grab your attention, the first chapter hooks you and I promise the following chapters will keep you going. The processes, best practices and templates make you want to test them …

you'll be asking do they really work, and finally, you'll see they do and will be satisfied with *Cult of Analytics*.

"In *Cult of Analytics* Steve provides you with an exploration into a leading analyst's mind and reveals the diamonds he has been uncovering over many years. I honestly believe this book reveals some of the secrets of web analytics consultancy that needed to be published."

Kalle Heinonen, CEO of Superanalytics

"In *Cult of Analytics* Steve Jackson gives you an all-encompassing overview of web analytics, everything from tool selection to segmentation. This book will help you come up to speed and is the perfect guide for those new to the subject, but even experts will learn something new. Steve's work focuses on how to build an organisation that will promote actions from web analytics data, how to develop KPIs, and how to combine clickstream data with other data sources. I have experience in applying several of the techniques used in the book and can testify that they work. Building on years of experience Steve makes the case for analytics-driven business."

Lars Johansson, content director for Internet Marketing Conference, and web analytics blogger at WebAnalysts.Info

"Steve Jackson draws on years of experience to show how to improve results from digital channels without being overly technical."

Dave Chaffey, author, and consultant at Marketing Insights

Cult of Analytics

Data analytics for marketing

Second edition

Steve Jackson

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Avinash Kaushik for his books *Web Analytics An Hour A Day* and *Web Analytics 2.0* (another must-have), as well as his blog, which is a fantastic resource that has grown even better since the first edition of the book.

I can't thank people I have learned from without thanking Jim Sterne, an absolute gentleman in our industry, co-founder of the Web Analytics Association (now the digital analytics association) and true thought leader. His eMetrics summits are where authors, consultants and practitioners go to learn this stuff.

To bring the thanks to a close Dave Chaffey deserves the credit for giving me the kick I needed to write the first edition of this book and Nicola Cupit at my publishers who was patient enough to get me through the second edition.

Finally thanks to one I haven't mentioned. You are still not forgotten.

Foreword

Back in the 1990s one of the common questions asked of online professionals was, "what were you doing before the Internet?" The answers were many and varied and helped understand the drift of the ensuing conversation. Today, the most common answer is, "I was in school", delivered with a quizzical look as if wondering what on Earth people did "before" the Internet.

By the time I met Steve Jackson he had already started Aboavista and was happily posting to the Conversion Chronicles – long before we knew it was going to be called a blog. This was after he had tackled total quality management, KAIZEN and measuring standard deviations of the processes and end products within the automobile industry. It was after he had spent the later 1990s as a freelancer designing websites for companies that wanted online brochures. It was after doing A/B/C/D split test print runs of 10,000 direct marketing pieces in preparation of sending out a million letters.

Steve then shifted his focus to email marketing – a perfect melding of his previous experience. From there, he switched to the publishing world and was transformed. The telling moment in this book is where Steve tells us, "I decided to delve into the data", and "I then dug deeper … and noticed a very important trend." He was hooked.

There was no turning back for him then. He had been bitten by the data bug. He had caught the analytics virus and it would change his career, his life, his dinner table conversation. His passion for the possibilities is infectious and this book is his transmission technique.

When Steve moved to Finland (yes, there are a girl involved), he found a job with a company running online lead generation tests and, by the way, increasing their conversion rate by 800 per cent – worth approximately €2

million a year in new business. From there, he created his own consulting company, started publishing the *Conversion Chronicles*, became a frequent speaker at my eMetrics Marketing Optimisation Summit and a very active member of the Web Analytics Association.

So when I asked him what he had done before the Internet, his answer was, "getting ready for it".

Steve assumed nothing about the reader when he wrote *Cult of Analytics*. But he doesn't waste any time either. He starts with the roles, responsibilities and the organisational structure of an analytics-driven company. He then points the way to understanding the ins and outs of analysis by using the relationship between company and customer as a framework for collecting and examining your online success.

So if you're interested in identifying clear goals for your organisation, convincing those around you that you're onto something, truly understanding where the data comes from, what the tools do, and then how to delve into the data, you've come to the right place.

But wait – there's more.

Steve doesn't stop there. He hands you the keys to the process of becoming an analytics-driven company and shows you how to implement a cult of analytics in your organisation.

Steve has been there and done that. He's helped countless companies improve their customers' online experience, improved their conversion rates and improved their bottom line. Who wouldn't want to read this book?

Jim Sterne
Producer, eMetrics Marketing Optimisation Summit
Chairman, Digital Analytics Association

Introduction

The day that changed my life

I made a mistake back in 1998. It was a job-threatening mistake. I'd only been on the job a month and I was still on trial.

I was working for a media company running websites with readerships in the millions daily so we needed to be careful with what went out to our public.

One of my first jobs was to develop an application for the websites I was in charge of. The editors of the various publications (my bosses) had asked if I could get people to sign up to have the paper delivered.

We decided to use a voucher system where people buying the paper could sign up using the voucher for daily home delivery, the idea being that people buying a newspaper could sign up online and by using the voucher could get 10 per cent off.

I created the sign up pages, 7 of them, one for each website. There was a voucher form field (among five or six other fields, like name, address, email, etc.).

In those days we didn't have any sophisticated ideas about analytics we just measured the number of signups we got. We didn't measure conversion rates.

The bombshell

I got a call about a month later from a colleague working in IT who was responsible for subscribing folks to the newspaper who'd signed up on the web.

He asked, “Hi Steve, this online subscription thing is a pain in the neck, can we automate it somehow?”

I asked why.

He then dropped the bombshell.

He said, “I’m getting too many enquiries from abroad, South Africa, New Zealand, Australia, the USA, it’s too much to handle credit card details by phone when I get like 10 enquiries a day.”

As soon as he said “enquires from abroad” I remember putting my head in my hands wondering how that was possible.

I knew I’d made a mistake because the form should not be able to be completed by anyone outside of the UK.

You needed a voucher from the newspaper itself and the newspaper couldn’t be bought abroad. It should’ve been impossible to register outside of the UK.

Looking at the publishing system confirmed that the voucher field was missing from one of the seven websites.

At least the problem only existed on one site I told myself. It’s unlikely to be a big problem if I make the changes quietly.

But then I remembered a previous conversation a week earlier from one of the editors about taking the form off the site altogether.

Too many queries

I wondered why one site was getting too many queries and another was getting too few to warrant marketing it in the newspaper.

I asked my direct boss who was in charge of web development where he got his figures, how he knew that millions of visitors a day visited the websites.

“Oh,” he replies, “Yes, we have a system on our server called Web Trends, take a look if you like.”

Upon logging in, the whole thing looked new and scary to me.

Being the persevering type and having enough technical knowledge to know it would be difficult for me to break, I looked for the pages in question.

I learned that each page was sorted by its web address and that you could get the amount of page views over time.

Delving and digging into data

I decided to delve into the data looking for the form pages. Pretty quickly I had seven numbers in front of me a number representing the amount of form pages viewed for each website. I then dug deeper to find the number of submission thank you pages and stumbled across something important.

One site had five times more submission thank you pages in comparison to the others. It was the site I had made a mistake with.

In other words, the form conversion rate was five times higher without the voucher field.

I still didn't understand why but remembered that the original problem was overseas visitors (South Africa, New Zealand, USA, Australia).

I began digging again and a full day later – it was my first time with Web Trends – I found that over 60 per cent of the visitors hitting the form pages were people living abroad who couldn't get the paper where they lived but presumably would like it delivered. My hunch was that they were ex-pats, people who had moved away from their roots.

Discoveries

I'd also discovered that over 40 per cent of all the visitors hitting the whole website were from abroad. To me this was huge. I didn't think anyone in the company knew this.

I asked my boss how he used Web Trends but he only ever logged in to find out the visitor counts.

I asked my colleagues from the different newspapers who only ever used the sign up information as a measure. They relied on my boss for the weekly and monthly visitor counts.

That was all that they measured at the time.

I asked all the editors how many subscribers had signed up and most of them said two to four a day from local people but that was about it. "Waste

of time" was a common theme throughout the conversations I had with the print ad people who wanted the voucher print space for ads.

It quickly became clear to me that this was a huge opportunity, but it didn't get around the fact I'd made a mistake and I was still worried about my job. Primarily at this point in my career I was a young web editor and I shouldn't have made a mistake of that magnitude.

Testing?

I approached my boss with the line, "I did a test which has uncovered some extremely interesting results. I left the voucher field out of one of the forms we put online in order to compare them to the others …"

Self-serving? Of course, but in truth no-one cared whether I'd dropped the form field by mistake or not, the main finding here was the uncovering of something important:

1 We weren't serving our overseas visitors as well as we could and there was a demand for newspaper subscriptions from them.
2 Additionally the marketing effort locally was a wasted effort according to all the data I had (and a lot of that 'data' was simply discussions I'd had with editors).
3 Newspaper advertising revenue was worth more than the voucher space in the newspapers.
4 Serving the ex-pats would be very cheap to maintain once the system to process credit cards was in place.

Getting the buy-in and running the tests

Once I'd explained all this to senior management everyone was impressed.

We would run some phased tests dropping the offline vouchers from the newspapers and dropping the voucher form fields from the websites. If sales from ex-pats went up at the same level from the other sites we would invest in an online booking subscription system.

Dropping the newspaper vouchers delighted the editors who wanted the space for advertising and the bottom line results from the ex-pats sign ups

cleared the way from a happy senior management to invest in credit card clearing facilities online.

This also pleased my boss because his unit suddenly took on new importance, having the ability to process orders online meant ideas about how to start ecommerce operations flowed through the company.

Even the original colleague who complained about too much work from processing credit card orders from ex-pats was happy because he got what he wanted in the end, an automated system saving him processing the orders manually.

My first quick win

I'd made my first quick win with analytics, but more importantly I'd talked to the editors in a language they could understand.

I'd used analytics to change a business process. Something I was used to doing years before in my quality assurance days, but something I'd never been able to apply to online marketing and business thinking. This to me was the life changing experience. I didn't realise then how it would change my life, but I recognised that I would never look at web marketing in the same way again and I would always use whatever tools I could to back up my arguments.

There are a number of reasons that my first quick win turned out the way it did:

1 A lot of it was down to luck in leaving out the voucher field that started the chain reaction but I believe quick wins can and should be planned and Chapter 4 will deal with that a little later on.
2 Another reason is that everything I was doing related to the business and it was easy to explain to senior management why I wanted them to listen. This in itself is an important skill and is covered in Chapter 8. This early in my analytics career I didn't know what to call the various metrics in Webtrends so I was actually speaking in a language everyone understood. The language of money and the best use of allocating resources.
3 I was also one of the people at the centre of the business. I was running all the websites and therefore I talked to everyone involved with the

online channel regularly.

4 There was a tool in place that could give me the data I needed.

5 The whole thing was a test anyway, the vouchers were just an idea the management came up with as a test to increase sales that failed locally but actually worked internationally so no one minded. Had there been someone whose "baby" it was to come up with offline vouchers I might have had more problems.

This leads me to what I see as the biggest obstacle to getting web analytics grounded firmly into the daily lives of every company employee.

There is an abundance of data. There are hundreds of tools. But there is a fundamental lack of analytics culture. There are few mechanisms in place that will help you or your colleagues take advantage of the tools or data at your disposal. Since I wrote the first edition of this book in 2008 I've helped dozens of companies get the most out of their data and take their analytics to the next level.

Throughout the book I'll use a case I'm calling ACME plc. This is a totally fictitious company, but it's based on situations I've worked on in companies since I wrote the first book. By telling their stories I am hoping I'll be able to directly relate the theories from the first edition of cult to actual situations that exist in your business or project.

1 The hub and spoke method

We must become the change we want to see.

Mahatma Gandhi

1.1 How the Internet has changed our world

Unless you live in a cave or somewhere completely untouched by technology, the way you do business has changed enormously since the arrival of the Internet as a practical business tool.

Consumer empowerment

Anyone can use the Internet to find out about products and services. This has huge implications for your business. First, your competition is out there. Search engines like Google have empowered people with the ability to find out anything written that has appeared on the public domain. This includes the products and services your competition are talking about. Now, unlike any other time in history, there is the ability to compare all this information quickly and easily. I can look at your product or service in Europe as well as a similar offering from the USA or Australia in less time than it takes to make a cup of coffee. The consumer is now more likely to ignore traditional push marketing methods in favour of doing online research when it suits them, not when it suits you.

Communication

No aspect of the Internet has changed our lives or the way we live and work more than Email. It has enabled 24-hour global business to become a reality. It has also revolutionised internal communication within companies. Mobile access to email and other forms of communication has also helped to make the world a smaller place. There are also public forums in which customers or competitors can openly talk about the products and services you offer in a positive and a negative way. Think for example how influential TripAdvisor (www.tripadvisor.com) has become when planning a holiday. The first thing a lot of people do is look for customer reviews via services like TripAdvisor before they make their bookings. In addition to being able to read peer reviews about your products and services your customers can also publish any information they want quickly and easily via services like Wordpress (blogs and articles) YouTube (videos) and a wide range of social media applications like Facebook. In this environment it can be easy for you to lose the control of your brand but again this is double edged, you can also use this to great advantage.

Data abundance

Businesses have begun to understand the abundance of data, which can be easily gathered by analytics tools for **all kinds of business purposes**. It is now easier than it has ever been to track the effectiveness of a newspaper advert because of the Internet. Online you can measure clicks, visits and visitors, pages viewed, where people come from, keywords, paths, track campaigns and a whole lot more. All vital data that can help you improve your business.

Big Data is a term which first came to light in the 1940s but became very popular in 2011–12[1] when the growth in processing power and integration meant all kinds of data could be queried, stored and used. You can get real time data about the weather and change business communication on a website to suit.

In fact the **amount of information** you can get is often **overwhelming** and this is the paradox. Never before have we had so much data about the behaviour of our customers and prospects. Part of the problem now is that the people who know the data is there don't know how to capitalise on it. The opportunity here is obvious, with this much data you should be able to

take real advantage of it. The threat is that you can't figure it out and your competitors do.

1.2 How the information revolution changed your business

Your business has now got all the communication tools, at least (email, Internet access, global networks, website), and perhaps you're even shifting the focus of your marketing to make it more and more customer-centric to take advantage of the information revolution.

There are reasons that you have embraced the technology so prolifically and come to terms with it in the way that you have at the speed you have. These reasons are more to do with culture than with the actual technology.

1 **Urgency.** The mid 1990s was the Internet's "wild west" period when everyone was rushing to be online. If you weren't online you were missing out on the "*information superhighway*". Companies were appearing that didn't exist before, everything from consultancies to help you get online, online marketing companies to help you get "traffic" and *"hits"* meant everything. You were missing out if you weren't online, thousands of people from all over the world had money to buy your goods and services so you needed to get online and quick or miss out on the gold rush. This sense of urgency was slightly misplaced but the effect was the same, people sat up, took notice and took action.
2 **Lead top down.** The charge wasn't led by a geek in glasses. The charge was led by people with titles like CEO, founder, chairman of the board of directors. These folks understood the sense of urgency and had built up enthusiastic executive teams to make the changes happen.
3 **Vision.** The vision was there that the Internet could cut costs, open up new markets for a lot of businesses and there were some extremely big successes to look at as proof (for instance Amazon in 1996/7). The vision of the success you *might* or *could* achieve existed in the mind of every leader of every business and that was enough to get them excited enough to invest and use the technology.

4 **Internal communication.** The practical communication aspect of the Internet cannot be overstated. The fact that you could communicate easily over large distances via email and get the message in seconds meant the vision of the Internet was a very easy sell. A CEO could now email everyone in his business the same message, essentially for free. This was probably the biggest catalyst for change, the fact that the communication medium was so strong. The vision was easy for the management to communicate because everyone could easily understand the benefit.

5 **Actionable.** Many of the Internet's tools actually fitted in nicely with the way most businesses worked. For instance, communication was made easier by email, chat rooms, bulletin boards and later things like video satellite links, Skype and other Voice over IP technology. Marketing had the potential to improve – but it took longer to work like it does now because people had to learn a new way to do things. Cost cutting happened very quickly, admin procedures which required routine and expensive faxes were quickly replaced by email, telephone bills were considerably lower, why speak to the guy at the other side of the world when you could drop him a line? The Internet was very quickly actionable and effective in a business sense and therefore very quickly adopted.

6 **Quick wins.** Most businesses could quickly produce short-term wins based around the communication power of the Internet. Cost cutting could quickly be produced as proof that the people leading the charge were right. Every time a new deal was signed from somewhere abroad it was hailed as a "victory for the online strategy". Because of the new communication ability these quick wins were often communicated by email to all staff and so the march toward a cultural change continued.

7 **Continuous organisational improvement.** Because the Internet worked so well all the changes were built upon. Systems got better, suppliers got slicker, employees who learned quickly and were positive got moved up the ladder often to new positions created by the management leading the charge. Roles like web designer, digital marketing manager, online marketing manager, search engine marketing specialist, VP online sales, director of online research, chief information officer and many more appeared as the new roles were required. These new roles meant that

even more culture was built around the Internet tools being used in the business.

8 **Routinely used.** When was the last day you can remember where you were offline all day? When was the last time you sent an email or at least thought about checking your email?Yesterday? Last week? Or an hour ago? It is simply a part of your daily life, accepted and normal. Actually the thought of not being able to access this routine is now worrying to us. Unless you live in a remote part of the world that is not yet developed and has no Internet access the thought of a few days without net access is at the very least an inconvenience.

Looking at these powerful cultural drivers it's easy to see why the Internet has been so successfully integrated into most businesses in the developed world today. Without the cultural drivers behind the adoption of the Internet as a business tool we might still have a small network that existed for the use of governments and military.

In order to fully benefit from analytics tools you need to change the culture in your company to that of a data driven culture. In order to do that you should follow the 8 steps that you've just seen demonstrated that worked for the Internet.

1.3 The cultural factors present from my introduction to this book

According to John P. Kotter, a leading consultant in the change management field, there is an eight-step process in change management. For more information on change processes within your company I recommend you read Kotter's book, titled *Leading Change*.[2] He goes into a lot more depth about an overall change process.

I've believe his eight-step process works well with building an analytics culture. Hopefully I will demonstrate how this method combined with a lot of best practices developed over the past 15 years at more than 60 companies can help you to build a data guided organisation yourself.

I have combined many approaches and used a fictitious company ACME plc as the case study throughout the book to tie it altogether. There are

simply too many non-disclosure agreements I've signed over the years to do this any other way.

To begin, if you haven't already, please read the first example I described in the introduction. I didn't recognise it at the time but as you can see every part of John Kotter's process was followed. I believe the, urgency, vision and the leadership were the biggest factors in helping the analytics culture develop roots in the organisation.

1 **Urgency.** Newspaper sales were going down because web readership was going up. This was – and still is – a huge threat and had blasted away the complacency in the publishing industry. Later when I discussed the "test" I made to my boss I talked about how much it was costing us and how much we could benefit by testing take up from ex-pats. He passed this onto senior management that meant point two (below) was initiated.
2 **Lead top down.** The managers and senior guys after seeing my results made the decision to test and spend resources seeing if they could improve their business results. They understood the urgency and had the vision already.
3 **Vision.** The vision was to get the online channel working for the business so that losses from sales of newspapers to the online channel could be re-claimed. Because my findings fit with that vision it was an easy decision to continue with the testing. The vision was *not* about analytics it was about the bottom line.
4 **Internal communication.** I was right at the heart of the business in a central role. I took advantage of this by communicating with everyone involved. I knew for instance that one website was ready to drop the coupon idea and knew that others were saying they could get more money from the advert space in the publication. I also knew that one site was vastly out-performing the others in subscription sales from overseas. There is important. A central unit functioning for the business as a whole but independent of the business units can be very useful in breaking down information silos.
5 **Actionable.** One of the most important things I did was go to the management team with an actionable plan. It was very easy for them to say yes to a phased roll out of dropping the coupon fields on each website. I was playing every positive I had to get them to agree. I knew

that some parts of the business were dissatisfied so we could start roll outs with them. I didn't communicate it with technical jargon, I communicated it by saying "Because of the figures I'm seeing, I think we can bring *x* amount of extra revenue through each site and return to our original advertising revenues in the print publication." They liked that because my plan was simple and easy to understand.

6 **Quick win.** The whole thing would be proved as a quick win in 1 month or else we would scrap the idea. I had given them a timeframe to prove my findings and the whole win was very low risk from a cost point of view. I had "planned" the quick win and everyone was waiting for me to produce the results.

7 **Organisational improvement.** The management elevated my position and also elevated the status of the online team because we now had more ecommerce functionality in house because of the credit card capabilities.

8 **Routinely used.** Web analytics became one of my weekly tasks and reporting/actions were part of the process. I promised editors one insight a month that they could use.

In the words of Catherine Courage:

> Change is always hard, and it can be scary if you're not sure if your executive team supports the changes. When we started working outside the product organizations, we discovered that many teams were very risk averse. They would come up with great ideas for different ways to do things, but when the rubber hit the road, they would revert back to their old ways. However, some teams found it easier to embrace change, and what they had in common was that they received explicit support and encouragement from their executive managers. It didn't require much to make a difference—just that an executive on the team would come in and say, "Guys, fantastic ideas. This is the behavior I want to see. Keep pushing forward!"[3]

1.4 Leadership, vision and steering the ship at ACME

ACME is a hardware and software manufacturer with a turnover of around $1.2 billion, about 5000 employees and headquartered in the silicon valley. John Bull, the CEO of ACME, had been a successful VP at one of the top 10 companies in the world. He was 38 and had a 15-year career in his previous position moving up through field sales positions to mid-management and then fast track success when leading a major software business.

The company he worked at previously were only 20 years old and had grown up in the Internet generation. Indeed they (and John) had helped shape a lot of the technology and software we take for granted today. His unit was a team that generated over $1 billion in profit over the 3 years he was VP of that division (approximately 500 people).

He'd known nothing but success prior to ACME. When he was appointed as CEO by the board 5 years ago he'd been tasked with three major challenges in his first 5 years:

1 Get the company back to profitability. That meant cutting costs. They were spending too much time, money and resources on things that were not driving profitability.
2 Find out what the company could be number 1 or 2 at in its market.
3 Devise a plan to become 1 or 2 in those markets that he'd defined.

Point 2 he'd figured out by the end of year 1. Points 1 and 3 were a lot harder to execute. He'd had to cut staff by 15 per cent, merge certain divisions into one another, drop others altogether and start changing the culture of his organisation to be much more agile as a result of having less staff.

In his first year he made a loss of nearly 10 per cent on revenue. A big failure in his own eyes but actually better than the financial markets expected. The second year he broke even which he saw as his goal for year 1. The third to fifth years he'd made back the money ACME had lost in the first year of his tenure.

John was unsatisfied with the turnaround. As far as he was concerned he was still 2 years behind. When he compared how he ran things in ACME to his previous experiences there was a lot of work still to do.

Changing the culture was the biggest challenge

John needed help. His board still backed him – after all he'd achieved the targets that they had set him– but in his last meeting they suggested a consultant might help with change management challenge.

In his previous role John hadn't needed to instigate huge changes as the whole culture was agile enough that if something was a good profitable idea they would move quickly to capitalise.

In ACME it was different. No one seemed to understand what urgency was, why changes had to happen and why the competition were out there taking their pie. Things moved slowly and only when a committee of managers had agreed how to do things.

He engaged an external consultant to help him some up with a change management plan.

Three months and a big bill later the insights came:

1 The company was still largely driven by personalities rather than data driven. Many of the managers interviewed cited a lack of information or insight that led to indecision. Indecision led to nothing happening unless there was a personality behind the division prepared to take risks. In a company that had just cut 15 per cent of its staff this was a rare breed and often the ones that stood out did so because they failed at something.
2 Politics internally was rife. Failure was jumped on as a reason for budgets to be cut or re-allocated. Personal agendas were often the real reasons things happened rather than good business. Fear in senior management was a big factor in people keeping their thoughts private.
3 Silos existed. Marketing had a job which was to drive traffic to the website and generate external awareness. Then it was over to sales to sell (on the website and in trade events). There was little if any real business driven connection between the two. Marketing and sales didn't even operate in the same offices. Hardware worked with the software teams very well, but neither worked well with marketing. Sales (usually direct sales) worked very well with hardware but not well with software.

The recommendations were simple but required decisions across the business:

1 Start taking advantage of data and turning it into good information. ACME had a 10-person data analytics team that currently reported to the whole business but their insights were largely looking back and driving very little action. John had considered culling the whole lot in his last cost cutting exercise. But he also knew from experience that data was key to agility and in ACME it was a process problem not a people problem. Used more effectively and managed from the top down, they could become key to serving the entire business effectively.
2 Focus on increasing cross functionality. Marketing and sales should work together. Software and hardware should make sales more aware of software features that could be sold. Customer care and after sales should be working closely with marketing and sales and at present there was no way aside from the odd meeting here and there to share information.
3 Reward people that have the guts to try things with budgets. Call out the successes and failures and report this across the business when the efforts make sense. Weed out the politicians and either move them to somewhere they can use their skills more efficiently for the business or fire them.
4 Try to make it their idea. This was key to changes making sense. John had already fired a number of executives in his cost cutting cull and there had been very few hires since. The management team were a little afraid they would be next, so he needed to get some people on board that would help him make the moves they needed to make.

The vision

He called a strategy meeting of his executive team. The CFO (Karen), the CMO (Carol), the CTO (James), director of software (Artur), director of hardware (Tim), director of customer care (Lisa) and director of sales (Greg) were all in attendance.

In the meeting John outlined that all of the people in the room had his trust that ACME could truly turn the business around to be the number 1 or 2 in each of the markets they had previously defined. They were currently 50 per cent of the way there and they just needed better information and better teamwork to get the rest of the way.

He pointed out one of the key findings from his change management study had been a lack of information.

Then he suggested that this could be the catalyst for the changes to happen. By adding a strong analytics function in the business with the sole purpose of getting good information to their teams they could all do better business.

Carol, Artur and Tim immediately jumped in with dozens of ideas about how to improve things. This sparked off a good discussion which warmed all parties to the idea and John had a list of items that he saw as action points for improving the operations.

In the end it was decided unanimously that a new director of analytics be appointed to ACME's management team who'd be responsible for improving all their respective business functions. A tough job to say the least.

The vision wasn't about analytics, it was about getting to number 1 or 2 in each market that they were competing in by providing the best hardware and software in the market. That required analytics as a source of information but it also required teamwork across all the business functions and John now had a clearly defined role for the analytics unit that was currently a reporting function as well as a role for the leadership of that team.

Steering the ship

John looked at the team and he knew no-one in the analytics unit had the necessary skills to step up to the executive team. No-one had the necessary strategic experience. So he was looking for a person who could talk the talk but also walk the walk to get good information flowing through the company.

After a number of interviews he made a new hire 3 months later: Bill Brannon, the new director of analytics. His job? Use data to help him turn his business from average to world class.

1.5 The hub and spoke method at ACME

Many big companies acquire smaller companies in order to develop a competence or a business area. If they don't acquire they set-up business

units to develop the competence or business area themselves. ACME plc is one such company that has done both acquiring businesses (it's software division) and building up its own competences in other areas.

At ACME one part of the company handles core products like hardware production while another part of the company develops software to work with the hardware. Then the surrounding support functions such as customer care, Sales, Marketing all sit in their own departments with their own budgets.

In the best companies in the world these units or separate entities communicate well with one another and work as a team. At ACME this wasn't the case.

ACME was very much a departmental business. The left hand didn't know what the right hand was doing. People in marketing worked largely independently from sales with different targets and performance goals. The core business (hardware) was very sales driven meaning that sales worked very well with the hardware folks, but they were often uninformed of what went on with new software developments. These information silos led to inefficiencies.

Information silos are the enemy of any company trying to change into a data guided business.

This does *not* mean you need to go and spend millions on tools that vendors tell you will break down the silos. It means you need to organise your people to be able to effectively share information, work together and all pull in the same direction. That's different to buying a new business intelligence system.

ACME plc wanted to improve their efficiency in internal communications between departments and become driven by the data they knew they had at their disposal.

First they needed to decide upon the key drivers of the hub and spoke model they decided made sense. Figure 1.1 shows the model they adopted. At the centre of ACME is the information hub.

The hub is in its essence simply a method to help break down silos. By organising current tools, processes and people around useful information derived from data ACME could not only drive the business on data but also improve internal collaboration.

On the outside are the spokes. The spokes are the different business units, areas or competences that your business has. ACME management had defined hardware, software, sales, marketing and customer support functions as their major business areas. The job of the hub is supply the spokes with the information they need via processes designed to get the most out of their existing tools. (We'll come to the tools ACME use in Chapter 5).

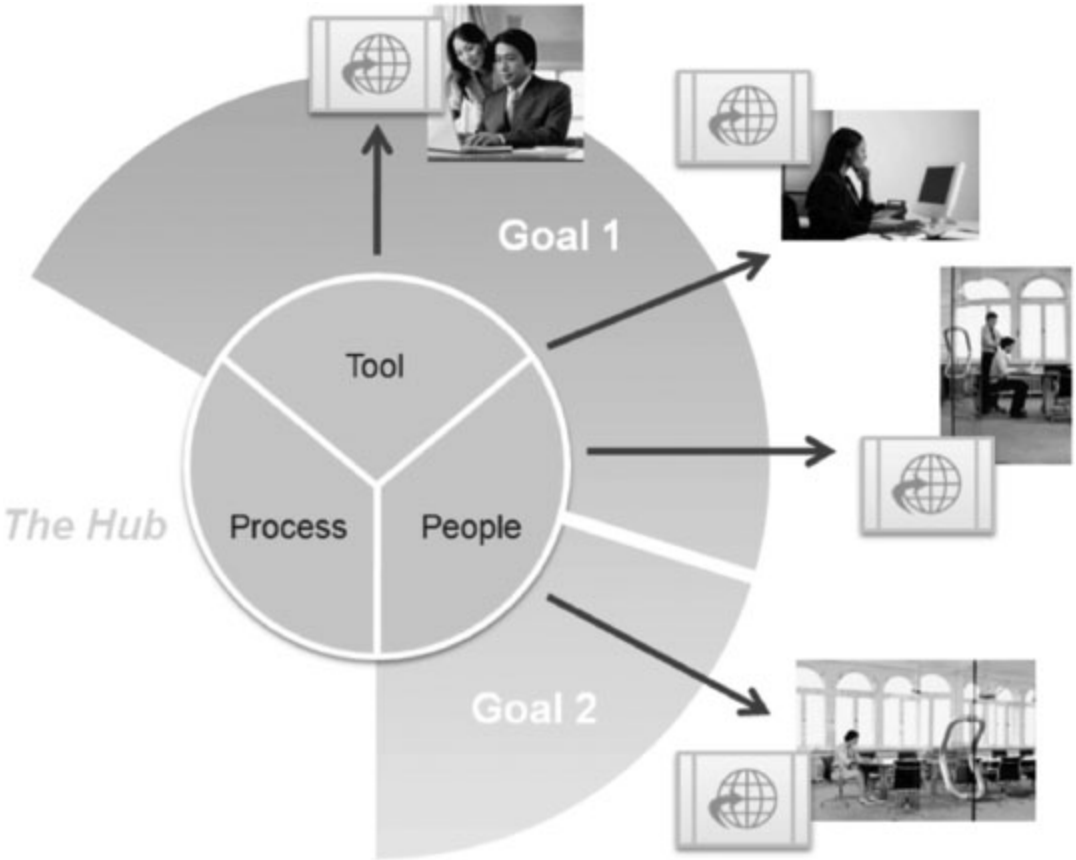


Figure 1.1 The hub and spoke model

The hub people

The hub include people like senior business analysts, technical specialists, data miners, business intelligence people, marketing analysts, web analytics specialists and report directly to someone like a director of analytics who reports directly to the CEO and the executive steering group.

In ACME there was a 10-person analytics function serving the whole business. They now reported to a new director of Analytics who was responsible for the reorganisation of this team and reported to the board of the company.

The main job of this team before the director got there was to supply reporting and insights for the whole company across a variety of different functions. They literally did everything they were asked as a consultative service to everyone but hardly ever met them.

Now things were changing due to a new focus on information and communication that came from the very top of the business.

ACME's director of analytics (Bill)

Bill had 20 years of working experience in various analytics and strategic leadership positions. He had been brought in reporting directly to the CEO in order to structure the team in a way that brought the most value to the company. He wanted to reorganise the business so that data was actionable and used rather than simply a reporting function.

Bill was a contact point between the management team and the analytics teams business. He focused on providing input from the management as to the most important projects the hub should focus on and provided guidance around how to measure success most effectively. He was an analyst himself in a previous life and as such could pass on his experience, very often getting hands on with the tools the organisation used such as SPSS, Excel, web analytics, BI and reporting visualisation tools. This helped him gain the respect of his staff, who saw this wasn't just some yes man exec put in place to manage them.

Additionally Bill was quickly able to communicate with all the major stakeholders around the different services ACME ran. He wasn't just a geek. He spoke their language and made his decisions around organising his team of specialists on the best initial opportunities. What follows is how Bill helped re-organise the marketing spoke using his 10-person team.

Senior analytics specialist (Jan)

Jan had worked for ACME for 6 years as a data analyst and had just been promoted by Bill to a more senior position because he quickly recognised her talents. Her job initially was to provide strategic direction and insights for the marketing team. However Bill decided to place her into the marketing team for 4 days of her 5-day week. Her key role became finding out what stops marketing campaigns working as well as they should, help build performance benchmarks and most importantly optimise and develop the marketing part of the business.

All information was fed up to Bill and the CMO Carol. Having Jan embedded with the marketing team meant they uncovered the insights together and were far more practical with getting things fixed.

Jan being experienced with web technologies knew about things like tag management tools that help with digital tool deployments. She conducted a study with the help of Bill and Tommi to point out some inefficiencies and also monetised the whole thing. It was costing the business at least $10 million a year.

The marketing team were using email that wasn't directly connected to the CRM system. There was no automation going on around search engine marketing or display and along with her junior analyst (Tommi) she immediately pinpointed a lot of ways to spend marketing money better.

Because of this Jan and Tommi became trusted go-to researchers for the marketing team.

Web analytics specialist (Tommi)

Tommi also sat for 4 days a week in marketing reporting to Jan as a part-Javascript programmer, part-analyst supporting act. He basically got her the data she asked for, made sure tagging on the website worked and handled all the reporting for the marketing function.

The reporting was evolving rapidly as the needs became more refined. Tommi actually stopped doing reporting in Excel or PowerPoint and relied upon his analytics tools as he was right there with the marketing team he could simply pull up the data from the system as it was needed.

In his previous 2 years in ACME when he sat in the analytics team he had to supply everything in PowerPoint decks with explanations but now he could simply pull the data when asked and answer the question

immediately. So there was much less reporting in the traditional sense going on.

This meant the whole operation became more agile and Tommi could work more on mining data, finding answers to questions and doing business optimisation rather than providing reports which take time to deliver.

Tommi's biggest marketing issue became tagging the website and various marketing campaigns and he was pushing Jan to help him streamline this process using a tag management tool.

Platforms manager (Harri)

Harri had been with ACME for 10 years and knew everything about the platforms used to drive the digital business. He was an IT function that was partly responsible for the marketing platforms, but also the analytics tools used by ACME.

Jan and Tommi approached Harri about the first phase in an ambitious plan to automate part of the marketing ACME was doing.

They knew if they started using a tag management system (TMS) they could alleviate the time to deploy marketing tags on websites from weeks to hours.

They also knew that if they connected the CRM to email we could use the TMS to help us trigger emails to potential prospects and clients. The TMS could also help ACME deploy remarketing tags meaning if people visited ACME's website product pages they could present ads about those products at relevant times.

For instance one of the company's products was software so if the visitor was browsing "software weekly" a popular web site dedicated to software we could automate the placing of an advert on that website about the software ACME was selling that the prospect was previously interested in. (More about marketing automation in Chapter 2).

Harri valued the business case provided and since the cost of deployment was basically only time (the TMS we used was Google's free version) then testing the validity of the case was a no brainer.

The spoke (marketing)

The head of customer relationship management (CRM), Joan, was also responsible for email marketing. Jan and Tommi had come to her and explained a proposal that would require some resources be allocated to a pilot project automating some email marketing that could be run for prospects that had downloaded white papers.

Initially Joan was against the idea. There were technical problems that had stopped her doing this before. It would require new rules being set-up in the email system, new segmentation of prospects and she wasn't sure whether this was even possible. The solution required technical tags be placed on web pages to trigger the message to be sent. Those tags weren't there and she knew it took months to get that kind of thing done across the business.

She knew that would be hard to convince Harri in IT to approve this because she could never replicate the idea at scale quickly and efficiently.

However, when Jan presented the idea about the tag management system and presented the way the marketing team had thought about segmentation, she started to warm to the idea. When Tommi then showed how the rules would work and the triggers would be fired from the TMS she started to be impressed. When Jan said they had already showed Harri and he was up for the test she was convinced.

She set up a meeting with Carol (the chief marketing officer), Jan, Tommi and Bill. Jan and Tommi explained the idea, Carol and Bill approved it and everything was set in motion.

The outcome

Before explaining the outcome look at how this pilot came about.

1 **Urgency.** In her presentation Jan had pointed out all the inefficiencies in their current marketing as well as potential loss which she'd monetised conservatively at $10 million a year. This was something that could be fixed without any cost to IT platforms and therefore was an easy decision to make.
2 **Led top down.** Bill, Carol, Joan and Harri supported Jan's idea. That was two of the management team and two senior heads of department who were supporting the idea.

3 **Vision.** Since Bill was supported by the management team (where Carol also sat) any ideas he brought up in his early tenure were readily accepted. Indeed something like this was Bill's remit to decide. In a way this was part of Bill's vision.
4 **Internal communication.** Jan and Tommi worked with the marketing team to come up with the solutions. It was everyone's idea not just theirs.
5 **Actionable.** It could be done, they had a plan and they knew how to execute. It was also a measurable activity.
6 **Quick win.** It would take less than 1 month to implement if all parties worked together.
7 **Organisational improvement.** By doing this the email marketing team would have fewer campaigns to create as much of their work became always on. Other campaign tagging processes was also streamlined by the TMS so they could be launched quickly. Additionally because we were using white papers as the trigger, the sales spoke content development team got interested in the initiative and started developing different more suitable white papers fit for this purpose. There were four organisational improvements in one action.
8 **Routinely used.** The marketing team started focusing on better segmentation strategy, and Tommi used the TMS to handle all the website tagging for marketing.

All of these things together resulted in much better working practice. The whole idea was shared between four departments and was a success defined by return on investment in time.

There was approximately $17 million in new sales generated in a 6-month period that came from either retargeted search and display ads or triggered email. The win was a sign to the management team that what Bill was instigating was worth the investment.

Indeed Bill (on our suggestion) had also put a senior and junior analyst in each of his other four major focus areas, all of which had vast improvements in terms of acting on the data. Instead of servicing 5000 people Bill was now only servicing approximately 2000. But he was doing so in a much more meaningful and measurable manner.

He also had a real case for hiring more people to service the rest of the business that needed it to.

1.6 Checklist and exercises

As demonstrated by ACME, there are a number of drivers (mechanisms) that helped make things happen as we've already outlined. In order to make it work for you we've devised a checklist of items and potential ways for you to create a similar situation in your business or scenario. Any change initiative needs to be led from the top and in order to get buy in you need urgency.

1 How do you create urgency?

If you don't have an urgent need it's more likely to be put on the back burner as something to do when the urgent things are out of the way. In ACME's case the urgency came from the board and the CEO so it was an ideal backbone for the change to happen. However in your case the urgency may need to be identified from one of the following risks that businesses face.

A Missing out on an opportunity. The easiest way to create urgency is get everyone enthusiastic about a great opportunity. How are you going to change the world even in a small way? In ACME's case being 1 or 2 in the market across their major product lines would mean they were earning close to four times the amount. That's a massive opportunity that is difficult to do but attainable.
B Losing money/market share. If you can identify waste or inefficiencies that cause your business to lose money or market share. This could be a trigger to create an urgent situation.
C Bad feedback from customers. Are customers giving negative feedback? Have you identified ways to solicit feedback from customers in an effective manner?
D Competitor moves in the market. Have the competition taken a direction that threatens your business or gives you a new opportunity to focus on a gap they have left in the market?
E New Insights. It's something you can fix yourself by re-allocating resources that have been previously unidentified.

Its all really then about how you put a dollar value to the opportunity for your business.

2 How do you get leadership buy in?

Only three things are required to get leaders interested in making a change.

A There is a significant monetised opportunity that the business hasn't yet capitalised on. On other words there is a good business case.
B It's plausible (not impossible) with resources at the leaderships disposal.
C It fits with the company's vision.

If you have satisfied those three criteria a good leader will back the initiative. A bad leader won't. Consider your options.

3 What about the vision?

All good companies have some kind of vision. Analytics is never the vision. Analytics is a tool to help you achieve the dream. What you need to identify are all the ways in which analytics can align with the vision by helping the business.

A Marketing: how does the company go to market? Can analytics help with that? How?
B Lead generation: how does the company sell? Can analytics help with that? How?
C How does the company do research and business development (R&D)? Can analytics help with that? How?
D How does the company keep their customers happy? Can analytics help with that? How?
E Have you considered all external data? Like the weather? Does the weather impact your business? Can knowing that impact what you sell? Can analytics (big data) help with that?

4 How do you communicate well?

Simple is best. Don't over complicate when you can simplify. Roy Williams[4] once said "Speak to the dog, in the language of the dog, about what's in the heart of the dog." What he meant was if you try to persuade the dog that's interested in the steak on your plate that a piece of lettuce is better, you've lost the battle before it's even begun. On the other hand you won't have much problem persuading the dog that steak is good.

All communication is the same. If you're talking to the right person, in the right language at the right time you have a far better chance of persuading them to do what you need them to do.

A Have you identified the people that are stakeholders who can help you accomplish what you need to accomplish?
B Have you learned what their goals and objectives are in the business? Can analytics help them? How?
C Are they working on anything specific? Can analytics help them? How?
D Look people up on LinkedIn and Facebook. Do they publicly state what their professional or personal interests are? Can analytics help them? How?
E Communicate simply with easily shareable media (email, newsletters, intranets, PowerPoint, PDFs, Excel, etc.).

Get a meeting with people whom you feel are important to know and ask, you don't have to go in with an immediate plan. Become their friend if you think you can get on with them. They're your colleagues first but making friends is a sure fire way to earn trust so if you like the people you meet, what's the harm? (But don't make friends for the wrong reason).

5 How do you know if it's actionable?

If it's something you can measure the success of its actionable. Its then about whether the effort is worth it. If it's going to take six months it might be actionable with a long term plan. But that might require greater buy in and usually the 6-month plan has a lot of smaller steps where you can measure progress. So point to the little steps along the way. The way you define if something is actionable relates to the never ending search for the best use of resources.

A Does the plan save or make money if implemented well?
B Does it make a system or process easier for everyone?
C Can you break down a complex longer term goal into measurable actions?

6 How do you define quick wins?

Building business cases or quick wins is covered in Chapter 4, but in short, the idea is to get operational with your analytics tools and find something that is valuable to the business.

A Determine your REAN framework business goals (see Chapter 2).
B Reach/Conversion: what is the best source of traffic in terms of volume and conversion? What is the worst?
C Run a 'what's changed?' report. Are there any differences over time? Days? Weeks? Months?Years?
D Run a disengagement report. Are there any processes on your website stopping your visitor from converting? What is the difference usability wise from best to worst?
E Check the 25–50–25 rule (see Chapter 4).
F Form a hypothesis.

Once you have your idea, you know its actionable and you can help the business make money you have a business case.

7 How do you identify the organisational improvements?

It's possible to identify common themes in the business. For instance if there is a chief marketing officer on the management team that's because the board thought it would improve the business to have marketing close to the people running the company. That's an organisational improvement (or at least a change). If suppliers start being measured differently (cost versus value) or if budgets start getting allocated in a more positive way (based on ROI instead of size of department), these are organisational improvements.

A Have new titles or departments been created as a result of something you have done?

B Have new hires been made as a result of the changes?
C Have people been removed from redundant positions?
D Have suppliers changed (with lower costs or more streamlined services?)
E Are budgets within the business allocated differently as a result of analytics?

8 How do you know if something is used routinely?

Basically, if an organisational change becomes a permanent fixture used by the business regularly, it can be classified as routinely used (for instance, if generated reports are acted upon every month).

A Is the approach regularly used (daily, weekly or monthly) across all parts of the business?

Exercise

Either you're reading this as a student/academic or you're working for a business. In the first case you need to relate the ideas mentioned to a hypothetical company.

Make it as real as possible. If you're working in a business it's ideal. Try to relate the ideas mentioned to your own company.

1 Try asking yourself the questions in sections 1–3 of the checklist above. Can you come up with a way to create urgency, get leadership buy in and align it with the vision of the company (or hypothetical company) you're working for?
2 Then read Chapters 2, 3 and 4 before coming back to this checklist.

After running through the checklist above, have you determined an approach that you can follow to move the needle for the business you're working for up a notch? You shouldn't have all the answers, but you should have a plan to help you get there.

Notes

1 See “A Very Short History of Big Data”, *Forbes*, www.forbes.com/sites/gilpress/2013/05/09/a-very-short-history-of-big-data.
2 Published by Harvard Business Review Press in 2012. See also “Kotter’s 8-Step Process for Leading Change” at www.kotterinternational.com/our-principles/changesteps.
3 From McKinsey & Company, “Applying Design Thinking Across the Business: An Interview with Citrix’s Catherine Courage”, February 2015, www.mckinsey.com/insights/business_technology/an_interview_with_citrixs_catherine_courage .Quoted with permission from Catherine Courage.
4 See “The Wizard of Ads”, www.rhw.com.

2 Reach, engage, activate and nurture

> If we plan, then we can effectively measure; and if we measure, we can effectively improve.
>
> John Quarto-von Tivadar (Future Now Inc.)

2.1 Developing a REAN model

In order to understand what the outcomes of your business efforts are you might need a common framework to be able to effectively measure it.

A few years ago one such framework was created called REAN. It's an acronym that stands for:

- **R**each
- **E**ngage
- **A**ctivate
- **N**urture.

I've found this an extremely valuable training and planning tool for almost a decade in any business, big or small.

REAN helps you to visualise what you're measuring and why for planning purposes. All KPIs and metrics fall into at least one of the four dimensions of REAN.

By using a framework like this from the beginning you eliminate all kinds of discussions around "what engagement is" for instance, as REAN clearly defines engagement across all business functions.

Introducing REAN

The history of this model goes back to 2006 when an ex-colleague of mine, Xavier Blanc, came up with REAN framework with help from another colleague, Leevi Kokko. I then helped develop the model further for measurement purposes. Xavier and Leevi's comments in the box below provide a good introduction to the concept.

I came up with this loose framework while I was working at Satama. The basic point was to get a large client of mine grasp the very simple idea that there's no point at all in investing a lot of money in creating cool websites if no one was looking at building traffic for these sites in the first place, if no one was then defining what to do with these hard-won contacts, how to leverage them as part of an entire branding, marketing or sales effort, now and in the future, online, but and this is very important, also offline … (because as much as it hurts to recognise when you're working in a web agency, there IS life outside the web and guess what, most of it still happens THERE!).

In other words, the two messages I wanted to deliver were:

1 Break the silo walls: it's not SEM + website + CRM, it's about creating EXPERIENCES that flow from the very first moment a contact is established down the lifecycle of that contact. This often implies, especially in large companies, difficult organisational alignment efforts and clear leadership challenges.
2 Define KPIs: Define what your business goals and objectives are. Think hard, before you do anything and invest any dollars. Only then does it become possible to design effective and efficient marketing ARCHITECTURES. Creatives are cool and certainly needed, but without a good, solid architecture, they're pointless.

Initially, this framework was more educational material than anything else. However, we instantly started to use it as a tool to structure our work and ground our metrics. This was my contribution.

Xavier Blanc, managing director, Fjord (www.fjord.co.uk)

Having seen Steve present the KONE case at eMetrics Stockholm, I was impressed by how the initial REAN model was improved and how it was utilised very cleverly in the analytics process.

I remember the first realisation very well, and from the very first project it was created for it really turned out to be an invaluable tool in customer communications and planning in many ways. I'm personally using a variant of the model with practically all of my clients now."

Leevi Kokko, head of analytics, YLE (www.yle.fi)

Defining REAN

Every product or service can use all aspects of REAN as part of a framework.

All products or services need to **reach** an audience of potential customers.

Then the sales teams, copywriters or creative advertisers need to **engage** with the prospects in some way. That could be talking to them, teaching them, helping them or persuading them that their product or service is what they need.

Then they need to **activate** them that is to take the first step, buy the first product, give the sales person permission to contact again or simply say yes.

Finally once your customer has said yes at least once you need to **nurture** them, in other words encourage them to come back again and again and consume more.

When it comes to the measurement of REAN with analytics we define these four factors in the following way:

R Reach: the set of activities needed to raise prospects' attention for your brand, product or service.

E Engage: the gradual, typically multi-channel, often recursive set of activities needed to engage the prospects you just won.

A Activate: the activities needed for your prospects to take, eventually, the actions you wanted them to take.

N Nurture: the activities needed to nurture the customer relationship you just managed to create.

You can then directly apply measurement as an application of the framework.

R Reach: measures the methods you use to attract traffic to your website or online presence and measures how people find your brand, product or service.
E Engage: measures the click depth and time spent interacting with your online creative elements and processes.
A Activate: is a measure of the effectiveness of visitors that have taken an action on your website, preferably one that you had pre-defined and wanted them to take.
N Nurture: measures the effectiveness of the way you actively encourage your activated visitors to come back and consume more of your website content.

Framework thinking

It's not really about REAN – it's about using a framework that everyone understands. The REAN acronym is just an acronym. In Nokia, for instance, they used AECR instead of REAN as the terms internally were better understood: *acquisition*, *engagement*, *conversion* and *retention*.

Google Analytics is now set up to measure *acquisition*, *behaviour* and *conversion* (ABC). Google measures all nurture activities as acquisition (email and CRM are classed as acquisition campaigns in Google).

The purpose is the same but the acronyms are different. I'd like to think that the first edition of this book and the REAN framework included there has influenced the way many companies have evolved their thinking.

Many references to REAN exist and it always delights me to see someone using it. But feel free to rename it to something your own company culture understands. It's far easier to adapt REAN to your corporate culture than to adapt your corporate culture to using REAN.

Of course if you have no framework then REAN could be used. But the point I want to be clear about is that your efforts shouldn't be directed at

getting an acronym adopted in your company. It's not about making REAN famous. It's about getting your company to use a framework to define their goals, objectives and measurement strategy in the same common terms across the business.

Doing that helps you create a culture that works in the same direction.

How REAN helps build the process

Reach helps you define and understand how to use analytics tools to measure the effect of everything from a brochure, fliers, TV and radio commercials, corporate events as well as all online marketing.

Engage helps you understand all the places that customers or prospects might interact with your brand so you can develop a solid understanding of how to begin thinking about your assets or content strategy.

By understanding what you want your customers to do, when you want them to do it and why, you can attempt to persuade them to become **active** participants in your business. You persuade them to actively consume your products, services or content. So you begin to think about your persuasion strategy.

Finally once they have consumed, what's the next step? How do you **nurture** them to consume more? This is using different reach channels to satisfy and inform your customers of things you're doing that they may be interested in, or helping them to use your products and services to the maximum.

How REAN works

In the first edition of this book I outlined a number of different definitions of different elements within the REAN plan. In 2015 I think we've moved beyond needing a clear explanation of what SEO and SEM is. If you don't know what any of the elements are in the upcoming frameworks, Google is your friend.

In short, though, the first thing to do in your REAN model is make a mind map.

How REAN helps people understand the ecosystem

Once you understand how the REAN model works you can easily visualise what you need to do in terms of tactical operations like website tagging requirements, campaign measurement requirements as well as understand how platforms work together. It will help identify bottlenecks and issues that could be causing operational inefficiency.

It nurtures discussion around ideas that might never have happened before.

REAN allows you to quickly develop visualisations within your organisation that both marketers and technicians understand.

What follows is how ACME formed their first REAN model.

2.2 The ACME REAN model – departmental goals

On the one day of the week all his team were working directly in the analytics unit of ACME (or the Hub as they affectionately referred to it), Bill Brannon called together his senior specialists.

He had Jan who worked in marketing, Fredrico who worked in sales, Mari who worked in software, John from hardware and Sam who worked in customer support. Each was the senior analyst that after a couple of months working directly for their new teams had revolutionised the way each department used information.

There was still a lot to do, however, and a problem that Bill had identified. There was no common terminology between each department and Bill was having trouble understanding what his businesses common objectives were. He was trying to find a way to unify the business processes and get them working more closely together. To do that he opened a discussion with his team about the goals of each department.

Different activities in different departments

In the meeting Bill asked his colleagues what the key drivers across each department were. What did the everyday conversations look like? What were the things that people focused around and how did it help the business. The team came up with the following.

Marketing

In Marketing everyone talked about campaigns and were very tactical about things like doing Google adwords, SEO and driving traffic from display traffic sources. There were discussions about how to tie in offline campaigns to online marketing initiatives and other such ideas to get people into the sales funnel.

The customer target markets were also segmented into different verticals such as telecoms, finance, retail, gaming, etc., and campaigns were targeted to the different verticals.

Sales

In sales they were talking about classifying leads. They had a scoring model called PSM (prospect scoring model). If you were PSM level 1 (psm1) you were scored lower from a sales leads perspective than if you were psm3.

Psm1 Lead name/contact method known and the prospects company falls into one of our target markets.

Psm2 As Psm1 but also a known business interest has been categorised (such as a white paper had been downloaded with a specific angle).

Psm3 As Psm2 but also sales have opened up a discussion which has led to an opportunity (sales meeting or call booked).

The sales team were focused on turning all prospects that were Psm2 into opportunities. The problem here was that some opportunities were better than others and there was no clear way see which sources the opportunities came from as that was largely handled by marketing or in some cases research and development cases.

Hardware and software

In hardware and software (H&S) it was all about production, engagement and information. They were looking to help sales get leads by helping people find relevant white papers, but mainly they were trying to hit serious deadlines on production and sales goals.

For instance their cloud services team had direct eCommerce activities they needed to meet targets on. They were moving a lot of their customers away from in house hardware ownership to a cloud based model that allowed ACME to hit a more global customer base (one of the key improvements since the new CEO took the reins) and hardware was developing in a different direction than it had previously taken.

Software was now all available as direct downloads (often with free trials) where before they were part of the hardware sales packages. So the goals of H&S were clearly aligned with one another. In a way they were responsible for leads, trials and both direct (online) and indirect (offline) sales.

Offline sales were often sold via distributors and partners so there was a different H&S team called ACME partnerships that handled those. There was an online lead generation system for this as well and ACME partnerships was growing.

H&S were naturally tied together on the main product lines, but there were other issues that often caused problems. H&S strategy was such a big part of the business that research and development (R&D) also got involved and would often push their own marketing initiatives without involving anyone else in H&S.

This would often conflict with other H&S targets. The head of R&D (Ellen) saw her part of H&S as a separate function that ruled the future growth of the business. Indeed she'd written an internal newsletter that was widely distributed and read by many in the company and been relatively successful in getting face time with the executive management on a number of occasions to get her ideas approved.

The problem was that she was a bit self-serving. She always had the business best interest at heart but was ambitious and experienced with marketing so could push her own agenda with her own budgets and used her own third-party agencies. Unlike many others in H&S she would often use data to back up her arguments, something the CEO had insisted on since his appointment. She knew how to use the system to her advantage.

Customer care

The focus of customer care was to build long term relationships with some of the bigger clients. Like sales there was a scoring method for clients so that customer care specialists knew how to allocate their time.

The business largely fell into three areas: small to medium enterprises (SMEs), mid-range and enterprise level.

- **SME**: Employed less than 100 people and less than $250K in annual revenues. Over 35 per cent of the business were in this segment, but because of cloud services this was the fastest growth area.
- **Mid-range**: Employed 100–1000 people and generated $250K to $10M in annual revenues. 50 per cent of the business was in this range and they were the majority of the customer care standard focus area.
- **Enterprise**: Employed 1000+ and generated more than $10M in annual revenues.15 per cent of the business was in this segment and each account had its own dedicated team in customer care.

Not only did they segment their customers based on importance to the business but they also used an recency frequency (RF) model to predict when they should reach out to their clients with email or calls. They found that if a customer had not been contacted in 6 months with some kind of relevant messaging they were more likely to defect to a competitor.

Customer care had two major headaches. The first was supporting software and hardware problems. Typically in the early stages of the relationship with their client they would run into various issues that needed solving. This would be a period where support was absolutely necessary but consumed a lot of resources (calls generally) that led to high costs. One of the goals was to reduce that cost.

The other major headache was learning how and when to contact customers to reduce churn to competitors. The wide variety of software and hardware sales in the company didn't help that process. Some were very small customer lifecycle that needed to be contacted quickly whilst some were very long relationships that needed a lot of support. The key was nurturing their customers at the right time with the right message for each one.

2.3 The ACME REAN model – definition

As Bill listened to his analysts go through their various issues and measurement challenges he realised he needed a framework to pull all of these functions together. He needed to break down the barriers that existed between the divisions and get them working together on the same blueprint.

In his previous job as a strategic consultant focusing on analytics he'd used a model called REAN to visualise different functions from each and he decided that together with his team he'd develop one for ACME. This could be used to help the management team understand how the business could work together and get internal buy in to drive the changes.

After his team had told their various stories he got up and started drawing on his whiteboard first dividing it up into four vertical sections and naming each section in turn. Reach, Engage, Activate and Nurture.

He drew a line through the centre of the 4 vertical lines and on the left hand side in the top segment wrote "online" and in the bottom segment "offline".

His team looked on curiously.

Bill introduced the REAN concept as a customer lifecycle framework by which ACME could map out the various touch points that prospects or customers interacted with across their business.

He started with **reach**.

Reach

He explained that reach would be all the ways on and offline that ACME reaches prospects. Not customers at this point, just the people they were anonymous to them.

The brainstormed what they knew, and (without getting too tactical at this point) Bill grouped all the ways they reached new people or generated awareness of the ACME brand.

Online:

- paid search engines
- paid display media
- paid social media

- paid email
- organic search engines
- organic social
- organic referrals
- direct traffic (branded).

Offline:
- paid events (trade shows)
- paid seminars (speaking engagements)
- paid leads – call centre requests for information (RFIs)
- print ads in trade magazines
- direct "snail mailed" catalogues to distributers
- billboards, logos and branding on vehicles/property.

Once Bill had finished with the Reach section he moved onto the Engage part of his white board.

Engage

The next phase was to map out all the touch points that ACME interacted with their customers. Again after a quick brainstorming discussion with his five colleagues Bill had another top level list of things they could identify as touch points with a potential customer.

Online:
- website home/corporate page views
- hardware page views
- software page views
- solutions page views
- ecommerce sales funnel page views
- video/webinar plays/views
- social media likes/follows.

Offline:
- mobile app views
- telephone calls

- snail mail catalogues
- business cards
- trade magazines.

Activate

After engagement had been defined the next discussion focused around how people acted upon the various engagement touchpoints. Again after a quick brainstorming discussion with his five colleagues Bill had another top level list of things they could identify as conversion points.

Online:
- downloads of software trials
- downloads of hardware specifications
- downloads of software configuration specifications
- downloads of solutions white papers
- online sales of SaaS (software as a service)
- social media shares/retweets.

Offline:
- mobile app activations
- telephone (telesales) recorded leads (into the CRM)
- snail mail catalogue enquires
- business cards leads (into the CRM)
- trade magazine voucher applications for free trials.

Nurture

After the high level conversion points were defined as activations the Nurture touch points needed to be defined. Sam from customer support was especially helpful in this part of the brainstorming session. Again at the end of the discussion Bill had filled up his whiteboard with the following.

Online:
- customer email campaigns
- software support forum
- hardware support forum

- sales after-care forum/chat service
- CRM (customer relationship management) platform
- social media real-time support (Twitter).

Offline:
- customer support call centre
- after-sales call centre
- account manager direct contacts.

The outcome

The team Bill had assembled had in a few short months began to understand how the whole business worked. By being embedded in each function they had a unique perspective about what the goals and objectives of each team were. He could never have pulled that off had they been doing reporting functions as they were before. When Bill had finished, his whiteboard looked something like Figure 2.1.

Bill now had a picture of how the business connected together on a very high level as well as a few ideas on measurement points. The next thing he needed to do was make it actionable at scale. There was no point him pushing this from the top down he needed each respective unit to really buy into the idea. He knew that each department needed to make the idea their own to get the whole system to start working together. They had to become the change they wanted to see.

After the brainstorming sessions the senior analysts were hyped about what they could do to get this to work but Bill explained the next step was to get the REAN concept out to the divisions they were working in.

Bill gave Jan, Fredrico, Mari, John and Sam a new task when they went back to their respective teams. They would need to turn the REAN model concept into a practical and measurable tool that each unit could use.

The way that they could accomplish this was to set-up KPI workshops across each division. Chapter 3 covers this in more detail.

2.4 Checklist and exercises

REAN does two things very well.

First, it gives you a very clear indication of the measurement challenges you might have when breaking your strategy down into its component parts. Tactics can be decided from the ACME example (like Figure 2.1). If you understand that you're going to be driving traffic via social media, you have to put tactics in place to measure it that are different to the tactics you use for measuring visitors from Google.

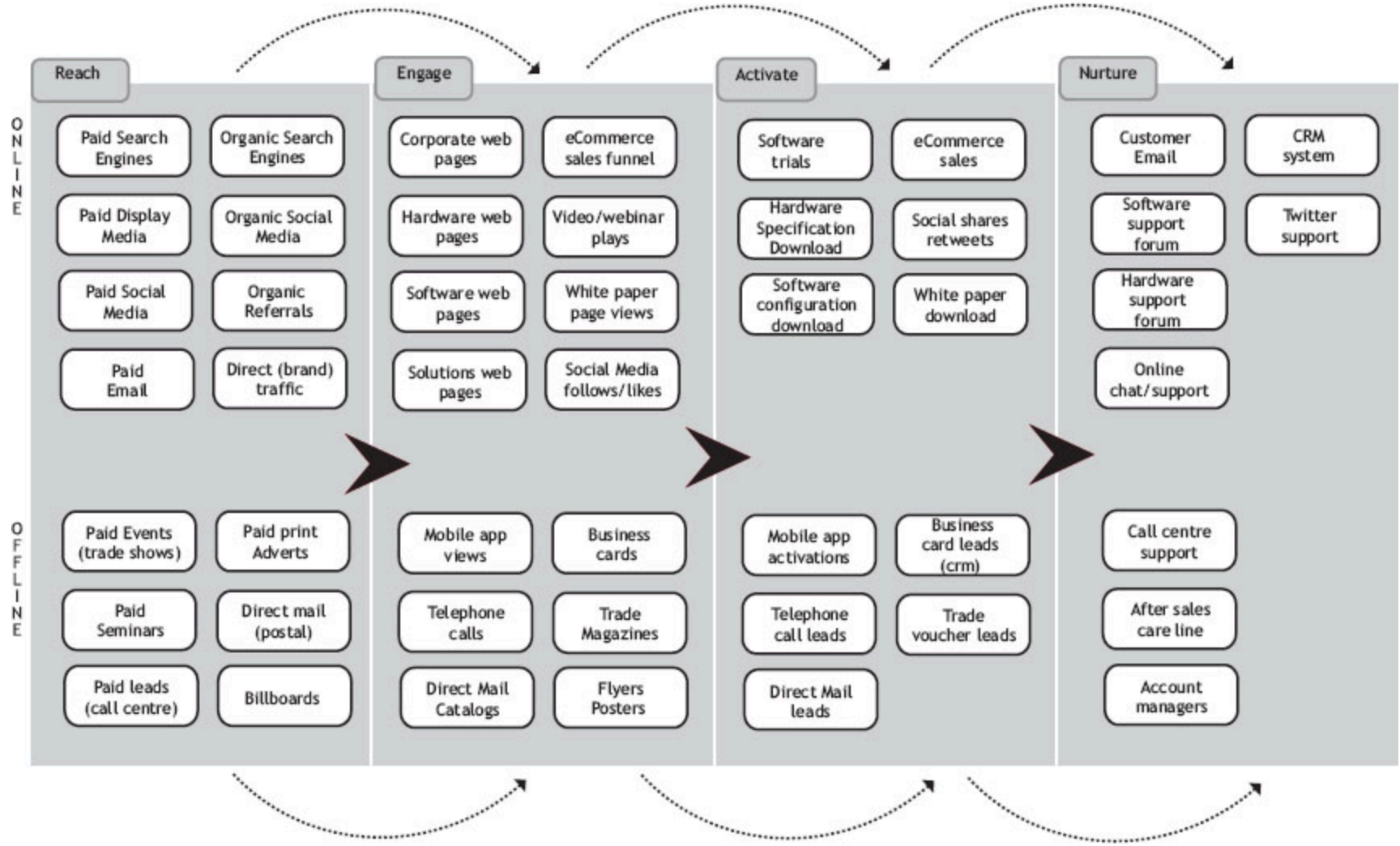

Reach
Engage
Activate
Nurture
ONLINE
OFFLINE
Paid Search Engines
Organic Search Engines
Paid Display Media
Organic Social Media
Paid Social Media
Organic Referrals
Paid Email
Direct (brand) traffic
Corporate web pages
eCommerce sales funnel
Hardware web pages
Video/webinar plays
Software web pages
White paper page views
Solutions web pages
Social Media follows/likes
Software trials
eCommerce sales
Hardware Specification Download
Social shares retweets
Software configuration download
White paper download
Customer Email
CRM system
Software support forum
Twitter support
Hardware support forum
Online chat/support
Paid Events (trade shows)
Paid print Adverts
Paid Seminars
Direct mail (postal)
Paid leads (call centre)
Billboards
Mobile app views
Business cards
Telephone calls
Trade Magazines
Direct Mail Catalogs
Flyers Posters
Mobile app activations
Business card leads (crm)
Telephone call leads
Trade voucher leads
Direct Mail leads
Call centre support
After sales care line
Account managers

Figure 2.1 ACME high-level REAN model

Second, it can be used to help you combine process, platform (tool) and people strategy. By looking at REAN even on a high level you can start to realise where it would be useful to use different tools your organisation has, the people you'll need to do it and the processes you'll need to create.

We'll use REAN a lot more in Chapter 3 when we come to define KPIs which is the part where we involve people with the process. Before getting to that point though there are a number of drivers of the REAN model you could identify within your own business. Again you should develop this for your hypothetical company or your own business as an exercise to see if you can derive value from a REAN framework. You could do this alone or in a workshop/brainstorming environment (preferred) and the outcome should be a mind-map like Figure 2.1.

1 Identify a division (IE a product or service line) within your business.
2 Identify the activities they're most focused on.
3 List the top 10 on- or offline most important reach sources to this business.
4 List the top 10 on- or offline most important engagement or interaction points with *potential* customers.
5 What are the actions you want people to take? List the top 10 on- or offline most important conversion points with potential customers.
6 How do you get customers to consume again? List the top 10 on- or offline most important engagement or interaction points with *current* customers.
7 Develop the model into a mind map visualisation. I use Adobe Illustrator or Photoshop, but there are free tools such as MindNode[1] and others to create your visualisation very quickly.
8 Create a graphic representation of the REAN model that can be shared via PowerPoint or email.

Note

1 See Life Hacker's five best mind map tools at http://lifehacker.com/five-best-mind-mapping-tools-476534555.

3 Key performance indicators

> Everything that can be counted does not necessarily count; everything that counts cannot necessarily be counted.
>
> Albert Einstein

3.1 What are KPIs?

According to Einstein, success is not the opposite of failure. What he meant was you have to figure out why things fail with as much clarity as figuring out why things work and this is what key performance indicators (KPIs) help you to do.

There are two types of KPI: the visionary KPI and the tactical KPI.

Visionary KPI

The visionary KPI reflects what your company is trying to achieve. It's the big hairy audacious goal (BHAG)[1] defined as a metric.

In his book *The One Thing You Need To Know* (Free Press, 2005), Marcus Buckingham described this as the "Core Score." In it he asked Sir David Ramsbotham, then the chief inspector of Her Majesty's prisons, what their key metric was. It turns out that in Sir David's prison reform process he changed the metric from "number of escapees" to "number of repeat offenders". In doing so he completely changed the culture of the prison system. This visionary KPI is one that drives change or helps your company build its culture. This has to be defined by your leaders as part of the strategy.

Tactical KPI

The tactical KPIs depend on the goals and objectives of spokes – covered in Chapter 1 and here in Chapter 3. This chapter will cover how to develop management level KPIs starting with their goals and objectives, applying industry thinking and a workshop process.

Digital Analytics Association definition

The Digital Analytics Association (DAA) has developed standard terms and definitions around KPIs. Since the purpose of the DAA is to standardise the approach to using web analytics I would advise you follow their definition in your tactical workshops.

There are three types of analytics metrics – counts, ratios, and KPIs:

- **Count** – the most basic unit of measure; a single number, not a ratio. Often a whole number (visits = 12,398), but not necessarily (total sales = $52,126.37.).
- **Ratio** – typically, a count divided by a count, although a ratio can use either a count or a ratio in the numerator or denominator. (An example of a ratio fabricated from ratios is "stickiness.") Usually, it is not a whole number. Because it's a ratio, "per" is typically in the name, such as "page views per visit". A ratio's definition defines the ratio itself, as well as any underlying metrics.
- **KPI** – while a KPI can be either a count or a ratio, it is frequently a ratio. While basic counts and ratios can be used by all Website types, a KPI is infused with business strategy – hence the term, "Key" – and therefore the set of appropriate KPIs typically differs between site and process types.

A fourth type of definition is included for terms that describe concepts instead of numbers:

- **Dimension** –A general source of data that can be used to define various types of segments or counts and represents a fundamental dimension of visitor behaviour or site dynamics. Some examples are event and referrer. They can be interpreted the same as counts above, but typically they must be further qualified or segmented to be of actual interest. Therefore these

define a more general class of metrics and represent a dimension of data that can be associated with each individual visitor. Metrics are measured across the dimensions (REAN represents four such conceptual dimensions).

A metric can apply to three different universes:

- **Aggregate** – Total site traffic for a defined period of time.
- **Segmented** – A subset of the site traffic for a defined period of time, filtered in some way to gain greater analytical insight: e.g., by campaign (e-mail, banner, PPC, affiliate), by visitor type (new vs. returning, repeat buyers, high value), by referrer.
- **Individual** – Activity of a single Web visitor for a defined period of time.

The definitions set out by the DAA are now quite widely accepted as industry standard and you can get a copy of the whole file from the DAA website.[2] At the time of writing you can still get this free document at the address listed in the footnotes although now the rest of the DAA site requires membership to view the contents and become part of the community.

REAN explained in Chapter 2 are some "dimensions" referred to by the DAA, a conceptual way to define different counts or ratios, such as referrers (part of reach) and conversion rate (part of activate).

They define the KPI as either a count or a ratio, and go on to say "KPI is infused with business strategy – hence the term 'Key' – and therefore the set of appropriate KPIs typically differs between site and process types."

Infusing KPIs into business strategy

In order to infuse KPIs into business strategy four attributes are assigned to the metric that mean actions are taken depending on the performance being indicated. If you can apply all four of these attributes to your metrics then you have KPIs. If you can't then what you have is a count (number) or a ratio (a calculation of two numbers).

Every KPI should have the following attributes assigned to it.

1 The metric has a timescale associated to it (i.e. it is reported once a month).

2 The metric has a benchmark (see standard deviations later in this chapter).
3 The metric has a reason to be reported to an actor (we call them actors because they "act" on the information).
4 The metric has an associated action surrounding it if a problem occurs. (If metric = xxx an action is taken to rectify the situation).

The DAA defined *unique visitor*, *visit* and *page* views as the "big 3" counts. This is because nearly all ratios and KPIs include at least one of these three metrics. You can also substitute page views with clicks or events in a rich media environment where pages themselves don't exist. The point that the DAA are making is that there are people (unique visitors), doing things (page views) in a time frame (session or visit).

Taking the DAA example of a ratio – page views per visit – a typical number reported by all web analytics tools you can see that on its own it doesn't really help you do that much. However, if you said page views per visit was a KPI the attributes and business logic would apply.

1 *Time period:* You would be looking for trends over a time period (such as a month).
2 *Benchmark:* You would start with the first figure (5 page views per visit) and monitor the figure. If the figure deviated (depending on your standard deviation), for instance, either plus or minus 20 per cent from 4–6 you would be flagged to take action.
3 *Actor:* The analyst is flagged if the benchmark indicated there was a reason to investigate further.
4 *Action:* If the page views per visit have deviated you would conduct a study to examine what had occurred to make such a big difference. It may be for example that users have found some content that they were confused by and meant they were clicking through a lot more pages to find the content they were looking for. The action then would be to fix the problem either by taking action directly or passing the information to the steering group to take a decision.

A real-life KPI illustration

MTV3 are a media company in Finland. They run one of the most popular websites in the country regularly recording over 1.5 billion page views per

month. The numbers are publicly available on the TNS Gallup website[3] which includes all the top media websites in Finland (the list shown gives a weekly count). While the list includes only the Finnish sites with the TNS metric tags (in other words Facebook, Google and Bing are missing) it does have quite a comprehensive list of all the natively grown media sites in Finland.

It's vitally important that MTV3 retain the high page view levels they currently enjoy so that they can sell advertising space on their website.

One of the most important things is the MTV3 front page.

One of the editors of the site asked me "How do we know *when* to change the content of our front page?"This is a great question because the front page of the site is a very important entry point for a lot of their visitors. The point is if they put content on the front page which doesn't sit well with the visitors then they need to know when to change it to try and improve click through and therefore number of pages viewed.

The editors working on the page need to monitor this in real time. Because the time of the day affects the number of visits to the website ratios not counts are required.

In order to answer the question on a page level, I might suggest two KPIs: bounce rate and page time index.

Bounce rate of the index page is the amount of visits (sessions) that visited only once and left the page without doing anything further as a ratio of total visits to the page. The page time index is a sum of the total time spent on the page divided by the total visits to the page as a ratio. Alone, these numbers are just ratios, as we already discussed. However, if we infused business logic, thus making them a KPI, this example scenario might play out as follows (the numbers below are not accurate reflections of current MTV3 figures and are used for illustrative purposes only).

Front page bounce rate

1 *Time period:* Hourly
2 *Benchmark:* 15 per cent and 20 per cent deviation (deviation upwards is the action flag). This means that the actor knows that on average the page bounces at 15 per cent. If his page scores below that there is no problem.

If he scores above that (by more than a 20 per cent deviation, so 18 per cent or more) he needs to take the action described.

3 *Actor:* Editors working on the page.

4 *Action taken:* First, look to see if the page time index is dropping at a comparable rate. If so it is time to change the content as visits looking at the page are viewing it with a worse than average time and a more than average "lack of interest". If the page time index is not dropping it could mean that the article needs tweaking (the headline may need to be more attractive for instance) or there may not be enough links to related pages embedded in the article.

As you can see from this the page time index is the secondary factor in the decision being made to take action and what action is taken. It's no less important however than the combination with bounce rate for answering the business question and you should use the same approach.

Front page time index

1 *Time period:* Hourly

2 *Benchmark:* 85 seconds and 20 per cent deviation (deviation downwards is the action flag). This means that the actor knows that on average each visit views the page for 85 seconds. If this time drops by 17 seconds or more (20 per cent) action needs to be taken as it is deviating away from average levels. If the time spent is lower it simply means sessions are spending less time on the page but moving throughout the site, which also meets the business objective, therefore no action required. If the time spent is higher than 85 second it means sessions are engaging longer than average with the page and actions described should be taken.

3 *Actor:* Editors working on the page.

4 *Action taken:* If lower than the deviation first check to see the bounce rate. If this is deviating at similar levels then change the page content. If it is higher than 85 but the bounce rate is Ok (or lower) then it may indicate a usability problem. Please inform the hub analysts to investigate further.

The outcome might be that those 2 KPIs answer the business questions that people working for MTV3 have about their website and how they should

improve it.

What follows is a full description for ACME on the KPI workshop process that will help you design tactical KPIs.

3.2 The ACME visionary KPI

When the CEO John Bull arrived at ACME the business was driven by KPIs relating to revenues, profits and operational costs. Shareholder value was the main KPI that the owners of the company wanted to turn the dial in a positive direction.

ACME wasn't the best in the world at anything they did and John believed that a vision had to relate to more than operating profit. In order to turn the business around they needed a big hairy audacious goal that everyone could get excited about.

John, the board and his management team decided to critically look at all the businesses they were in. They then decided to focus on the ones where they believed they could be number 1 or 2 in that particular market.

By being the biggest and best in the market places they competed in ACME could confidently say they were "the best" or "world class" and actually mean it. By giving the businesses in ACME a solid growth target and by mapping out investment plans to become number 1 or 2 they knew financially what they had to do (both from a cost cutting and investment perspective). They also either sold off businesses they couldn't compete in or discontinued them in order to raise investment capital for the focus areas. People wanted to be the best and got enthusiastic about what they were doing again.

The job took 3 years to complete in terms of acquisitions, mergers and sales across their business but now they were starting to see the benefits in terms of hitting their targets. They had become number 1 in one business area that drove 35 per cent of the annual revenues. The growth had helped them turn a profit last year for the first time in 5 years which was encouraging.

But John still realised he needed the culture to work as one team despite the numerous businesses they operated and in a company employing 5000 people it was challenging. John hired change management consultants to

help him understand the key areas he could improve the whole business (see section 1.3) and it helped him define his 'information challenge'.

When he decided to hire an Analytics director to join his management team and help him resolve the issues he had getting the right information, to the right people, at the right time, he'd been impressed with Bill Brannon's answers in the interview.

Bill had told him about how the REAN framework had helped him establish cultural change, helped him define KPIs that aligned business models and focused efforts on the right things across business function.

KPIs across the business

In sections 2.2 and 2.3, the process Bill went through with his senior analysts helped define how they could go back and re-create the same situation in their respective units. Their teams had to become the change they wanted to make. According to Bill there was no point forcing the idea on people, the people in the respective spokes had to fully understand, buy into and own the ideas.

As soon as Jan left Bill's meeting she went back to her unit and started working on arranging a workshop for the marketing unit.

As an experienced analyst and communicator Jan had run dozens of workshops before. Bill had explained you needed to involve a small team of the right people that consisted of the main stakeholders.

People

Marketing analysts

You'd need the workshop facilitator and an assistant, Jan herself would facilitate the workshop, she was going to organise and run it. Tommi her back up in the marketing unit would be the assistant. It would be the first time Tommi had seen a KPI workshop being run so it would be good experience for him. His main job was to take all the notes while Jan ran the workshop but he was also a good source of information as he worked with the analytics systems daily pulling out insights for the marketing team.

Chief marketing officer

It was imperative that Carol could give the management perspective to the workshop as she would be the main budget owner for the whole of marketing. She knew what the main goals of the business were from a marketing perspective and would also lend real credibility to the whole exercise. If she was there it meant everyone's boss in the marketing team was paying attention and so other people were more likely to take the whole thing seriously.

Carol was busy but Jan found a slot she could attend and decided to re-arrange other peoples calendars around this date.

Campaign managers

Jan also needed the people that did the day to day campaign work to be involved so they could give their perspective. She picked people responsible for different marketing areas of hardware and software campaigns. The more mixtures in their responsibilities the better this would be. There would be 4 campaign managers that could make the workshop that represented most of the campaign marketing in the business. James was in charge of Software as a service (ecommerce) marketing, Jack was in charge of inbound trials (software), Helen was in charge of Hardware services marketing and Fred was the main man for inbound hardware leads marketing.

Technicians

The main campaign tagging work was handled by Tommi since his new role in marketing had been established, however there were still programmers, content management specialists and developers that worked for the marketing team that had responsibilities such as making sure the website performed and that content was updated.

Jan invited the platform's head of IT, Harri, as he could offer the best business-wide perspective.

Tools

Stationery/office equipment

All the best workshops involve interaction of all participants. There is nothing worse than sitting doing nothing but listen to someone else for 3 hours so Jan had planned in some interactive parts to her workshop. For that she needed Post-it notes, flip charts, magic markers pens/pencils etc.

She also needed a prepared PowerPoint presentation that could explain the KPI workshop concept as quickly as possible and get people thinking. An internet connection to a big screen view of the website was also made ready.

There were also light snacks and drinks provided, plus coffee and a 15-minute break planned in the middle of the workshop. Jan understood that people only have a limited attention span and there needs to be time to think, take a toilet break, have a cigarette or a coffee (or both). She wanted the workshop light and interactive, not a heavy 3-hour session with no breaks.

The introduction

On the day of the workshop Jan quickly got everyone settled in, coffees were distributed, jokes about the weather were cracked and pleasantries exchanged before she got down to business.

She quickly introduced the Hub and Spoke concept. The cross functional team approach that was being adopted by ACME. She mentioned that Tommi and herself were part of the hub people but also embedded with the Marketing department and that this was an information sharing strategy. She also mentioned that all the other major business functions in ACME were following the same model and would be arranging workshops just like this one. There were nods of approval around the room. This made sense to everyone, as they'd all benefitted from having Jan and Tommi in the marketing team.

Jan then opened up the stages they would go through in today's session. The slide shown in Figure 3.1 was up on the screen as she described how the workshop would be conducted.

The actor definition

Jan set about defining the "actors" the main stakeholders of the marketing unit, in other words everyone sitting in the room. She pointed out that the

reason they had been selected is that any KPI they developed together would then have someone to act upon the indicator direction. If the performance indicator went up, great, what was the action? Spend more, for instance, to earn more? If the indicator went down, why? Could rectifying actions help stop the problem?

For instance, if a campaign manager saw negative results they may “act” and stop a campaign from wasting more money.

The first 5 minutes of the workshop was then determining what levels of action people could take.

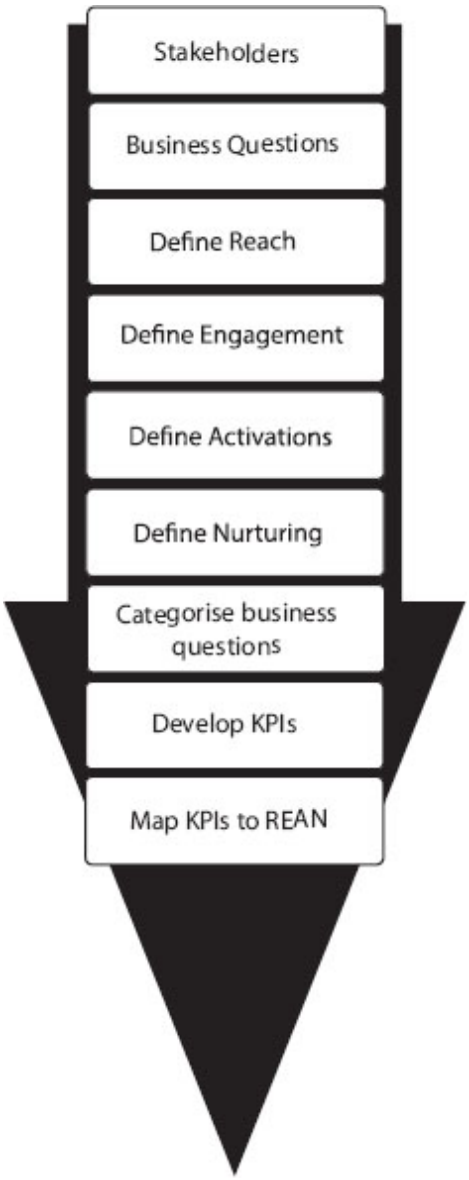


Figure 3.1 The KPI development process

The CMO

Carol had the most power to stop or start any campaign or initiative, but she would not act unless the information was conclusive one way or the other.

The campaign managers

All the campaign managers had the budget control and power to stop or re-allocate funds as they saw fit. However again if the data was not telling them something conclusive then they would not make any radical changes.

The technicians

The tech guys had the control over the platforms the digital marketing activities were run on such as the websites and landing pages. If they saw something wrong they could move to fix anything technical. As with all technical problems it could take minutes to fix or days depending on the issue, but there was a simple ticketing process for all issues technical.

The analysts

As such Jan and Tommi had the job of supplying actionable information. The flags were the information that the whole team needed in order to function so it was vital that the information flowed freely and was as conclusive as possible.

Once Jan had finished this part of the discussion and Tommi had made notes about the main challenges that stopped actions from happening they moved to the next part of the process.

The business questions

Jan introduced the second part of the workshop as one she'd run before in other departments and her record was 27 business questions asked. The subtle reason for this was that this gave the people in the room a benchmark for what was expected of them. She emphasised that the point of this part of

the workshop was simply to uncover business questions they may have that they want answering regarding the everyday work they do.

Jan asked "What's hard about your job and why? What information would you need to do your job better?"

It has to be run in a perfect world scenario. It shouldn't matter what tools you have or what kind of potential tool you could have as your analytics system.

Jan said, "If there is a business question you want answered lets write it down and later we'll see if we can come up with KPIs to match it."

Jan then kicked off by taking up a magic marker and writing the first question on her flip pad, saying "This is a question I get all the time in the marketing department."

1 What is the best source of traffic in terms of volume and sales?

She then explained that by knowing this they could re-direct marketing funds to get more volume or more sales depending on the goal. At first it was a little quiet as everyone thought about the concept.

Jan asked Carol directly what she is often asked by other colleagues in management.

She answered "Which channels are the most productive?" and when Jan asked her to explain she was writing the second question down on the whiteboard:

2 2 Which channels are the most productive?

Carol explained that executive management would often want to know how to spend their money at different times, so not just for sales or volume, but for awareness of a new product line. What kind of campaign worked in different environments? So what for instance was the best campaign channel to reach new customers versus old customers?

Jan then wrote down questions 3 and 4:

3 What kind of campaign worked in different environments?

4 So what for instance was the best campaign channel to reach new customers versus old customers?

Once Carol started others joined in and within 30 minutes or so people were all throwing in lots of ideas:

5 Is it good or bad that the content in our site changes?
6 How can we get prospects to buy?
7 How many prospects make the purchase decision online but buy offline and vice versa?
8 Where are our visitors coming from?
9 Which channels are overlapping – are reaching the same visitor/visitors?
10 Which landing page converts the best? (And why?
11 Do the registered users buy more than others?
12 What's the most searched product or service on our site?
13 How many software download trials do we get?
14 What is the value of a free download to ACME?
15 How many leads become sales?
16 What is the average response time for lead response?
17 What's the average lead value?
18 What is best, getting people to buy direct or getting people to come back and buy?
19 How engaged are our visitors with our content?
20 What are the top paths through our site?
21 How do we know when something isn't working on our website?

Once people started running out of questions Jan started going through the list and categorising them based on the REAN model (see Chapter 2) always explaining why she did so to the team who would either agree or come up with angles Jan hadn't thought of. Her final list looked like this:

1 What is the best source of traffic in terms of volume and sales? (Reach/Activate)
2 Which channels are the most productive? (Reach)
3 What kind of campaign worked in different environments? (Reach/Engage/Activate/Nurture)
4 So what for instance was the best campaign channel to reach new customers versus old customers? (Reach/Activate/Nurture)
5 Is it good or bad that the content in our site changes? (Engage)

6 How can we get prospects to buy? (Engage/Activate)
7 How many prospects make the purchase decision online but buy offline and vice versa? (Reach/Engage/Activate)
8 Where are our visitors coming from? (Reach)
9 Which channels are overlapping – are reaching the same visitor/visitors? (Reach)
10 Which landing page converts the best? (And why?) (Engage/Activate/Qualitative)
11 Do the registered users buy more than others? (Engage/Activate)
12 What's the most searched product or service on our site? (Engage)
13 How many software download trials do we get? (Activate)
14 What is the value of a free download to ACME? (Calculation – see Chapter 8 on monetisation)
15 How many leads become sales? (Activate/Nurture)
16 How many visits does it take before a lead becomes a lead? (Engage)
17 What's the average lead value? (Calculation)
18 What is best, getting people to buy direct or getting people to come back and buy? (Activate/Nurture)
19 How engaged are our visitors with our content? (Engage)
20 What are the top paths through our site? (Engage)
21 How do we know when something isn't working on our website? (Engage)

You'll notice that Jan added a couple of things as she went; for instance, on question 10 she added "qualitative" to the categories. This was because the landing page that converts the best can be reported by using engage/activate metrics (landing page conversions for instance as a list) but the only way to know why someone converted better on one page than another was to add qualitative data. You could either develop a hypothesis based on heuristic analysis of the pages or ask the people that signed up via a survey. Both require qualitative data.

Also, question 17 ("What is the average lead value?") is a calculation extrapolated from sales value and the amount of leads over a given time frame.

By pointing these things out as she went, Jan was helping her colleagues understand how she was going to frame the questions into KPIs. Not

something obvious from the discussion.

Workshop technique

What you want is as many business questions you can get.

Jan got the ball rolling by writing "What is the best source of traffic in terms of volume and sales?"

The she asked for more things from everyone in the room. She went to the most vocal or senior person first (Carol), who started the ball rolling.

Once a few people there in some ideas, Jan encouraged the room with helpful comments. She made sure every question was written on the flipchart. This served two purposes: first, it made sure people don't repeat the same issues in the workshop, and second, it stimulated thinking, when people can see the progress they are making, they start to feel really involved.

Facilitating business questions

As a facilitator of the workshop you can do the following to help this stage of the workshop run smoothly:

1 Start off the questions. One I always ask is "Which source of traffic is best in terms of volume (number of visitors) and cost?" I then explain the thought process behind asking this question. I'd say something like "By knowing the answer to this question we'd be able to direct our marketing spend to the sources of traffic that were most effective in terms of value for money and numbers".
2 Write the questions down. When you start it off, write it on the flip chart. When someone else asks a question which hasn't been asked, write it on the flipchart. This shows the people in the workshop that they are making progress and also helps to get them thinking.
3 Give the actors a benchmark. 49 questions defined is the record I have had from a workshop. When I mention that you would be surprised how many workshop teams try to beat it. Most of them don't (the average is 25-30 questions) but it really gets people going.
4 Encourage questions by agreeing with the actor and help justify the question where possible. For instance if the question "What is the value of

a download to our organisation?" was asked by a workshop actor, I'd answer with "Great question, by knowing the answer to that we can help justify our marketing spend on getting people to download as well as start monetising our website" while writing it on the flipchart. This helps the atmosphere and also improves the chances of a shy actor asking a question.

5 If the process is slow to start, ask the senior guys in the room specifically what they think would be useful to know. You could ask open questions like "Carol, what's the hard part about your job? What do you want to know about the online part of your business? How could knowing something about the behaviour of our visitors online help you in your work?"The senior people are in senior positions because they have opinions and thoughts about these kinds of things. They will get the party started.

Once you have exhausted the room and they can't think of anything else then you have done a major part of the workshop. You now move into the last phase defining REAN so this is a good time to have your coffee break.

After a short break Jan now moved into the final phase of the workshop itself. Defining the REAN model ideas for ACME's Marketing unit.

On one sheet of her flip chart Jan drew two lines. One from the middle of the chart vertically down all the way to the bottom and one horizontally across the middle so that she was left with four squares on the flipchart. She wrote Reach in the top left, Engage in the top right, Activate in the bottom left and Nurture in the bottom right.

Everyone in the room had Post-it notes and pens.

Defining reach

Jan asked everyone to write down the top five most important reach sources (on or offline) for ACME. There should only be one source per note, so each person should have finished with five Post-it notes. She asked everyone to do this alone, it wasn't a group discussion.

Once each person had at least five Post-it notes she collected all the notes and started sticking them to the top left box called reach. Where similar ideas appear she started grouping them together.

From 35 Post-it notes she had the following groups of similar answers:

Online:
- Paid search engines. (6)
- Paid display ads/banners. (6)
- Organic search engines. (6)
- Direct traffic (branded). (6)
- Paid social media. (4)
- Organic social media. (1)
- Organic referrals. (1)

Offline:
- Paid events (trade shows). (3)
- Paid leads (call centre RFIs). (2)

By not allowing discussion prior to writing the notes Jan had made a point of showing that everyone in the room knew the importance of Google paid search campaigns, display, free search engine traffic and direct or traffic they called branded. These would be the top areas to look at when determining reach KPIs.

Jan allowed a brief discussion around the reach sources to glean any further insights before moving onto defining engagement.

Defining engagement

Jan pulled up some of the business questions everyone had already worked on earlier in the workshop and identified the questions that had an element of Engage in them.

She then went through each question one by one and asked the group to explain how a prospect would typically interact with ACME in each case. She then also wanted to understand what a successful interaction would be.

The questions were:

3 What kind of campaign worked in different environments? (Reach/Engage/Activate/Nurture)

Answer: Jan ruled out question 3 because it was too broad a question, as it encompassed everything ACME did.

5 Is it good or bad that the content in our site changes? (Engage)

Answer: Content in software and hardware landing pages changes frequently (often by using A/B testing tools), especially on campaign landing pages. Success would be clicking the call to action from the landing page resulting in at least two pages being viewed.

6 How can we get prospects to buy? (Engage/Activate)

Answer: They either bought online directly or became a lead allowing sales teams to follow up. The success would be a sign up (via the online shopping cart) or a lead form being filled in. This would be either five pages being viewed (cart) or three (sign-up form process).

7 How many prospects make the purchase decision online but buy offline and vice versa? (Reach/Engage/Activate)

Answer: The only way that ACME could at answer this was to understand what registered users did. A registered user was someone who'd signed up as a lead (at least three pages viewed) and then would have downloaded a specific product sheet (minimum two more pages).

10 Which landing page converts the best? (And why?) (Engage/Activate/Qualitative)

Answer: Similar to question 5, the success was determining which from the landing pages resulted in calls to actions being clicked (minimum two views).

11 Do the registered users buy more than others? (Engage/Activate)

Answer: Similar to question 7. A five-page success metric ending in at least two conversions.

12 What's the most searched product or service on our site? (Engage)

Answer: This required that they mine the data of keywords entered into the search engine. The minimum success for this was two pages being viewed (the landing page with the search engine and the successful search results page).

16 How many visits does it take before a lead becomes a lead? (Engage)

Answer: Jan asked Tommi "how many leads were returning visits?"Tommi pulled this figure up from their analytics system and they discovered on average people had to visit five times before they signed up or bought anything.

19 How engaged are our visitors with our content? (Engage)

Answer: Jan again asked Tommi what the average page views per visit was. This would determine at least on a high level how people looked at the content on an aggregate level. Tommi showed that on average people looked at 2.3 page views per visit and an average of 3 minutes per visit.

20 What are the top paths through our site? (Engage)

Answer: ACME defined the top paths they wanted people to take were the Shopping cart, (five pages) the sign up process for software trials (three pages) and the registration process for hardware solutions and services (three pages). What people actually did was look at hundreds of different paths but that was something ACME couldn't control.

21 How do we know when something isn't working on our website? (Engage)

Answer: Related to all the engage answers it seemed that the ideal situation for the visitor and ACME would be a 3–6-page visit with a conversion of some nature. If something wasn't working it was probably because they couldn't find something that might lead to more pages being browsed/searched for. Tommi again dug into the data to look at people who'd viewed more than 8 pages per visit to see if they had issues. He noticed a number of new visitors that had looked at 12 or more pages had

also searched for a specific product. When he went to that product page there was a 404 error message (page missing). After a brief discussion they settled on 10-page visits being a potential "warning flag" for new visitors for when something might not be working.

Segmentation

All traffic is not equal and the better web analytics tools today can segment (examine different parts of your traffic) based on visitor behaviour (see also Chapter 6 on this topic). Jan had used this session for ACME to determine some segments that were the most useful from an engagement standpoint. Going through the findings from the questions above the group determined four segments that could help ACME identify opportunities or problems:

- New people viewing 1–2 pages were "not engaged" as defined by ACME. These people may have come to the wrong place. (Not engaged segment)
- People returning 5 times or more could be interesting to ACME to turn into a lead or customer based on past averages. (Prospect segment)
- Returning people that looked at 3-10 pages may be a good prospect and had more potential to become customers than the "not engaged" people. (Potential prospect segment)
- Non-registered people that looked at 10 or more pages were potentially lost (Flagged segment).

Segmenting their analytics tools allowed ACME to see which sources of traffic drive "engaged" visitors.

Those visitors that stay more than a few seconds, or move through the steps in your shopping cart, or enter your lead generation process are more valuable to you than visitors that see your page for a few seconds and leave.

Jan listed what the four segments were in the engage part of the flip chart and moved on to activate.

Defining activations

Jan went back to her flip chart and asked the group to do the same as they did with reach to define the define the activations or conversion points.

Jan asked everyone to write down the top 5 most important conversions (on or offline) for ACME. There should only be 1 conversion per note, so each person should have finished with 5 Post-it notes. As before she asked everyone to do this alone, it wasn't a group discussion.

Once each person had at least 5 Post-it notes she collected all the notes and started sticking them to the bottom left box called Activate. Where similar ideas appear she started grouping them together.

From 35 Post-it notes she had the following groups of similar answers:

Online:
- Downloads of software trials (7).
- Online sales of SaaS (software as a service) (7).
- Downloads of hardware specifications (5).
- Downloads of software configuration specifications (5).
- Downloads of solutions white papers (5).
- Social media shares/retweets (1).

Offline:
- Telephone (telesales) recorded leads (into the CRM). (4)
- Mobile app activations. (1)

What this did was determine what the ACME group thought the most important conversions were thus helping Jan define the KPIs.

Once the activations were defined Jan moved onto the last part of the activities for the workshop group. Defining methods of nurturing your visitors so that they become loyal to your brand.

Defining nurture

Jan again pulled up all the questions from the earlier part of the workshop that had a nurture element. She then went through each question one by one and asked the group to explain how a customer would typically interact with ACME in each case. She then also wanted to understand what a successful customer interaction would be.

The questions were:

3 What kind of campaign worked in different environments? (Reach/Engage/Activate/Nurture)

Answer: Jan again ruled out question 3 because it was too broad a question as it encompassed everything ACME did. It was too complicated to answer in this workshop alone.

4 So what for instance was the best campaign channel to reach new customers versus old customers? (Reach/Activate/Nurture)

Answer: There were 4 major ways ACME reached their customers. Email campaigns, The software and hardware forums for registered users, chat/aftercare services online and the social media real-time support service. The success depended on how many customers became repeat customers and over what timeframe. They also wanted to stop customers defecting to competitors and so would like to measure churn rates in order to know when to contact people again.

15 How many leads become sales? (Activate/Nurture)

Answer: This could be measured via their CRM system. A lead was registered in the CRM system when someone signed up for a download or a support forum membership. Simply measuring how many of those people became customers was key to answering the question.

18 What is best, getting people to buy direct or getting people to come back and buy? (Activate/Nurture)

Answer: Related to question 4, ACME could determine this by learning if new customers or repeat customers were more valuable in terms of revenue over the customer lifecycle.

Now the group had been working together for 3 hours on the workshop and had had a lot of useful discussion Jan pulled up the slide she showed at the beginning of the day (Figure 3.1, page XXX).

She explained that their work was now done and Tommi and herself would go and develop KPIs from the information gathered in the room. The

KPIs themselves would then be used across Marketing to start optimising and improving the unit performance.

What follows is a checklist to run the workshop in your own company.

REAN workshop checklist

1 On one sheet of your flip chart draw two lines. One from the middle of the chart vertically down all the way to the bottom and one horizontally across the middle so that you have four squares on the flipchart. Write "Reach" in the top left, "Engage" in the top right, "Activate" in the bottom left and "Nurture" in the bottom right.

Reach

2 Ask everyone to write the five most important reach sources on a Post-it note. One reach source per Post-it note is required and the objective is to get each actors opinion (not a collective discussion). So people should do this alone.
3 Collect everyone's Post-it notes together and group similar sources together by sticking the Post-its to each other in the "reach segment" of the flip chart.

What this does is determine what everyone in the room thinks are the most important sources of traffic and helps the analyst when defining the KPIs. Usually we find that 5–10 reach sources are defined with a clear top three or four that everyone agrees on. This is the point of the exercise.

Engage

4 Go back through the business questions to identify the processes which exist on your website. Work out how many clicks/pages it takes to complete the process and how long this process takes in seconds. This could be one set of criteria you could define in your segment. For example if it takes five pages and 60 seconds on average to buy a product you could use this as engagement criteria, because people that stayed this long potentially could have bought a product.

5 Look at the average time spent/pages viewed on your website from the tools you have. This could be used as a benchmark for engagement.
6 Define which pages are the most important. For instance you might want to segment purely by all the folks that saw your services page. This in itself could be a segment.
7 Again write everything down on the flip chart.

Activation

8 In the same way as you asked with the Reach exercise earlier ask everyone to write the five most important activations on a Post-it note. One activation per Post-it note is required and the objective is to get each actors opinion (not a collective discussion). So people should do this alone.
9 Collect everyone's Post-it notes together and group similar activations together by sticking the Post-its to each other in the "activation segment"

Nurture

10 Go back through the business questions and pull out ones that you think might relate to nurturing. Remember that nurturing is all about helping your customer or your visitor become happy with your brand so that they buy more of your products or services.
11 Nurturing activities usually happen after "activation" or "conversion" of one sort or another. It may be that a visitor has subscribed to a newsletter and you are keeping in touch by sending the news for instance. It may be that you're running customer service activities, helping your customer with frequently asked questions or difficulties they have with your product/service. Both of these occasions have come as a result of an activation (a newsletter subscription and a purchase). Try to think of things that could happen after you have sold or converted someone in some way.
12 Repeat visits, purchases (or repeat activations) are also nurturing activities. Look for things that happen two or three times from the same customer/visitor as potential nurturing factors.

After the workshop was over Jan and Tommi gathered together all the notes and made sure they were all correctly recorded digitally so they could refer to them easily.

Jan had already categorised most of the questions in the workshop and explained as she went to the people in the workshop, but it's worth going over how she did that.

Using REAN again

First, Jan took the information provided by the stakeholders and categorised them into reach, engage, activate and nurture questions. By doing this she hopes to reduce the amount of KPIs required and also makes it clearer what KPIs need designing. The idea is not to create a KPI per question but to create five or six KPIs that cover all or as many questions as possible. This allows people to focus on a few things rather than everything.

In order to demonstrate how she did this we'll take a few questions from her workshop discussions with which to go through the process:

1 What is the best source of traffic in terms of volume and sales?
3 What kind of campaign worked in different environments?
6 How can we get prospects to buy?
18 What is best, getting people to buy direct or getting people to come back and buy?

Jan was thinking in terms of the REAN framework. She broke the question down based on the elements within the question that fit the framework. For instance:

R **Reach:** measures the methods you use to attract traffic to your website or online presence and measures how people find your brand, product or service. In other words Jan looked for questions that involve how you attract people to your offers, their areas, their locations, demographics and categorised them as reach questions. Offline Jan was looking for sources like flyers, direct mail, branding on the sides of transports, event leads, business cards etc.

E **Engage:** measures the click depth and time spent interacting with your online creative elements and processes. So here Jan was looking for

anything that might show how visitors interact with ACME. How they move through processes is an engage question, the landing page they hit is the first engagement, the time they spend looking at pages is engagement, the number of pages or clicks they make are engagement questions. Also offline just speaking to someone on the phone is engagement or talking to someone at an event. The point being that an interaction has taken place. The difference between reach and engagement is the interaction. Reach is the channel that a prospect might see, engagement is the interaction with the channel.

A **Activate:** is a measure of the effectiveness of visitors that have taken an action on your website, preferably one that you had pre-defined and wanted them to take. Activate questions are those which assume an action has been taken. How many downloads, purchases, subscriptions, logins and such like are activate questions. Anything that includes some form of conversion is an activation. The difference between engagement and activation is that engagement may not lead to a conversion, an activation is always a conversion. When Jan can easily identify it's something that ACME wanted their visitors to do, she knew it should drop in the activate category. It should also be the FIRST time that an action has been recorded by that particular person, otherwise it goes into the Nurture category.

N **Nurture:** measures the effectiveness of the way you actively encourage your activated visitors to come back and consume more of your website content. Nurture questions are those that generally mean that the visitor is a customer, a lead, a subscriber, or made contact with you in some way so that you know who the person is. They have already activated at least once and this is the key difference between Nurture and Reach or Nurture and Activate. Nurture questions generally include things like how often do our visitors login to our extranet, or how many leads become sales.

Categorising business questions

1 What is the best source of traffic in terms of volume and sales?

Jan identified 3 elements in the question. Source of traffic, volumes and sales. Source of traffic is a reach dimension, and volume is simply a measure

of the elements in the question. Sales however is an activation as it assumes a sale has been made. Therefore Jan determined this was a Reach and activate question.

3 What kind of campaign worked in different environments?

Jan identified that because the question was about campaigns it was a reach question. However in the workshop she'd asked Carol what she meant by "environments". Carol explained that executive management would often want to know how to spend their money at different times, so not just for sales or volume, but for awareness of a new product line.

This meant she was talking about standard campaigns for sales (Reach/activate).

It also meant she was talking about awareness where the point was simply visiting a campaign site or reading about a new product (Reach/Engage and perhaps activate).

She could also be talking about reaching current customers (nurture). So this question encompassed the entire REAN framework.

6 How can we get prospects to buy?

Jan identified 2 elements prospects and purchases. So a typical reach/activate question.

18 What is best, getting people to buy direct or getting people to come back and buy?

With this question Jan identified "buy direct" or "come back and buy" as key to the statement. It was clear to her that by measuring activation on the first visit as oppose activations on the second+ visit she would be able to answer the question. Getting people to come back was a nurture issue and therefore the question was categorised Activate/Nurture.

3.3 Developing the REAN model

After the workshop was completed, she'd double checked and categorised all her questions Jan then set about making a mindmap of her REAN model for the marketing spoke. She took all the elements she had defined and mapped them out as shown in Figure 3.2.

By mapping out a REAN framework as shown Jan could see how the various things they did in ACME marketing fit together. She could now start to create the KPIs that would measure how well they worked in correspondence to each other.

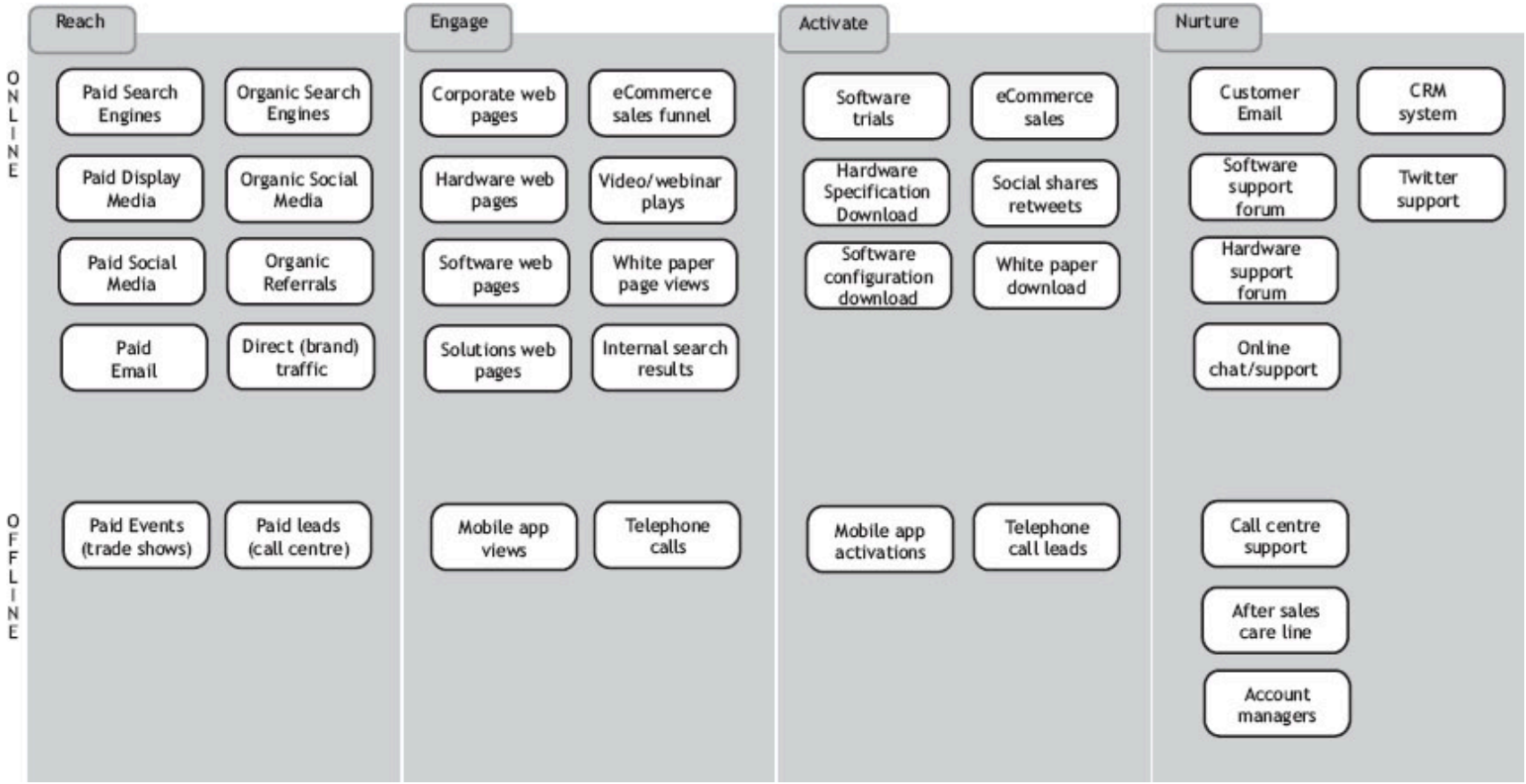

Reach
Engage
Activate
Nurture
ONLINE
OFFLINE
Paid Search Engines
Organic Search Engines
Paid Display Media
Organic Social Media
Paid Social Media
Organic Referrals
Paid Email
Direct (brand) traffic
Paid Events (trade shows)
Paid leads (call centre)
Corporate web pages
eCommerce sales funnel
Hardware web pages
Video/webinar plays
Software web pages
White paper page views
Solutions web pages
Internal search results
Mobile app views
Telephone calls
Software trials
eCommerce sales
Hardware Specification Download
Social shares retweets
Software configuration download
White paper download
Mobile app activations
Telephone call leads
Customer Email
CRM system
Software support forum
Twitter support
Hardware support forum
Online chat/support
Call centre support
After sales care line
Account managers

Figure 3.2 ACME marketing REAN model

Jan was now at the stage where she had modelled the marketing units REAN framework. To create the metrics to measure the success of the framework she now needed determine some measurements that would work to answer the stakeholder questions.

This is the process Jan followed:

1 First she determined which counts or ratios (see section 3.1) were required to answer the question.
2 Second she looked for ways to enrich her data to answer the question.
3 Then she looked at reporting requirements.
4 Finally she applied some benchmarks (25–50–25; see Chapter 4).

Step 1 Counts and ratios

Most business questions have a number as the answer. These numbers are usually made up of counts or ratios of unique visitors (or people), visits (to a website/app or physical location) and page views (or a hit/event, or dwell time in an offline environment). When looking to answer the business questions Jan identified which counts, or ratios she needed to examine. She knew a count made more sense when she was answering a volume question. 1000 visits is better than 100 as a simple example. However she also knew a ratio like conversion rate might be better measure of performance of a channel 100 visitors from 1000 converting on search (10 per cent) being better than 50 from 1000 from direct channels (5 per cent).

Step 2 Enriching data

Jan knew that not all the data she had at her disposal. There were external data sources that she might be able to use that enriched her ability to answer the business question more conclusively. Things such as costs (like labour costs) that might need adding to her data, CRM data with identified visitors, qualitative data such as survey data, a heuristic analysis or other offline data which could be compared to online figures.

Step 3 Determine reporting requirements

Jan determined how quickly and frequently her stakeholders could take action on the results. She recognised this as the one question she needed to answer in order to make the KPI relevant. If the marketing people couldn't act on the data there was a process problem so she'd have to look at whether it was worth fixing the process. If the business question could be answered at scale then it was worth fixing the process. If the question was so obscure that it couldn't be answered for every market they were in with a KPI then they would do a one off analysis if the question was important enough, but not scale it department wide. She also recognised the time frame of a KPI would help her know how often to report the numbers and develop benchmarks.

Step 4 Determine benchmarks

Jan knew the best way to develop benchmarks was with ACME's own data. External benchmarks such as published figures she could find from vendors aggregating a wide range of customers and industry conversion rates are available. However published statistics don't take into account the vast array of variables that apply to your business and it's difficult to accurately benchmark for instance a conversion rate. They are worth looking at to get a general idea of what the rest of the world might be doing, but that is all they should be used for.

Using statistical analysis to determine benchmarks

Jan knew a standard deviation is the most common measure of a statistical dispersion, in other words measuring how widely spread the values in a data set are. If many data points are close to the average (or mean), then the standard deviation is small; if many data points are far from the mean, then the standard deviation is large. If all the data values are equal, then the standard deviation is zero.

She could use standard deviations to spot deviations in data from normal levels and determine her benchmarks but she needed a simple flagging mechanism that said "this is good, this is bad and this is OK" to marketers not versed in statistics or data mining principles.

She used a simple 25–50–25 percentile rule (from Excel), where the top 25 per cent performers would be green lights, the bottom 25 per cent would

be red lights and the 50 per cent in the middle would be orange lights. This traffic light mechanism allowed her colleagues to quickly understand what was good and what was bad so they could focus their efforts on eliminating the bad stuff. It's not as exact as using a set of standard deviations, but it's close, and it's much easier to explain. See Chapter 4 for more on the 20–50–25 rule.

Back to the business questions

Jan listed all the questions from the workshop.

Step 1 Counts and ratios

1 What is the best **source** of traffic in terms of **volume and sales?** (Reach/Activate)
2 Which **channels** are the **most productive?** (Reach)
3 Is it good or bad that the **content** in our site **changes?** (Engage)
4 How can we get **prospects to buy?** (Engage/Activate)
5 **How many prospects** make the purchase **decision online** but **buy offline** and vice versa? (Reach/Engage/Activate)
6 **Where** are our visitors coming from? (Reach)
7 Which **landing page** converts the best? (And why?) (Engage/Activate/Qualitative)
8 Do the registered **users buy more** than others? (Engage/Activate)
9 What's the **most searched** product or service on our site? (Engage)
10 **How many software download** trials do we get? (Activate)
11 What is **best**, getting people to **buy direct** or getting people to **come back** and buy? (Activate/Nurture)
12 **How engaged** are our visitors with our **content?** (Engage)
13 What are the **top paths** through our site? (Engage)
14 How do we know when **something isn't working** on our website? (Engage)

She identified all the common themes. All the reach questions had a source or channel element. Stakeholders wanted to know where people came from. Second they wanted to know volumes and performance. Volume questions

referred to “how many” or “most”. Many of the questions referred to “best” and “most productive” which she determined would be best served by conversion ratio from a channel sales perspective, or what she knew as a bounce rate from a traffic volume perspective.

Bounce rate measured performance of landing page engagement, or simply the ability to persuade someone to click through to the next page and therefore something marketing could focus their optimisation efforts around.

She therefore had identified two ratios she could use that would help answer many of the questions when applied specifically to the question:

i Bounce rate.
ii Conversion ratio.

This would be in addition to reporting volumes and developing segments.

9 **Which channels** are overlapping – are **reaching the same** visitor/visitors? (Reach)

Question 9 was of particular interest. This was difficult to answer especially when people often used multiple channels to visit the same website and also multiple devices like a desktop, a mobile or a tablet. So she needed to break down the segments even further.

Step 2 Enriching data

4 So what for instance was the **best campaign channel** to **reach new customers** versus **old customers?** (Reach/Activate/Nurture)
14 What is the **value** of a free download to ACME? (Calculation)
15 **How many leads** become sales? (Activate/Nurture)
16 **How many** visits does it take before a **lead** becomes a **lead?** (Engage)
17 What’s the average lead **value?** (Calculation)

Jan knew that most of the marketing work they did was paid media channels therefore she knew she should add costs into the equation. The other part was customer relationship marketing. She had to somehow correlate how customers bought versus how prospects bought from ACME. Calculations such as questions 14 and 17 could also be scaled so would be useful to add.

She realised in order to answer the questions in any depth that she would need segments or channel metrics as well as just overall volumes.

She also needed to understand how lead generation worked from a channel perspective and generally how content was consumed. She figured that by showing popular content usage as a ratio of all content costumed would help marketers understand what worked from a campaign perspective.

Finally repeat visitors and customers coming back were also key questions that needed to be answered so she figured some kind of return visitor user segment compared across channels would also help. The users were identified earlier in this chapter, and returning visitors as a segment would suffice.

She had identified that into addition to reporting conversion rate and bounce rate another 3 metrics from the business questions.

iii Content consumption ratio.
iv Cost per user.
v Value per user.

Step 3 Reporting requirements

3 What kind of campaign worked in different environments? (Reach/Engage/Activate/Nurture)

She hoped she could partly answer question 3 by showing how to optimise in real time, versus longer term, but really question 3 was all encompassing. It required analysis that looked across campaigns run in the four dimensions of REAN, rather than simply prospect acquisition. She figured that by understanding the value per user across all campaign channels she could completely answer question 3.

In addition the marketing mater would need real time data for campaigns, but this was better used in a big screen view on the wall of the marketing department than as a report. All of the stakeholders would find value in all the KPIs but the campaign managers were the ones that needed to take immediate actions. Carol and the others could live with bi-weekly updates and monthly roll-up reports for planning purposes.

Step 4 Benchmarking

Jan benchmarked by segmenting the total audience into its individual components and modelled the best and worst from those in terms of percentiles. As described more thoroughly in Chapter 5, her three segment types were users, devices and channels broken down into the various REAN segments already defined above.

Taking just the channel segment as an example with conversion rates for software trials, Jan broke down the conversion rates to form a benchmark in the following way.

First she gathered all conversion rates per month from every channel. There were 9 channels and 12 results per channel, one representing each month. From her 108 results she ran two simple Excel formulas on the results.[4]

The excel formula for the top 25 per cent was =PERCENTILE(B2:B109;75%) and on the bottom 25 per cent is was =PERCENTILE(B2:B109;25%).

The bottom percentile (lower than 25 per cent) was anything below 3,55 per cent conversion rate. Her upper limit was 8,25 per cent. That meant she had a benchmark for trial signups. She repeated this across all the activations (conversion points).

She then repeated this process for all the segments and all the KPIs giving an internal benchmark based on a full years data across her 6 KPIs.

Communicating the results

She put everything in a simple table and mapped her thinking to make it clearer to the marketers how she'd relate KPIs to the questions (Table 3.1). She then added a flow of how the metrics connected to each other on the REAN mindmap she'd already created (Figure 3.3).

The final thing was to tie this back into the visionary objective. Jan thought about how best to answer this. She knew that to be the best or world class they had to be the number 1 or number 2 in each market. The business itself of course had to be good, but the management had already identified markets they thought they could win in. That meant to her marketing team

that they had to drive the highest value customers at the lowest possible cost to the business across those chosen markets.

In order to derive more value from their users they would need to direct them to the correct content, engage them with the most relevant materials, convert more of them to leads or to become repeat customers.

Jan figured this was a very clear alignment with the overall business strategy and she knew her colleagues working with other units were doing similar work to her.

She couldn't wait to show the work to her team.

3.4 Checklist and exercises

The following checklist can be followed to help you run a workshop for your hypothetical company or your own business to see if you can derive value from clarifying your KPIs.

1 Tools to run the workshop:
   - Stationery (Post-it notes/markers/pens).
   - Flipboard or marker board.
   - PowerPoint presentation of hub and spoke model/KPI workshop process.

Table 3.1 Mapping benchmarks and actions to KPIs

KPI	*Calculation*	*Helps answer question (purpose)*	*Actions marketing could take*	*Benchmarks marketing could use*
Channel volumes (channels defined in Reach part of workshop – measured as segments)	Number of uniques/channel	1, 2, 4, 7, 8, 9, 13 (shows which channel drives most/best volumes)	Apply marketing budgets where they perform best in terms of channel strategy	25–50–25 method across channels to determine best and worst performers in terms of leads/sales; improve worst 25%, increase budgets to best 25%
Bounce rate	Number of visits with a visit to only one page	1, 2, 5, 6, 10, 19, 21 (shows the channel, user and device type where disengagement with content happens from first entry point)	Apply testing to poorly performing landing pages to improve click through	25–50–25 method across landing pages to determine best and worst performers; start improving worst 25%, learn from best 25%
Conversion rate	Number of visits/number of conversions	1, 2, 3, 4, 5, 6, 7, 10, 11, 13, 15, 16 (shows which channel, user and device type is the most effective from a conversion perspective)	Apply testing to poorly performing landing pages, device type and type of visitor to improve click through, and poorly performing channels to improve acquisition strategy	25–50–25 method across channel, user type and device type to determine best and worst performing conversion scenarios; start improving worst 25%, learn from best 25%
Content consumption ratio (content segments defined in REAN workshop)	Number of page views/visit	5, 6, 7, 10, 11, 12, 16,19, 20, 21 (shows which channel, user or device type consumes the most content)	Determine the content that is interesting to your users and test/improve the content that isn't	25–50–25 method across channel, user type and device type to determine best and worst performing content; start improving worst 25%, learn from best 25%

Cost per user	Cost per channel or device/users per channel or device	2, 3, 4, 14, 17, 18 (shows the channels or device types that cost the least in terms of user type)	Determine the channels and device types that cost the most to work on reducing costs and improving effectiveness; start testing for different users	25–50–25 method across channel, user type and device type to determine best and worst performing cost scenarios; start improving worst 25%, learn from best 25%
Value per user	Value of channel or device / users per channel or device	2, 3, 4, 14, 17, 18 (shows the channels or device types that have the most value in terms of user type)	Determine the channels and device types that drive the most business value and start improving effectiveness; start testing for different users	25–50–25 method across channel, user type and device type to determine best user value scenarios; start improving worst 25%, learn from best 25%

2 Define the actors in your team. You will need the following:
   - A facilitator for the workshop and an assistant taking notes/gathering materials.
   - Defined the budget/executive owner.
   - Defined the managers/operations people.
   - Defined the technicians/IT support people.
3 Define business questions:
   - Set a benchmark for questions (the record is 49 business questions, you're aiming for as many relevant ones as possible to answer).
   - Facilitator starts with a simple question, such as "What is the best source of traffic in terms of volume and sales?"
   - Involve senior actors early to get the participants motivated.
   - Write all the questions down on the clipboard and have the assistant record them.

– Facilitator starts categorising questions as she goes into the REAN framework.

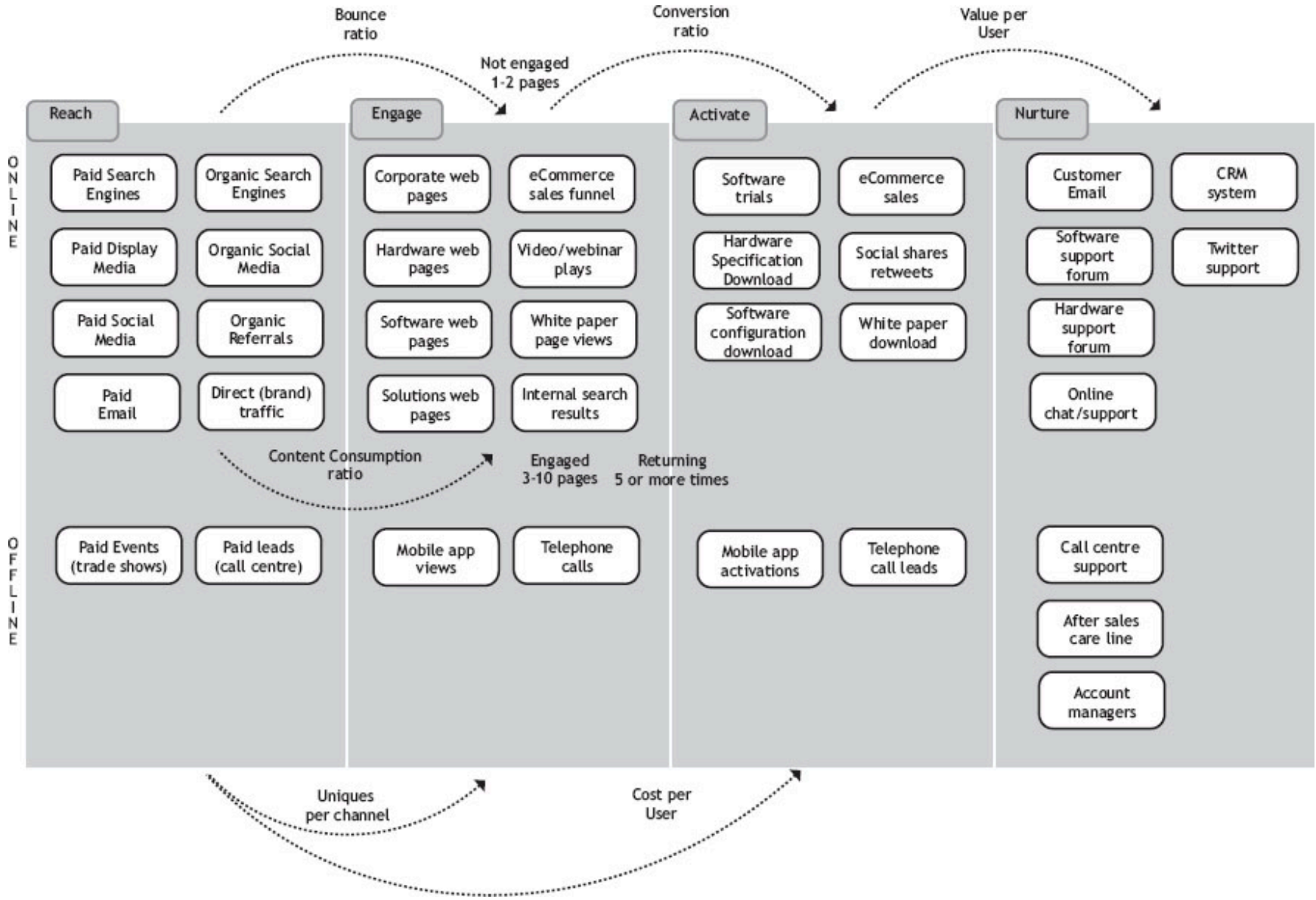

Bounce ratio
Not engaged 1-2 pages
Conversion ratio
Value per User
Reach
Engage
Activate
Nurture
ONLINE
Paid Search Engines
Organic Search Engines
Paid Display Media
Organic Social Media
Paid Social Media
Organic Referrals
Paid Email
Direct (brand) traffic
Corporate web pages
eCommerce sales funnel
Hardware web pages
Video/webinar plays
Software web pages
White paper page views
Solutions web pages
Internal search results
Software trials
eCommerce sales
Hardware Specification Download
Social shares retweets
Software configuration download
White paper download
Customer Email
CRM system
Software support forum
Twitter support
Hardware support forum
Online chat/support
Content Consumption ratio
Engaged 3-10 pages
Returning 5 or more times
OFFLINE
Paid Events (trade shows)
Paid leads (call centre)
Mobile app views
Telephone calls
Mobile app activations
Telephone call leads
Call centre support
After sales care line
Account managers
Uniques per channel
Cost per User

Figure 3.3 Final REAN mindmap

4 Define reach:
   - On one sheet of your flip chart draw two lines. One from the middle of the chart vertically down all the way to the bottom and one horizontally across the middle so that you have four squares on the flipchart. Write "Reach" in the top left, "Engage" in the top right, "Activate" in the bottom left and "Nurture" in the bottom right.
   - Ask everyone to write the five most important reach sources on a Post-it note. One reach source per Post-it note is required and the objective is to get each actors opinion (not a collective discussion).
   - Collect everyone's Post-it notes together and group similar sources together by sticking the Post-its to each other in the "reach segment" of the flip chart.
5 Define engage:
   - Go back through the business questions to identify the processes which exist on your website. Work out how many clicks/pages it takes to complete the process and how long this process takes in seconds.
   - Record the processes.
   - Look at the average time spent/pages viewed on your website from the tools you have. This could be used as a benchmark for engagement.
   - Define which pages are the most important (in terms of content you want them to see).
   - At the end you should have an idea of what a good visitor does versus what a bad visitor does (actively engaged with content versus lightly).
6 Define activations:
   - In the same way as you asked with the Reach exercise earlier ask everyone to write the five most important activations on a Post-it note.
   - One activation per Post-it note is required and the objective is to get each actors opinion (not a collective discussion).
   - Collect everyone's Post-it notes together and group similar activations together by sticking the Post-its to each other in the "activation segment".
7 Define nurturing activities:
   - Go back through the business questions and pull out ones that you think might relate to nurturing. Remember that nurturing is all about

helping your customer or your visitor become happy with your brand so that they buy more of your products or services.
   - In the same way as you asked with the Reach exercise earlier ask everyone to write the five most important nurturing activities on a Post-it note.
   - Collect everyone's Post-it notes together and group similar activities together by sticking the Post-its to each other in the "nurture segment".
8 This ends the workshop for the actors. However, end it by explaining the process again and what comes next.
9 Categorise the questions into the REAN Framework.
10 Develop a REAN mindmap.
   - See section 2.4 for resources you can use to help develop this.
11 Develop the KPIs
   - First, determine which counts or ratios (see section 3.1) were required to answer the question.
   - Second, look for ways to enrich your data to answer the question.
   - Third, look at reporting requirements.
   - Fourth, apply some benchmarks (25–50–25).
12 Re-present your findings to the workshop group and once everyone agrees to them and after any iterations implement the KPIs in your tools as your reporting requirements have defined. This last stage is important to continue to help your spoke feel like they are developing the KPIs with you.

Notes

1 See http://en.wikipedia.org/wiki/Big_Hairy_Audacious_Goal.
2 See www.digitalanalyticsassociation.org/Files/PDF_standards/WebAnalyticsDefinitionsVol1.pdf.
3 See http://tnsmetrix.tns-gallup.fi/public/?lang=en.
4 See Excel Sheet "Conversion Benchmark – Chapter 3" available at www.blackbeak.com/cult2.

4 Planning analytics business cases and wins you can share

> There are winners, there are losers and there are people who have not yet learned how to win.
>
> Les Brown, US author and speaker

4.1 What is a quick win?

One of the keys to changing the culture in your company to be more data driven, is to solidly prove that your system works to bring value to the business. Basically what you're doing when planning a quick win is building a business case.

Building business cases is part and parcel of any major corporations process for making major investments. Quick wins are business cases that are designed to be shared across the business to help build acceptance and get people interested.

Quick means that it's an insight that can be proven and acted on quickly to bring results.

This chapter will describe some easy techniques to do this and will present a number of examples from ACME's perspective. First, though, I'll describe the overall process and how to do this yourself in Google Analytics.

4.2 Process to plan the quick win

It's impossible to describe how to build a business case for your business. Only you know how the business you work for works. What this chapter

aims to do is give you a process you can follow to find your own business case to build using just web analytics tools.

Using REAN as the framework for your thought process

REAN can simplify the planning process by putting business questions you might need answering into a simple framework with a goal as an outcome. Simply go to your analytics system and develop answers to the following questions about your site or service.

1 **Reach.** Which traffic sources, campaigns or channels are the best performers in terms of volume and cost? Why? (Why is the hardest question). The goal here is greater volumes at reduced cost.
2 **Engage.** Which processes have high abandonment points? Which pages have too high bounce rates? Which keywords engage more visitors? Which pages have higher exit points than others?Which places on the website are not clicked very well when their main function is to attract click-through? Where on the site is there poor time spent? Is this justified (link pages?) or is this content you want to have read which is being ignored? The goal here is to determine the areas you interact with your customers that you might be able to improve.
3 **Activate.** Which reach sources are the best at converting traffic? Which keywords generate the most conversions? How do your process funnels convert? What are the most popular conversions? What are the highest value conversions? Can you monetise every single conversion on your website? Do you know how to spend more money to drive more value? The goal here being understanding what drives higher sales with the best value.
4 **Nurture.** How do you get people to come back? What's most effective to drive new sales? Does sending email drive up conversions? If so what's the business value? Would automated email or other communication work better than a campaign? Have you worked out the lifetime customer value (LTV)? What kind of campaign increases LTV and why? (The holy grail of marketing!) The goal here is to understand what kind of reach source drives the best customer lifetime value and when to select a channel to drive a customer.

Simply looking for cases by asking questions similar to the above and framing your thought process in the terms of the REAN model can lead you to a business case in any analytics system.

4.3 Determine potential quick wins

Now you've got some questions in your mind it's time to get down to business.

You have to demonstrate value, preferably monetary value. Once you have identified a potential win it's then up to you to go out take actions to improve things. You may need to run tests like A/B or MVT (see Chapter 7) to prove your case.

Pre-Study

Log in to your favourite analytics tool and start trying to find out answers to your questions. Start with reach engagement and conversion.

Reach, engagement and conversion

1 What is the best source of traffic in terms of volume and conversion? What is the worst? What is the difference usability wise from best to worst? Have you any hypothesis why one has outperformed the other?
2 Are there any differences over time? Days? Weeks? Months? Years?
3 Which channels are the most productive? (High level like search engines versus display?) Does one channel cost more than the other? Why? Are there reasons why your business does more marketing in one channel than another?
4 Are there any processes on your website stopping your visitor from converting? Pages where people leave without doing anything? Places where they stay on average a few seconds versus places where they will stay a while?

Make notes, form hypothesis keep digging.

Prove or disprove your hypothesis

Once you have dug up some ideas for improvement or building a business case you need to try and prove your ideas have merit or not. The following method is a standard process in experimentation. Ask yourself the following four questions about each issue you're examining:

(a) Have you determined the cause of the issue you're researching? (IE; a campaign peak)
(b) Have you identified the effect that the issue you're researching has had? (IE; a lift or a drop)
(c) Can you correlate with a degree of confidence that the cause has impacted the effect? (IE;When you ran a "what's changed" report you saw 90 per cent of the peak was attributed to the campaign)
(d) Are there any open questions you cannot answer? If so determine what you're missing, go back to step (a) and repeat until there are no open questions left.

Another thing you could do is base your pre-study on business questions you identified in your KPI workshop if you've managed to arrange one (see Chapter 3).

The following sections show some simple web analytics tactics you can use in order to find these improvement areas, starting with a 'how to' guide for Google Analytics.

4.4 How to use Google Analytics

Everyone uses Google Analytics or can use Google Analytics as it's free. So this section describes how I'd find the case using just this tool.

There is *always* a business case. I have yet to see a perfect website where I couldn't find a business case. Most cases we've found over the years we've have managed to monetise into 6–8 figure opportunities.

Also I'm not going to waste time teaching you about the interface. Half of the fun is figuring out how to do it yourself. Google is also your friend here. Everything I am saying has been written about 1000 times so if you don't know how to select a date range or look at an audience overview, Google it.

If you want training hire a GACP[1] or drop me a line and I'll charge you a fortune. And I'll expect coffee.

The steps

1 Select a 13-month date range for your data sample. Dates are found top right (at the time of writing). This allows us to compare the latest month of data with the same period from last year as well as all the months in between.
2 Look at the audience overview in GA (click around till you find audience then select overview). Do you notice any peaks or valleys in the trend line? If you do select a new date range that covers the time before the peak, during the peak and after the peak. For instance if the peak or valley happened in May and returned to more average levels of traffic in June select April, May and June. Run a "what's changed" report (see below). See if there are no peaks or valleys, or after you've run your "what's changed" report continue to step 3.
3 In GA select "Acquisition overview". You'll see the channels broken down with the most popular at the top and the least popular at the bottom. They show the acquisition metrics (reach sources) with a breakdown of whether they are new sessions or repeat sessions. Determine which traffic source drives the largest percentage of new and repeat sessions. Go to step 4.
4 Also while in the same view determine which traffic source has resulted in the most conversions and the lowest bounce rate. Make notes.
5 Take the top 4 traffic sources (in terms of visits, lowest bounce rates and best conversion rates) and add a segment for each one (see also Chapter 7 segmentation). Do this in GA by firstly selecting channels in the Acquisition drop down menu on the left. Around the top left of the page you'll see a round circle listed as "All Sessions". Next to this there is a place that says Add segment. On the right hand side you'll see a little search box in the window that appears. Type in the name of your traffic source IE; "Social" or "Search" and select the check box that appears. You'll be able to select 3 traffic sources by default and compare this to All sessions. But you can also remove the "All sessions" segment and replace it with a fourth segment so you can compare your top traffic sources alongside each other.
6 Segment in the same way as above the Mobile and Tablet visitors. *Hint:* if the segments don't exist in Google Analytics click +New Segment, name

it “Mobile visitors” or something similar, select “Technology”. Once selected put your cursor in the device category open field (mobile, tablet and desktop should appear). Simply select “Device category” contains mobile and you have a mobile segment. Repeat this process for tablets.

7 Remember your questions. You’re looking to understand from your data which source of traffic is best in terms of volume, conversion, where they come from and overall the most effective channels.

A How much do the channels cost? If you have one person paid 50K a year to manage your marketing and spend $50K across all your channels spread the cost of the person’s time across the activities and then add the media cost allocation. Work out cost per visit and cost per conversion per month. Now which ones are the most effective in terms of cost per visit and action? For more on monetising visitors see Chapter 8.

B Based on step A above, are there any channels that might warrant more spend? For instance is SEO cost per conversion a fraction of the price of Display marketing? Is the cost per visit of one channel way lower than another?

C Can you see any traffic sources that have high bounce or exit rates? Are the pages that these visitors are landing at difficult to navigate? Also run a funnel report if you have any booking engines, shopping carts or lead generation processes.

D What devices are people using to visit your website? Is mobile as effective as desktop at driving conversions? Similarly with tablets?

Have you found anything so far that peaks your interest? Write it down. With a pen if you like. Then move to the “what’s changed” reports.

Run a “what’s changed” report in Google Analytics

If a particular traffic source has driven a lot of visitors or conversions then knowing why is great ammunition for a business case. If, for instance, a Google paid search campaign has performed really well over a short period then a business case could be built around doing that again.

To run a “what’s changed” report in GA what you do is quite simple.

Select Acquisition → Overview from the left-hand dropdown menu. You should have selected the date where the peak or the valley occurs (a few

weeks before and after is fine). Then simply click the checkbox underneath the date selector which says "compare to previous period". What you then see is traffic sources compared over the two time periods.

This allows you to see where any gains or losses have come from. Once you've determined where your gains and losses are you can then ask the same questions (A to D above) for each channel.

Primarily, the "what's changed" report will show you how the major traffic channels changed from period to period and by drilling down to the data sources you can prove the value by monetising their cost and returns.

Run a "disengagement" report

Where people leave your website and abandon funnels are also opportunities. Shopping carts, booking engines and lead generation processes can be tracked in most analytics tools as a sequence. Typically to finish purchasing a product or signing up for a white paper the visitor has to go through the same set of steps as everyone else.

By tracking these "funnels" in sequence you can determine if people are leaving at the same spot every time. If that's the case there may be a usability issue or a technical problem that stops the user from completing the action you want them to take.

By finding these issues and monetising a potential lift you could make by improving the overall process you can create a business case or quick win that could be shared.

Setting up a funnel in Google Analytics is not complex:

1 You first need to make sure you have a destination goal set (in admin). This might be a confirmation page (thanks.html)
2 Check the "Funnel" to the On position.
3 Add the name of the step and the page it refers to. For instance "Registration page" and register.html.
4 Repeat for each page in the process by clicking "add another step". So if there are four pages in your registration process you simply add the four pages before the goal page.
5 Save the goal and exit.

Once you have your funnel report you can then see which of the pages have the highest number of people that abandon the process before completing it by opening the GA left hand menu; Conversions → Goals → Funnel Visualisation report.

Bounce rate reports

A key performance indicator used by many companies is bounce rate. People have written books about bounce rates. Companies have sprung up because of bounce rates! They're a great metric to use for business cases.

The reason is that a bounce is when a person enters your website at an entry page and leaves from the same page without interacting at all. That's bad in some cases.

Bounce rate is a perfect measure for this.

In GA this is done by clicking Behaviour → Site Content →All Pages, and sorting by bounce rate (just click on Bounce Rate). It's probably also worth filtering by more than X amount of pages so the list of pages you pull up has meaning. By default GA will sort by the worst bounce rate. By clicking the advanced filter field and selecting page views more than 1000 for instance it will sort the pages with the worst bounce rate that have more than 1000 pages viewed.

You now have a list of pages that have high bounce rates and lots of page views. By determining which of those pages are important to reduce the bounce rate of you have a list of optimisation opportunities. Again by monetising what a lower bounce rate would be (more click-throughs equating to more conversions (even at the same conversion rate) you will be able to show value.

Now one thing to note here is that bounce rate only tells that people left without doing anything by default for 30 minutes. It could be that they come back later, read again and are actually a valuable visit. So it's not necessarily the best optimisation point, but it's usually not a bad place to start. See Chapter 6 on segmentation, and Chapter 7 (specifically section 7.1) on defining your testing approach.

Exit rate reports

In a similar way to bounce rate, exit rate can be a good indicator of too much abandonment. You need to exercise good judgement on which pages are "good" exit pages (for instance thank you for signing up) and which are bad (like the homepage). However by seeing which pages people exit from following the same process as for bounce rates you can determine which pages you may need to work on and similarly monetise a lift percentage.

4.5 The 25–50–25 rule

In *The Sucking Manifesto*, which is freely available online, my co-author Markus and I explain the 25–50–25 rule and this is another good way to find business cases and ideas for optimisation. It's a general method because it doesn't matter which analytics system you use you just need an excel spreadsheet to find the results:

The Sucking Manifesto *model*

> The model is not like a law of science, it's more like driving a car. You put resources in (like petrol, water, oil) and you get the ability to move between point A & B. Of course to get between point A & B you have to follow the rules of the road like signals and traffic lights. So think of it like this, the petrol is your money, the car is your company or brand, the direction is what you want to do and the measurements you take are the signals and traffic lights.
>
> A traffic light has three lights: a green one, a yellow one and a red one. In our model, the green one represents the campaigns that are performing within a set of key performance indicators (KPIs). Yellow ones are performing worse than optimal, but are still within the tolerance and although they require attention and changes, they are redeemable. Red ones are beyond the point of no return and should be stopped immediately.[2]

How to use this method to find potential savings and value

Gather all the data for all the campaigns you've run for the last year. You're looking for the amount of impressions, bounced page views, visits, the number of conversions per campaign and the costs. Where possible get all the ROI numbers as well like value from each campaign in terms of sales or leads generated that can be monetised.

Once you have all that information in a spreadsheet you ned to equalise the data by selecting some KPIs which should be ratios that can be compared across the different campaign channels. Lets say we use click-through rate (CTR) as a **reach measure** of how effective a campaign has been at attracting a click, bounce rate (BR) an **engagement measure** of the number of bounced visits from all visits, conversion rate (the number of sales conversions or **activations** from all the visits), cost per acquisition as a measure of cost across each campaign and finally return on ad spend (ROAS) which is a ratio of euros earned from euros spent.

Scale always wins (despite ratios …)

ROAS has to be 100 per cent at least to break even and you should also keep things in context. You can have a very high ROAS but a very low spend and high return that will not build a good business case.

For instance, if you spend €1.5M, to get 7500 conversions, at €7.5M revenues, with a profit of €6M, that's a healthy ROAS of 500 per cent. The mathematics goes like this:

> 7500 × €100 price = €7.5M revenue. €7.5M/1.5M cost = 5 times (or 500% ROI).

However, if you spend €500 to get €10K revenues (a €9500 profit), that's a ROAS of 2000 per cent.

The first case is clearly the case your business manager wants to see not the second despite the excellent ROAS result, so remember to put it in the context that matters. Scale always wins.

Similarly with conversion rates. If CR is high (like 10–20 per cent) make sure the volumes are also high enough for that CR to be meaningful. 1000 sales from 10,000 visits is a great conversion rate. 1 sale from 10 isn't statistically significant. Just use common sense here, you want to build a case that really brings value to your business.

The traffic lights

Once you've got the information into excel you have a number of ways to organise the ratios. If you want to understand what was best at driving traffic you might look at CTR and organise them campaigns in a list from best CTR to worst. The best 25 per cent CTRs would be given a green light (you can sort this in excel using conditional formatting). The mid-range would be given an orange light and the worst 25 per cent would be given a red light. You can use the percentile function in Excel to calculate all the metrics.

The way you'd use this is to then see how the best ads compared with the worst. In the red light cases could it be the channel was bad? Could it be the ad creative was bad? How do the green light cases compare? And are the volumes significant?

Example with CPA

This is another example taken from *The Sucking Manifesto*:

> Richard has done 14 e-mail marketing campaigns for a particular lifecycle in the previous year. Most of his campaigns had a different average cost per customer acquisition and he wants to find out which ones were good and which were not. The campaign CPA's were $1, $2, $2, $4, $4, $7, $9, $12, $15, $15, $15, $18, $20, $25.
>
> We set the percentile function away and for the green lights we use 25 per cent because we want a low number to the green lights. This means if he got a customer for $4 or less he's really performing well. Then we run the percentile function for the range of numbers in the top 75 per cent. It means if he pays $15 or more he should stop the campaigns as they're too expensive.[3]

By running this kind of analysis across all campaigns you can compare reach channels, compare engagement, activation and how well you nurture your customers.

You can also total up the poorly performing campaigns and monetise a business case to re-direct funding to more profitable areas, or zoom in on a campaign type (like email) and make sure your campaign managers (at least example 4.5.1) have a top limit on CPA of $15.

If you could prove that the campaigns above $15 were worth $X million a year to the business in terms of lost money by continuing to run the campaigns you could simply kill them.

The idea being you use the top 25 per cent as your best practice, the mid-range 50 per cent as the optimisation opportunity and the worst 25 per cent as the leaky bucket which you simply stop doing.

4.6 Design the business case

After finding some opportunities you need to monetise them. Chapter 8 goes into this in more depth, but let's look at a brief illustration here.

In August 2013 you had a website renewal that completely re-vamped your business website and improved conversion by 10 per cent across the board.

The management team are very happy and everyone is patting each other on the back. However your analysis has uncovered that in the first 2 months organic search visibility is dropping rapidly which longer term will almost definitely result in lost sales.

The figures look like this:

	SEO visits	SEO conversions
Aug	10000	350
Sep	7000	245
Oct	5000	175

You have determined which pages are no longer getting SEO traffic by doing a what's changed report. Some of the pages no longer exist on your new website. The web design firm has made an error in not re-directing old page URLs to new pages and you know that by doing this you can restore the sites previously good ranking.

Monetise the direct loss

In 2 months you have lost a cumulative 280 conversions and you know each sale is worth €400 to the business. It means that despite the overall

conversion lift the opportunity cost has already been €112,000 and will continue to decline if something isn't done. You also know that today in November the opportunity cost is €70,000 per month.

Package the data in easy-to-share ways

Put this information into PowerPoint, Word, Excel or PDF formats – something that everyone can use and share.

Then book a meeting with the higher-ups and explain you've found a business opportunity costing the business €70K a month and have a solution to solve this. The solution being that you fix the error so that google picks up the new website pages immediately rather than having to wait until the site is indexed all over again.

Communicate the win

Easy to understand is the key to successfully conceptualising your win (see also Chapter 8 on writing your story). There is no point demonstrating that in August you got 10,000 unique visitors from SEO and in October you're only getting 5000 despite conversion rates across the board being great and reduced bounce rates. A management team doesn't want to hear that.

What they want to hear is an alarm bell that says we just lost €112K and are leaking €70K a month, and this is how to fix it. Management at the director level is all about making decisions. Make it a no brainer for them to make a decision and they'll do it.

If it costs 10K to stop a 70K a month leak that's a no-brainer. Demonstrate that you found this out by analysis of the data and need funding to fix/continue the work.

Publish to a wider audience

Once you have got your decision and done the job, distribute it throughout your organisation. Sing it. This is the part where you show off your results to the various stakeholders in other departments.

If you have a corporate Intranet put the win there and display it prominently with a strong headline. "€70K A Month Lift For A Few Days

Work!" is better than "Conversion Monthly analysis from Analytics Team – October".

Also don't forget your in house email lists, send the intranet link to all staff celebrating the win, and call the business to action asking for them to follow up with their own questions that could be answered by web analytics.

What follows are four cases that ACME use in their business starting with Jan in marketing.

4.7 ACME business to business case example

Jan had developed the KPIs for her marketing team but now she wanted to show the value when she presented the finalised metrics to her team and the CMO. She wanted to show the wow effect and blow away any scepticism that may have been present after the workshop, after all this was new to everyone, they needed proof. Rather than simply present the outcome of her work, she wanted to show a practical example of how the KPIs could work.

To begin with she started looking at the data with the questions in mind from the workshop.

What is the best source of traffic in terms of volume and conversion?

Which channels are the most productive?

The analysis

After selecting a 13-month trend Jan noticed a huge peak in the data over the period at the beginning of February.

Upon further investigation she saw that lots of people visiting a particular article that ACME had written that attracted a lot of interest about a particular facet of their software in Belgium. The traffic was all from social channels, Facebook, LinkedIn and Twitter. In this part of the year ACME had done some marketing on in Belgium that had been tremendously successful in terms of driving page views, visits and conversions. Jan saw this as a perfect example of how she could demonstrate how to answer the first two questions and demonstrate how KPIs could be used.

Social media as a channel had not been widely utilised by ACME and yet this initiative that one market had done resulted in excellent engagement and

conversion rates. The downloads of white papers and email addresses captured meant the customers could be re-contacted.

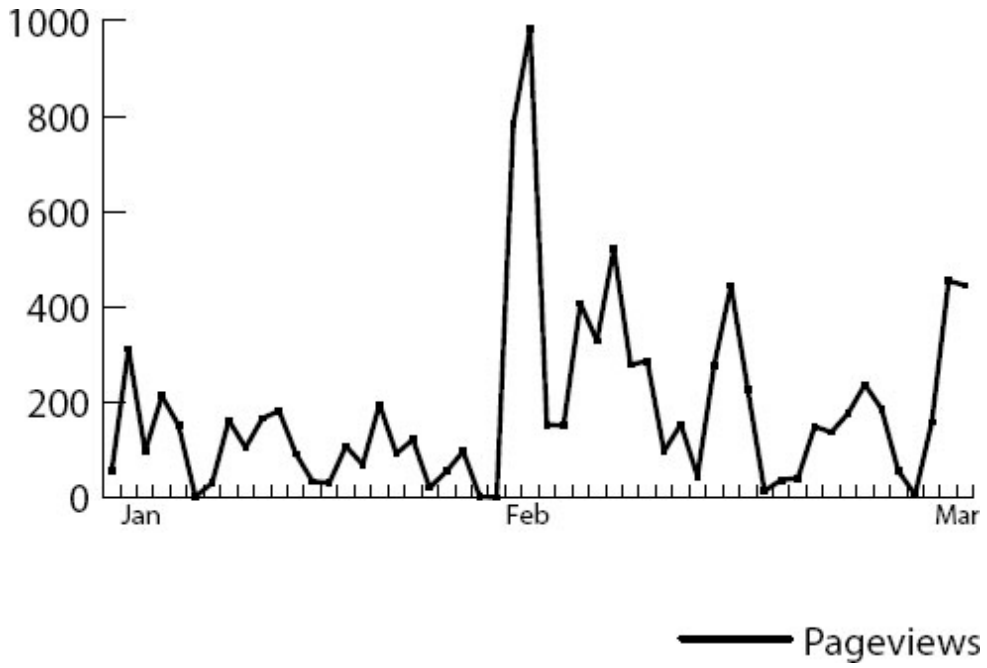


Figure 4.1 Data peak

She also mined the data over that period to see what people were using to view the site and noticed that 50 per cent of people were coming from mobile or tablets to this particular article. This was a huge percentage in comparison to other articles on the website. This particular country site was well optimised for mobile and the article was as easy to read on a mobile phone as it was on a desktop. The conversion point was a simple email address submission at the end of the article to get the white paper so not a massive problem for someone on a mobile phone.

Jan quickly realised the opportunity was to ramp this social activity up across different markets.

With enough data about the win gathered she moved onto building the case.

Monetisation

Jan had determined that the software the customer might buy from the articles was approximately €50K per year. She found out in 5 minutes by emailing her colleague Mari in the hub who was attached to the software team.

The website closed 375 new leads over this period when the peak happened from about 30,000 visitors (1.25 per cent conversion rate). The lead close rate was 10 per cent being booked to meetings within 3 months (again Jan found this out from Mari) and from there approximately 10 per cent would become customers. So it meant that it was fair to say that the lead conversion to sale was about 1 per cent. That meant that the lead value was €500 (1 per cent of 50,000).

Jan quickly worked out that this period of successful marketing was worth somewhere in the region of €187,500 in new business to the company. This was one market. This method of social marketing had only been tried in Belgium according to the analytics she was looking at and so by doing the same thing across 30 markets (literally translating the article and marketing it across social channels in the same way) the company could earn potentially €187K × 30, or €5.6 million in new business.

Communicating and sharing

Jan had used conversion rates (to understand the sales), volumes (to understand the potential), content consumption ratio (to show the content being consumed was higher) and worked out value per user as well as overall potential value.

But she wasn't satisfied. She reckoned 375 leads from 30,000 visitors was poor and by running a testing program to increase conversion and reducing bounce rate she could get an extra 1 per cent. That would mean 675 leads instead of 375 from the same 30,000 visitors. By applying her previous maths to the equation her single market value would have been €337,500. If that was ramped up across all markets you'd be looking at a €10 million opportunity.

She also worked out the cost. Again she got Mari to help by asking how long it had taken to develop the content in the first place as well as calling her translation agency and getting a quote. The cost per market worked out at about €2000 to localise and a day or so per market in time to publish –

this was time her colleagues could be doing something else so she factored it in.

Maximum cost therefore for doing the campaign globally was €90K. That meant a return on spend of €62–112 per €1 spent. *Huge!*

Jan added the case to her slides as a really strong case study for the point behind her workshop. She had found one case in which all the planets aligned to show how each KPI could be used to prove a business case. She also realised she was actually onto something really important and so created a two-page PDF that could be shared by email later.

Result

Jan assembled the stakeholders. She went through the various business questions, REAN and how they had come to the conclusions they had. This was met with approval and general nods of appreciation. After all everyone had been involved and they all saw the logic and clarity the KPIs brought to the process. The campaign managers liked the idea of having big screens that showed results in real time.

Carol the CMO, however, was less inclined to agree as the screens would cost money. This was when Jan held up her hand and said, "Yes, but Carol I haven't finished."

She demonstrated how the KPIs had shown her a way that if implemented could drive a potential revenue gain of €10 million. The room fell silent. Then erupted into chatter. "Is this real? Are we sure? How did you find this out? What the hell?"

Jack the campaign manager in charge of software trials excitedly started talking about pilot markets. James in charge of SaaS marketing was wondering if the same could be applied to his marketing concepts. Helen and Fred started deep discussions about potentially looking into similar initiatives an thought leaders that might help them with hardware content. Harri the IT manager simply smiled at his colleagues as they babbled away excitedly and nodded at Jan with a look that simply said "well done".

Carol had a huge grin on her face:

> That's fantastic. Simply fantastic. Ladies, gentlemen, implement these KPIs across our department as soon as is humanly possible. Jack, please

take the lead on identifying pilot markets to roll this out and get their buy-in. Also, contact Belgium and ask them what they learned. Is there any way we can do it even better? Harri – can we get your assistance putting those screens in place across our department? If something like this happens again we need to identify it immediately. Tommi, could you work with Harri to get country numbers, overall reach figures and those KPIs on screen 24/7. Jan, this is excellent work – we need to share these findings beyond our team. I'd like something short and compact, something I can show the management team and share over email to the whole company.

Jan smiled. "Something like this?" she said while showing a printed copy of the PDF she'd previously put together.

Carol read it and grinned wider. "Something exactly like that. Excellent stuff." She paused. "Ladies, gentlemen, you've done sterling work helping Jan and Tommi come up with KPIs that have helped us identify real business opportunities. This is exactly the behaviour we want to see. You should know that this will be reported to the CEO and the executive team and that were fully behind this. Really, people. Excellent work. Carry on."

With that she left.

The title of the email Jan sent out was "€5.6M business opportunity – we need your help to do it." Everyone in the different markets were sent the package over email and a teleconference was arranged where the case was explained for 45 minutes before opening up for questions.

Marketing was on its way.

4.8 ACME branding business case example

James was psyched. Jan had just shown an impressive way the company could make money using social media that he hadn't even thought of. That was something he'd definitely have to incorporate into his future campaign ideas. But right now he was curious about using the KPIs for his own campaigns. If Jan could find business like that with very little context about what was going on in day to day marketing surely he could uncover something from the KPIs too.

He was responsible for media campaigns and now he had something solid to guide his team's success or failure. His team were right now running

targeted online banners and search engine marketing. The main goal of the campaign was brand awareness (not sales at this stage) and the target was to get the brand seen by as many visitors as possible. The Software as a service (SaaS) he was running marketing for was pretty new and only existing customers knew about it. He needed to get the service out to the masses as well and let them know ACME had SaaS.

Remembering how Jan had started he looked at his business questions first.

What are the goals of the campaign? What is the best source of traffic in terms of volume with relation to the goal? And what is "awareness"? Could he use a KPI to say a visit is more likely to be aware than not?

The goals

The previous weeks James had been discussing with potential and current media agencies and asking them for advice about how to use different channels to generate awareness. His had a lot of responses from agencies:

- "SEM is great for sales or lead generation but poor for branding and awareness", "Banners carry branding value that text ads don't."
- "We create *brand experiences* by using a media mix."
- "SEM is the best form of marketing currently available as it's cost per click."

James knew that none of this was untrue. But he felt he needed more control of how he would guide his media agencies to spend his marketing budgets. He believed Jan had just given him the answer with the new KPIs.

Guiding the media agency

James's current media agency had suggested they use total number of ad impressions as a KPI. The point being that this is potentially branding if the visitor to the media site sees the ad. But James had concerns that there was no proof the visitor was "aware" and it would be too easy for the media agency to fulfil their end of the deal by simply putting ads on highly trafficked websites that get a lot of banner impressions.

If however James used a KPI that defined a branded visitor as someone who had consumed content and converted (something he now knew how to look for) he could give his media agency a clear target.

He asked Tommi to help him develop a benchmark for Content Consumption Ratio that would reflect users that "might buy". He would then pass this onto the media agencies. After running a 25–50–25 rule on the content consumed, Tommi determined that a minimum of 5–10 pages and 5 minutes spent on the site was enough to potentially lead to a sale.

James wanted to make it easier for the agencies so he simplified and said to them "awareness is defined as a click-through to our site and someone staying 5 minutes. That's what we're looking for. So if people stay less than five minutes from one campaign but stay longer in others, then you can allocate more spend to the highly engaging campaigns."

The media agencies across the markets weren't happy because this would require a lot of changes for them, the prices of the services they were running for ACME might have to go up. So James needed a business case.

Working with Tommi they identified the two old measures the agencies were used to using and the new one they were asking them to adopt.

- impressions
- clicks
- engagement.

Impressions

James knew that a media page loaded where your ad is presented was defined as an impression. Largely with ACME this was upon someone typing a keyword into a search engine or having the ad appear on media websites where ACME banners were placed. It doesn't mean that the visitor had seen an ACME ad, but potentially she could have.

Clicks

They knew a click is registered when an ACME ad is clicked. In this scenario someone either clicks ACME's text or banner ad and the analytics software records the response. A CTR could also be used to identify ad effectiveness (the higher the CTR, the more persuasive).

Engagement

This was the measure Tommi and James had defined as a click-through to the site and spending 5 minutes. Segmenting their traffic into users that have arrived from one of the media agencies ads and stayed on their site for more than 5 minutes for instance could do this. Tommi created the Engagement segment and got down into the data.

The analysis

By setting up a segment to filter out all those whom have spent five minutes or more on the website Tommi and James could compare how campaigns fared against each other.

The results are shown in Table 4.1.

From Google there were 1547 visits that were engaged from a total of 16,958 paid clicks. That meant 1547, stayed on the site for more than 5 minutes, meeting the success criteria and dropping into James index of "engaged visits".

On Yahoo, only 113 from 4351 visits were engaged. Coupled that with the fact that Yahoo was getting a very low CTR, James judged that the Yahoo visitors were not as interested or as well targeted as the Google visitors.

The Yahoo campaigns might therefore dropped and the money then re-directed to Google that was much the same price in terms of cost per click, but far more effective in reaching the goal.

What was even more interesting was that the banner campaigns placed on highly targeted sites were outperforming the search engines in this particular case.

This was interesting because James knew banners often have a bad reputation of being overly expensive and had very low CTRs. In this case the data proved this partially wrong. Click-through was very low at less than half that of Google and the media prices were much more expensive per click, however the engagement was considerably higher. Nearly double that of Google campaigns.

Table 4.1 Using an engagement index segment

Indicator	*Google*	*Yahoo*	*Banners*
Impressions	743,485	7,749,283	2,906,078
Clicks	16,958	4,351	2906
CTR	2.28%	0.06%	1%
Engaged visits	1547	113	503
Engagement index	9.12%	2.6%	17.31%

By showing these results James could see immediately that the banners worked much more effectively (nearly twice as effectively) as the best SEM campaign in terms of engaging the visitor.

Monetisation

Because the banners were 10 times more costly than search engine marketing to place with the media sites, James needed to weed out the banner campaigns that weren't working. What he needed to know was how he could improve the spending with the banner marketing to get a better return of engaged visitors to the website. This was done by drilling down into the campaigns and producing the same figures for each media website which displayed the company brand (Table 4.2).

As can be clearly seen the weakest performers in the banner campaigns were Media Site 4 and Media Site 1. The other campaigns all had higher engagement than 20 per cent. So the money spent on the banners could be re-directed to the campaigns which were the most effective like Media Sites 7 and 8 in this case.

By optimising in this way the spending was dramatically improved and the branding became accountable and justifiable.

James started doing some maths. Media Sites 4 and 1 had cost €276K over the course of the year. The average of the two was a cost per engaged visit (CPEV) of €285. However, Media Sites 7 and 8 were more niche. They only cost 50K per year. Their average CPEV was €141. If the money from Media Site 4 and 1 were re-allocated to Media Sites 7 and 8 the engagement index (people being made aware) would be much larger. The cost saving for the same return on Media Sites 7 and 8 would be €144 per engaged visitor.

In other words to get 968 more engaged visitors would only cost €136K rather than €276K an opportunity cost of €140K.

On average the SaaS business average annual sale was around 15K per year and Tommi had told him that from those visitors that hit that content consumption segment of 5 minutes the conversion ratio was 8 per cent.

Table 4.2 Using an engagement index segment to compare media sites

Description	*Responses*	*Engaged responses*	*Engagement index*
Media Site 1	4311	708	16.42%
Media Site 2	3969	920	23.18%
Media Site 3	2461	516	20.97%
Media Site 4	2042	260	12.73%
Media Site 5	1885	379	20.11%
Media Site 6	1575	349	22.16%
Media Site 7	697	199	28.55%
Media Site 8	613	154	25.12%

If that was halved (to be conservative) he could estimate future sales from his campaigns and create a business case for funding his agency spend as well as optimising how they worked in the future.

From just the two media sites in question with spend re-allocated to them he could expect 1321 engaged visitors at a cost of €186,261. From those he would get a 4 per cent conversion rate (or 53 sales). 53 sales at a €15K average order value was €792K in future business or a return on ad spend of 792/186 = €4.25 for every euro spent.

This wasn't fantastic, but wasn't bad either, and when you consider that this was an awareness campaign, that was something he could definitely justify internally.

Communication and sharing

James got backing from Carol after she saw his logic and she asked him to publicise the method in a marketing case studies best practices document. This was shared across the business by email and later in an all-hands

session where James met other stakeholders from other units and shared the findings face to face.

Result

The case showed ACME how to optimise campaign spend across all of their business activities. This changed the way the business did campaigns going forward, saved a lot of wasted spend and guided media agencies to be better at what they did. James was delighted.

4.9 ACME business to consumer shopping cart case example

Jan had recently shared the success she had with the marketing team and explained the story about how she went about developing the KPIs and indeed what KPIs she was using. John (the hardware analyst) had gone through a very similar process with his stakeholders in the hardware division and was now looking for a case he could describe to them in similar way Jan had done.

Hardware was split into two divisions. They had what they called “cloud infrastructure” and “in home”. Cloud infrastructure was mainly business to business sales (B2B) where as in home were a range of in home entertainment systems that connected to the cloud.

Things like amps, projectors, speakers, DVD players, music systems that could all be connected to a single cloud system handled by ACME at very low cost to the consumer. Hardware could be leased or bought depending on the consumer’s preference. Leasing plans were becoming quite popular as a service for around €100 per month, bundled with deals from other suppliers that provided movies and music for instance. The best thing was that a new ecommerce engine had been installed a year ago and the consumers could pick and choose the hardware, the delivery options and the installation options.

One of the business questions his stakeholders had asked in his workshop was “How do we get more leasing customers on the books?”This was where

John was sure that there was some room for improvement and where he planned to start.

The analysis

John had decided on a KPI called shopping cart abandonment rate to monitor the effectiveness of the process consumers had to go through to become a leasing customer.

He knew from previous studies he'd read that potentially a lot of money was being left on the table by poor shopping cart usability, poor product placement, poor copy and content and poor persuasion methods.

Disengagement report (shopping cart abandonment)

The first time he looked at the process for signing a leasing agreement, the percentages were as shown in Table 4.3.

As you can see, the "checkout 1" step in this process has a problem. The vast majority of visitors didn't continue from the "add to cart page" (engagement with the shopping cart process, adding a product to the cart), to the second step, the checkout 1 page. What this meant was that less than 1 per cent of people adding a product to their cart got to the checkout 1 page.

John knew this was a massive opportunity. It meant once he figured out why the abandonment was so high at this point, and rectified it, it may drive a lot of revenue.

First John did a heuristic analysis. He took a journey through the website as if he was a customer looking for a lease plan for a home entertainment system.

Walking in his customers' shoes

John developed a persona called Frank Butler, whom he used to do the analysis (see section 6.2 in Chapter 6). He figured Frank might already have heard of the brand through marketing and was more concerned about what happened once he knew ACME sold these products.

Table 4.3 Abandonment funnel

Process step	*Visitors that completed the step*
Adds to cart	100% (30,000)
Checkout 1 – Address info.	0.93% (279)
Checkout 2 – Subscription options	63.68% (177)
Checkout 3 – Credit card information	100% (177)
Confirm/deny	43.36% (76)

The home page was fine. There was a big clear call to action from ACME saying "Buy or Lease a Full Home Entertainment system with ACME from as little as €99 per month, click here to find out more". 37 per cent of those visitors from the home page then went to the consumer store front which was significant volume.

When "Frank" got to the store front there was a choice of three packages: standard, platinum or custom. Standard had a bundle of products. A 70 inch wide screen HDTV, a home stereo system, amp, speakers, DVD player and a cloud connection to on demand movie and music services all for £99 per month with a minimum three-year plan.

Platinum was pretty similar but an upgrade to standard. Instead of 70-inch wide-screen HDTV you had a HD projector with a 120" screen and cinema display, a top of the range home stereo system, amp, speakers, DVD player and a cloud connection to free movie and music services all for £349 per month with a minimum three-year plan.

Custom plans were available that allowed various choices or single items that could be purchased.

John looked at the data over the last 6 months and noticed that the conversions were generally coming from the standard packages or individual purchases through the custom choices. Almost no conversion at all from platinum and zero conversion from custom leases. He wondered why.

He tried ordering a platinum package and saw no problems. Upon adding to cart the purchase price was clear and the lease option was there. He just had to select one or the other. Standard packages converted at about 2 per cent of those who added to cart but platinum was not even 0.1 per cent. He was puzzled. Maybe it was simply the price or that the bundle option was too complex.

He was expecting a failure notice, a timeout or an error message because of the massive drop off rate at this point. However to his mind nothing at all happened.

He went to the custom packages and started adding various items. This was when he noticed an issue. The items added up the price as a purchase price but didn't have the leasing option till after you got to step 2. He checked the analytics. Custom options were by far the most popular choice because unless you didn't have any home entertainment system you would simply be adding hardware to a current collection.

He knew from Frank's persona that there was a great opportunity to test here.

After defining the persona gaps[4] he hypothesised people must have thought:

(a) Platinum was too expensive;
(b) you couldn't buy custom items over lease; and
(c) if you already had an HDTV or music system you might not want a standard or platinum package which meant only a small segment of users would take the next step.

Excited by this idea, he formed a test hypothesis.

What the shopping cart needed was a lease option being displayed as a calculation underneath the items added to the cart each time something was added rather that doing the calculation at checkout step 2.

He contacted Harry in IT and asked for a programming addition to the shopping cart. It was quite an easy process programmatically to change things around so the calculations were made as the consumer added their item to the custom checkout.

Monetisation

After the first day John saw a massive improvement (Table 4.4). The first tests showed a massive improvement in people getting from the "Adds to cart" page to the address information page and then incremental improvements all the way down to the confirmation page.

The result showed an impressive lift in the first week. The average order value (AOV) was €99 per month (or just under €1200 per year). In the first full month after the change the lease program was €169 per month or €2028 per year, but the big improvement was the amount of new lease programs.

Instead of approximately 30,000 per month adding to cart and having only 0.93 per cent move to the next step he had 45.5 per cent go to checkout stage 2. That meant instead of 250–300 people per month moving through each stage of the cart, he suddenly had 13,650.

From those 13,650 he had sales at the end of the funnel of 4884. That equated to annual sales of €60 million (€5.8 million per month). The previous month they had annual sales of €91K. This was a massive result.

Results

John purposely waited one month before assembling his stakeholders. By this time they had almost forgotten about the KPI workshop although after hearing about Jan's success they were enthusiastic about what John could show them.

In a similar way to Jan he went through all the business questions, explained the logic and showed the KPIs they would be using.

The director of hardware (Tim) was nodding quite interested and then said "What would be interesting is if we could see how we'd use one of these KPIs in action. I heard one of our colleagues at marketing, Jan, did some great work there. Did you manage to put anything together like that for us?"

Table 4.4 Abandonment funnel after the test

Process step	*Visitors that completed the step*
Adds to cart	100% (30,000)
Checkout 1 – Address info.	45.5% (13,650)
Checkout 2 – Subscription options	67.8% (9,254)
Checkout 3 – Credit card information	97.2% (8,995)
Confirm/deny	54.3% (4,884)

John smiled. “I went one better. One of the business questions was how do we get more leasing customers on the books. Well, I don’t know if anyone noticed but I made changes to our shopping cart process that got us 3 years performance in one month using the abandonment rate KPI.”

John went through his case study and showed what he did. As he was doing so Tim was furiously tapping at his keyboard pulling up sales figures from the past month. The room was stunned.

Tim was ecstatic. “That’s incredible, absolutely incredible. All because we didn’t have the leasing options shown a bit earlier? This just turned our division into one of the most important in the company. Holy shit. This is hundreds of millions per year!”

Tim called John Bull:

> John, you got to see this. We need funding. We’re sitting on a gold mine here. I need you to ramp up our In Home logistics capabilities. I need people night and day looking at this. Holy crap. Remember John, that guy from Analytics? …Yeah the fella in Bills team attached to our unit …Yeah well he just found a way that has got our shopping cart from producing 90K a year from new customers per month to 6M per year per month … Yeah that’s right, so we did about 70-100 sales per month in lease sales last year and didn’t know what the problem was online … John he just figured it out, we did 4484 last month. We were screwing up on our shopping process. I got orders from all over the planet here …Yeah, 60M per year from last month alone. Yeah. I’ll bring him now.

Tim was grinning ear to ear. “John come with me, we got a meeting with the CEO. Guys, thanks for the time today, make sure every one of John’s recommendations is implemented immediately.”

4.10 ACME customer service business case example

Sam was responsible for the analytics in customer care and after running his workshop and creating his teams KPIs he now needed to start getting the KPIs used routinely. He’d heard of Jan and especially John’s incredible

success and hoping he could find something similar he could use in customer care.

As discussed in section 2.2, customer care had two major problems. The first was supporting software installation and hardware implementation issues. The second was learning how to reduce customer churn.

Sam knew that by automating customer service on their website the company could save a lot of customer service time and gather vital intelligence on what customers needed. What he was looking to do was reduce the amount of man hours spent in answering customer queries and complaints. This might help with the first issue.

The KPIs he focused on was CTR to the support pages and conversion rate (customers who had their requests solved).

The pre-study

To begin Sam investigated the internal costs.

In the KPI workshop Sam had found out from Lisa (who directed customer care) that on a weekly basis the costs of after sales queries was €100,000.

This was calculated from time used by staff answering the phone and multiplying that by the average hourly rate of customer care specialists.

The analysis

Sam found out how many customer service queries were requested from the website and how many of these queries were actually completed. In proportion to the amount of total visitors to the site the number of visitors finding the pages there to help them was quite low.

Only 1.5 per cent of the visitors found the support pages designed to help answer their questions and put them through to after sales from the top three landing pages to the site including the home page.

Walking in his customers shoes

Sam ran the heuristic test with the task in mind of “looking to get support around software”. He started by looking for a customer support link in the top menu. He couldn’t find anything to click but there was a helpline number

on the top right of the page. This was very easy to see and very clear, so at least people could call.

When looking at the analytics he could see that only 10 per cent of all visitors even looked below the fold (the fold being the part of the screen an average browser could not see without scrolling).

Upon examining the website homepage there was very little mention of technical support (one tiny link in the bottom right hand corner). There were three big calls to action: the hardware section, software section (including SaaS) and the consumer section that led to the ecommerce part of the site.

This could be why the click-through was so low because customers either didn't see the link then went to the support contact points from the bottom right of the site, or they mistakenly went to hardware or software sections to seek further information.

On click-through to technical support, Figure 4.2 shows the page you were presented with.

Sam realised this request for information form had too much complexity for the average visitor who wanted to get help. It was long and it had lots of questions even the customer may or may not have information about.

The first big finding from the analytics was the 1.5 per cent click-through to this form.

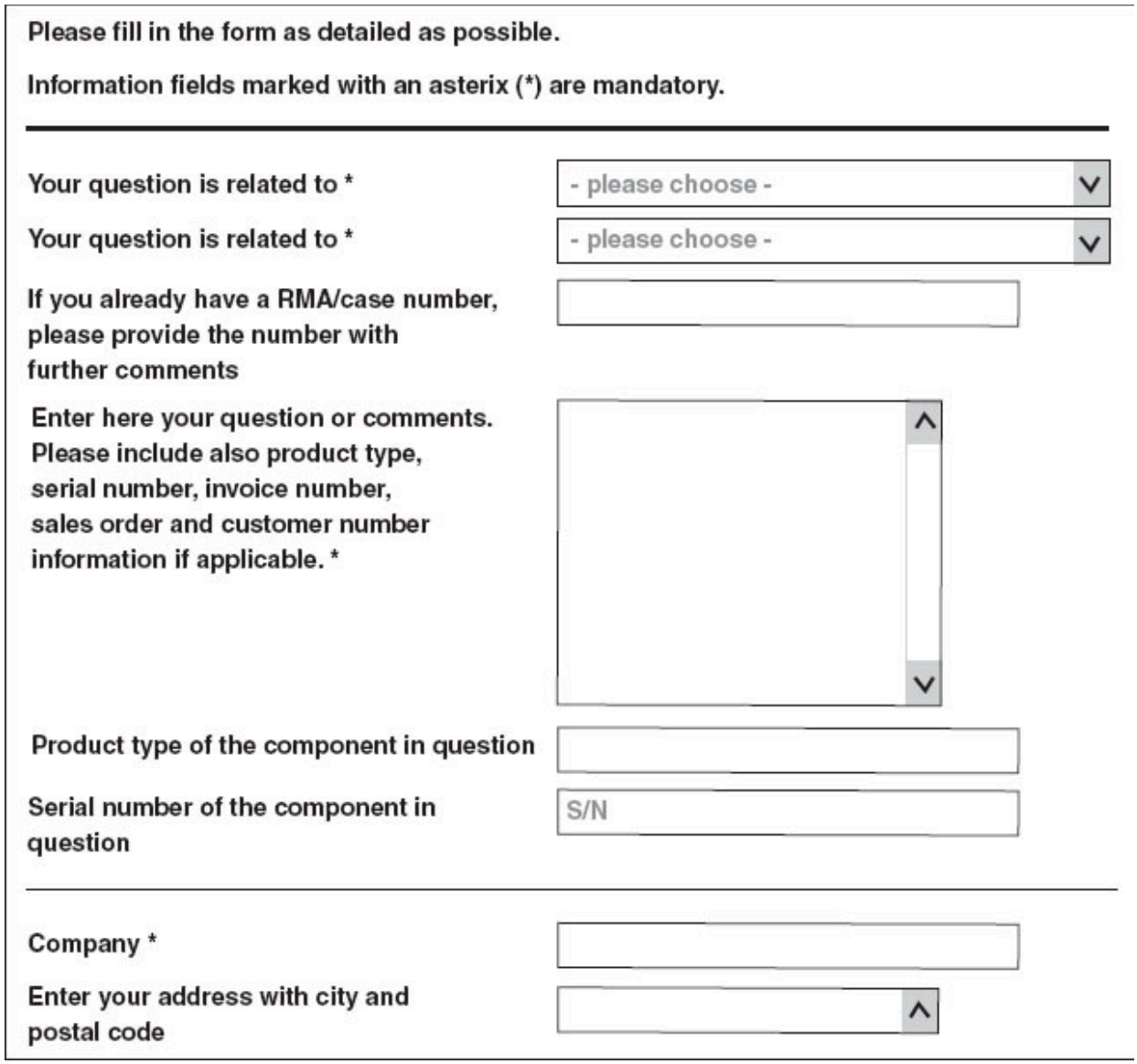


Figure 4.2 Too complex for the average visitor

This second finding was that over 95 per cent of those visitors were not happy with this support page including the form and either left immediately, went back to the homepage or sought answers elsewhere in the site.

Third from Analytics of the support search engine he'd found that the 3 major topics of concern were not being addressed at all on the website. This covered 60 per cent of the requests that came via the support calls.

At this point Sam needed to test his hypothesis. If ACME wanted to encourage people to use this support section and fill in this form it should be very easy to optimise firstly the click-through to this page and secondly

make the form less complex. Then by addressing the top three topics he might be able to cut the call costs.

Building his case

Sam put a meeting together to assemble the main KPI stakeholders, he also asked Harri from IT because he knew he was going to be asking for technical changes to the site. He started the meeting by saying:

> We're only servicing 1.5 per cent of our total visitors on our customer support section and 95 per cent of those don't get answers to their questions. Currently we're paying €5.2 million a year to service our support requests which come from the helpline number. I think we can considerably reduce that by using the online channel as one of our key support centres.
>
> I want to test 3 things: adding a link under the helpline number to get people to the support section, then I want to test making this request form easier to fill in, and finally I want to add some content around the 3 major problems we have requests for. Hypothetically this could save up to 60 per cent of our support calls, but let's be conservative and say 30 per cent. It still means we save €1.5M a year.

Lisa asked "What do the changes mean in terms of cost?"

Sam had anticipated the question and already knew what the customer care specialists cost per hour to create the extra content which he anticipated would be a few thousand. Then there was a little time from Harri's team to make the changes to the site.

Sam responded "Less than 5K total".

Lisa nodded. "This is a no brainer. Harri can we get a schedule on when you could make these changes and let's get this done? Good work Sam. Also one the changes have been made and we have the actual figures could you please put together a presentation deck we can share."

To actually follow this through various tests were carried out. First Sam tested the help section clicks by making the customer support links under the helpline number with an accompanying support image of someone wearing an earpiece and mike typical of those seen in telesales.

The second phase was then to be to improve the online help form to something that helped the user get a support ticket that could be answered rather than use the form to ask complex questions.

The third phase was to improve the support content so that customer service reps could direct people to precise answers to questions or they could find the answers themselves.

The results

ACME's support CTR improved to 7.5 per cent from 1.5 per cent. This meant they got 6 times as many support requests. The cost of that would have been approximately €30 million per annum using the previous methods, however they managed to handle all the requests with only 10 per cent more staff due to the better online delivery and ticketing. A saving of approximately €24M per year.

They also started to earn praise from their customers about how smooth the services were and one of the side effects was reduced customer churn. This equated to another €10 million per year in after sales business as compared to the previous year from customers they would have normally expected to lose.

Lisa was happy because she'd grown her business unit, increased customer satisfaction and improved effectiveness. The analytics was used to continue to feed content development so customers could fine their own answers to questions.

By knowing how to optimise and what actions to make Sam had shown what KPIs are designed for. Taking action in this case was not a clear cut decision, it was easy to say, change the front page of a website, but not easy to do when there are no concrete reasons to do so. By testing and experimenting with the options you learn what works and what needs to be improved.

However by making the decisions a no brainer for Lisa Sam had succeeded in effecting a useful change for the business.

Communication

Once the results came in Sam put out a newsletter that discussed the process and allowed everyone to download a PDF from a resource centre that was set-up as a kind of customer care best practices repository from the Hub.

This hub repository became one of the most popular intranet resources in the company.

4.11 Checklist and exercises

Follow the steps below to begin developing your own analytics business case.

1 Determine your REAN framework business goals

If you haven't developed your REAN framework business goals you could start with the examples provided.

- Reach = greater volumes at reduced cost.
- Engage = determine the areas you interact with your customers that you might be able to improve.
- Activate = understand what drives higher sales with the best value.
- Nurture = understand what kind of reach source drives the best customer lifetime value and when to select a channel to drive a customer.

2 Reach/conversion

What is the best source of traffic in terms of volume and conversion? What is the worst?

3 Run a "what's changed" report

Are there any differences over time? Days? Weeks? Months? Years?

4 Run a disengagement report

Are there any processes on your website stopping your visitor from converting? What is the difference usability wise from best to worst?

5 Check the 25–50–25 rule

6 Form a hypothesis

Once you have gone through steps 2, 3, 4 and 5 you should have formed an idea. The next step is to prove that idea is a strong one.

Can you prove or disprove whether your research is pointing to any particular issue or opportunity?

(a) Have you determined the cause of the issue you're researching (i.e.a campaign peak)?
(b) Have you identified the effect that the issue you're researching has had (i.e. a lift or a drop)?
(c) Can you correlate with a degree of confidence that the cause has impacted the effect (i.e. that when you ran a what's changed report you saw 90 per cent of the peak was attributed to the campaign)?
(d) Are there any open questions you cannot answer? If so determine what you're missing, go back to step (a) and repeat until there are no open questions left.

7 Once you have proven your hypotheses monetise the opportunity or the loss.

8 Create an easy way to share the information in the form of a simple to understand report.

9 Communicate the win in an easy to understand manner. Remember to speak in the language your audience will understand.

10 Publish the study as widely as possible in the organisation.

Notes

1 Google Analytics Certified Partner – see www.google.com/analytics/partners/search/all.
2 Excerpt from The Sucking Manifesto, available at http://thesuckingmanifesto.com/toc/the_model.
3 Excerpt from The Sucking Manifesto, available at http://thesuckingmanifesto.com/toc/the_model#example.
4 See the Persona Template file, available at www.blackbeak.com/cult2/files.

5 Tools of the trade

> Technology is nothing. What's important is that you have a faith in people, that they're basically good and smart, and if you give them tools, they'll do wonderful things with them.
>
> Steve Jobs

5.1 Data

There are five types of data people talk about when they refer to digital analytics:

- The first type is quantitative data measured automatically by analytics tools and is purely behavioural. This is often called *clickstream data*, outcomes data, video stream data, or visitor logs, and is traditionally what web analytics has always been about.
- Then there is data that comes directly from the *experiences* that the customer or prospect have about your offer, because of this it is also known as voice of customer data or attitudinal data.
- The third data type is *ecosystem* data. This could be where you rely on panel or ISP's to provide data about how well you're doing in comparison to your competition or enrich the datasets you have by adding relevant third-party data like the weather.
- The fourth type is *social data*. In the past 10 years companies like Facebook, LinkedIn and Twitter have established themselves as a major part of people's lives. There are over 1 billion people on Facebook. The reason they're successful is in large part due to their management of data.

- The fifth type is *big data*. This is usually a combination of the other 4 datasets but according to Wikipedia "Big data is a broad term for data sets so large or complex that traditional data processing applications are inadequate."[1]

What that means is that previous generations of databases (relational) have difficulty in storing and managing big datasets. The data handling requires running software on parallel servers often distributed geographically to better handle local requests of their databases.

5.2 Clickstream data

Measuring quantitative clickstream data and trying to make sense out of it all is the challenge when there is so much of it. This is why we have models like REAN (see Chapter 2) that can help considerably in planning mechanisms to capture data.

Clickstream data in terms of *reach* is finding out what the best source of traffic is by volumes or cost. It's about the numbers of visits or visitors that find your site.

In terms of *engagement* it's how the visitors behave, what processes they complete, what processes they abandon, how they interact with your website, how long they stay with you.

But just having them arrive and interact with you is not the point. You want them to *activate*, convert, to do something (see Chapter 3) so you measure the outcomes, the conversions of each reach source as well and start to make choices based on how the visitors behave.

In terms of seeing how you're *nurturing* your customers the clickstream data you're looking for is where your customers to purchase a second and third time. It's about measuring how your audience consumes more of your products or services and predicting when you should *react* or be *proactive* in your contact with them.

5.3 Experience data

As previously mentioned web analytics will tell you what people have done but it will never tell you *why they have done it*. One of the key differentiators between a very good analyst and an average analyst is the ability to find out why. When encountered with data that has deviated from normal levels a great analyst will investigate more deeply to try and find the root cause of the deviation. In many cases a great analyst can take actions based only quantitative web analytics data and one type of experience data – heuristic testing.

Heuristic testing

Heuristic testing requires that you put yourself into the mind of the visitor who is on your website and take an objective look at how difficult it is to complete a given task. There's a couple of examples of heuristic testing already in Chapter 4. See also Chapter 6 defining a persona for more information on getting into character. By looking at web analytics data you will be able to see where people are leaving the site for instance the top exit points in process funnels, top exit rates on pages or high bounce rates.

Then by putting yourself into the mindset of the prospect or customer and following their click through process online (especially at the exit points) you may gain some insight into why they leave. Heuristic can give you excellent insight especially if you have usability best practice knowledge and can combine that with what you've seen in your web analytics tool.

For example a typical heuristic test might be run on a shopping cart that had a high abandonment rate at a certain point. You go to the point where the visitor trend shows they have been having problems and do a heuristic test on a typical visitor's motivation around that point. It may be that by doing so you find a difficulty with the website that you previously did not know about, such as a problem with the checkout procedure for visitors using Firefox, for example.

Survey data

Another very valuable form of data is survey data. There are three forms of web survey data you can take advantage of:

- *Customer surveys* (post-activation survey) – Typically a customer survey is carried out from customers who have bought one of your products or services. Surveys of this kind usually focus around the product or service, the level of satisfaction and the quality of the customers' experience. The information from this data can be used in product strategy and website design.
- *Visitor surveys* (pre-activation survey) – Typically a survey is placed at strategic points on the website with an incentive for the visitor to complete a short questionnaire. Surveys of this kind usually focus on finding out how the visitor finds the site, how easy it is to complete tasks and what else the website could do to cater more effectively to the visitor. Companies also use surveys of this kind to segment out visitors that have answered positively and negatively using the so called net promoter score.[2] It's often helpful to be able to compare what loyal visitors are looking at versus unhappy visitors.
- *Exit surveys* (point of disengagement survey) – Similar to the visitor survey except that typically the survey is launched when the person leaves the website. Exit surveys are particularly useful to troubleshoot high abandonment points on the website. You're looking for trends that occur among the visitors who answer the questions to try and pinpoint root causes of abandonment.

A great free tool for visitor surveys is 4Q from iPerceptions (http://4q.iperceptions.com). One of the best paid tools I've seen is Survey Monkey (www.surveymonkey.com).

Lab usability data

Often called usability studies this is where you have enabled the conditions to observe what people do when they visit your website. For instance you can give participants a task to complete and ask them to go to a specially monitored PC where they try to complete the task at hand.

Their eye movements, hand movements, mouse movements, keystrokes and clicks are all recorded to be analysed later. This method can give you extremely valuable insights about how people do things on your website. Again you're looking for common trends from different participants so that

you can pick up a common way that people generally look for things on the site you're studying.

In essence experience data is any data you gather that gives you ideas about why visitors or customers do things. Surveys, usability testing, eye tracking tests, heuristic analysis and interviews are all types of experience data.

5.4 Ecosystem data

Neil Mason once commented in a post he wrote in 2005:

> Although growth is good, it isn't good enough. And that's the point. When markets are generally still growing organically, as they are in Europe, how do you know when good is good enough and that you're maximising potential? By tracking what the market is doing and benchmarking against your competitors.[3]

That's a timeless statement that is as valid today as it was then.

As Neil suggests growth alone is good, but alone not good enough. What happens if you grow at 15 per cent but your competition grows at 25 per cent? You'll soon find yourself being left behind.

You should also consider the data gathered by independent companies like Forrester, eMarketer, Marketing Sherpa and e-Consultancy who actually serve their customers by benchmarking best practices such as web analytics.

To look directly at your competition and compare yourself across different things you should look at as many types of competitive data you can find. There are lots of freely available and paid sources.

An example of such data can easily be found on Google trends (Figure 5.1). In October 2007 both were growing at a steady pace, more or less growing with the market. The Nokia brand was stronger at the time. In November 2007[4] the android operating system was launched with an initiative called the Open Handset Alliance. HTC, Samsung, Sony and wireless carriers formed a consortium to develop open mobile phone standards. Since then you can see year on year growth of the Samsung and year on year decline in Nokia who decided not to adopt Android and go the Windows route. Clearly a mistake, with the benefit of hindsight.

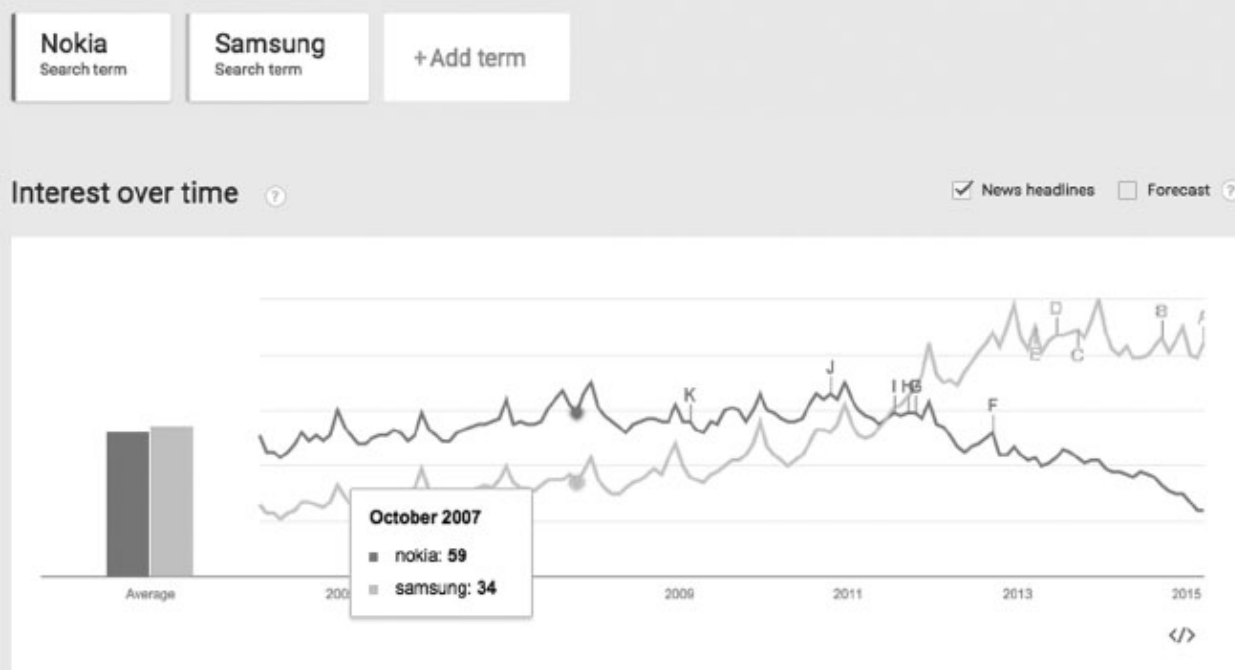


Figure 5.1 Google trends competitive data for Nokia and Samsung keywords

Search engine data

Google Analytics has a free competitive benchmarking service if you opt-in to their service.

The benchmark will allow you to look at related industry websites with a similar amount of traffic and see if you're doing better or worse than they are.

You could also look at Google webmaster tools data, Google Ad planner data or Google trends to see how consumers are using keywords to find related subjects to your business.

There are also alternative tools that use data like webmaster tools and make it more manageable. A good source I found was Pam Dyer, who has a list of free and paid search engine tools on her blog panorama.net.[5] I have personally used all of the ones Pam lists to identify what people do on search engines, compare my clients against the competition and help enrich other data sets. However, Google is your friend here. Do a search for top 50 Keyword research tools and you'll find a wide selection of free and paid services.

Audience measurement data

Often useful for benchmarking yourself against the competition, and there are a few ways to do it.

Panels

Panel data is collected from visitors who join a panel thus allowing vendors to track their movements across the Internet. This clickstream data is collected in a variety of ways.

Application service providers

The application service provider (ASP) is typically a company that will ask a participating website to be part of its panel by inserting Javascript code on its pages. An ISP usually has an agreement with a partner (like Hitwise) to supply the vendor with anonymous aggregated data to track trends.

Telecomms/internet service providers

An internet service provider (ISP) usually has an agreement with a partner (like Hitwise) to supply the vendor with anonymous aggregated data to track trends.

Examples

Compete (panel)

Compete (www.compete.com) allows you to compare yourself and your competition against each other in a similar way to Google Trends shown previously though with vastly more data and insight (it is a paid tool).

Compete.com is a form of (largely) panel based data though they do use other ways to collect and compile their data as shown.

Alexa (panel)

Alexa (www.alexa.com) is another largely US centric free panel based system offering a similar kind of service to compete.com. Alexa is an Amazon service so the panel is globally offered though there is a big US bias especially on smaller sites nearer the bottom of the top 100,000 sites in the Alexa list.

Comscore (panel/ASP)

Comscore (www.comscore.com) is another notable company in the panel space but offering a much more tailored service at a much higher premium. They have now combined audience data with their web analytics data directly. Panel data from the audience supplements the data directly on the company website tracked by Javascript.

TNS Gallup (ASP)

TNS Gallup (www.gallupweb.com/tnsmetrix) is very popular in Finland with publishers and advertisers because it gives an un-biased and fair view of how many visitors each major media website in Finland gets. Just about every media player of note is on the TNS list meaning the site is very well visited by advertisers and often quoted and published by the media companies.

FireClick Index (ASP)

FireClick (http://index.fireclick.com) is another ASP-based solution. It's actually a web analytics tool that aggregates its data and publishes it publicly. Again not perfect because you're relying on a small sample size with the majority of the geographical distribution of visitors from the USA.

Experian HitWise (ISP)

The best example of ISP data is HitWise (www.hitwise.com), which has deals with many of the world's largest ISPs to process aggregate data from subscriber web browsing behaviour. The advantage is that HitWise can tie in demographic data and, like Comscore, bring far better tailored services to large enterprises.

Quantcast (ASP)

Quantcast (www.quantcast.com) is a free service with data coming from mainly ASP sources but also part panel data. The interesting part about this solution is that the amount of data it is gathering with in 2014 over 100 Million web properties[6] now having adopted their technology.

Business data (mix)

There are lots of ways to get information about companies. All public companies have to report their yearly earnings. Many websites cover this information on stock trades but simply by looking at annual reports you can learn a lot about the business you're researching.

If you're a private business it's often harder but in some countries you can get good information or pay to get a credit report which will give all the numbers you need about a private business as well. One such example in Finland is Fonecta Finder (www.finder.fi).

5.5 Social data

I like to think about measuring social activities in three ways. You have research and development data, advertising data and competitive data (see also ecosystem data).

Research and development

Research and development (R&D) is data that helps you uncover something new about the product or service that you're working with. For instance, using a tool to crawl social networks looking for your brand keywords in order to see what users are saying about your product.

We once worked with a client that was selling boiled sweets and noticed that people were using the sweets in an unusual way. They were dropping them into vodka, which totally changed the way their vodka tasted. This wasn't the client's direct target market at the time and was a new insight for them. They developed a successful marketing campaign for their brand based on this new angle.

Advertising

Advertising on social channels is usually done in one of two ways.

First, by building up a community of users (customers or prospects) you may post things on your social networks that may interest them in the hope that they share the information with their social networks. This induces a viral effect. In order to increase your chances of engaging a larger audience you look for people commonly called "influencers" and hope your content persuades them to share your ideas. Influencers typically have a wide base of followers that means the reach is much higher than a typical push advertising campaign.

Second, you simply promote content on social channels. In a very similar way to a display advert or a search engine advert, you place an ad on a social channel. The advantage is that the ad can be targeted really well as the social network will typically have a great deal of data about their users. The disadvantage is often a lack of relevance with the users largely ignoring all but the best ad content.

Competitive and ecosystem

Social data can also be used to inform on what the competition are interested in, what they're doing and who they work with. For instance following people from your competitors on LinkedIn Sales Navigator allows you to see what they post, what things they like and who they connect with.

Tools like Sprout Social (www.sproutsocial.com) allow you to directly compare yourself against your competitors across key metrics like followers, engagement and influence for social channels like Twitter.

5.6 Big data

As mentioned earlier in this chapter, big data is a broad term for data sets so large or complex that traditional data processing applications are inadequate.

Big data therefore requires advanced technology to manage and analyse the data such as Hadoop[7] which was inspired by Google and developed by an ex Yahoo! Engineer. Such companies were the first to have to deal with big data. Keyword data coming in from all over the world is an example of

big data. According to Internet Live stats,[8] Google deals with 40,000 keyword searches per second. Or to put it another way 1.2 trillion searches per year worldwide (in 2012). In order to give a response to the user in milliseconds no matter where they are on the planet you need a big data architecture.

The tools behind web analytics are big data tools as well as they are dealing with data coming from billions of web pages and serving analysis results in real time.

Your job as an analyst isn't necessarily to learn how to create these tools, you have developers, coders and engineers who will do that for you. Your job as an analyst is to translate the data into meaningful information that drives action. Or your job as a marketer is to use third-party big data tools to make your companies life easier.[9]

The most powerful way of doing analytics comes from using data sources to find out "what" is going on, "why" people have done things in the way they have and where you stand in the whole ecosystem so you may plan for the future.

5.7 Data collection methods

All web analytics tools have a method to track the visitor. It's important to understand the different types of data collection and understand the main differences between the collection methods.

Beacons

Beacons are usually invisible 1 × 1 pixels that are embedded into the pages you want to track. Email marketers often use them to track when an email has been opened and ad serving networks use them to track campaigns across multiple websites. The way the technology works is that when an email or web page is opened by the visitor the web server sending the page back to your visitors browser also sends a "get request" to a third-party server (the ad network vendor tracking the activity for instance). As the page loads it executes the request for the image. This then sends anonymous data about the page view back to the vendor.

The disadvantage with using this kind of tracking is if the client software IE the web browser or email program has images turned off. If the image requests are turned off then you can't collect the data. Beacons also have to work with 3rd party ad networks (or email providers) that can mean the third party cookie is used to identify the anonymous visit as a new or returning visit. third-party cookies are often blocked by spyware and adware software on the visitors browser meaning the solution may not track all visits.

Tags/Javascript

Javascript tagging is one of the most popular ways to measure website traffic. The website being tracked has Javascript code inserted on every page. The Javascript (when loaded by the visitor) assigns each visitor a cookie and sends the information about the visitor and visitor session to a collection server. The technique does not record any personal data about the visitor it just uses the cookie as an identifier so that if the visitor comes back they can be classed as a repeat visitor rather than a new visitor. If the visitor has cookies disabled then the tracking will only record the visit information in most cases (some systems also revert to visitor IP address and browser version if a cookie is not available – though this has questionable repeat visit accuracy.)

The main disadvantage of tag based solutions is the fact that you have to tag every page you want to track. On an enterprise level when multiple implementation vendors are involved controlling the tagging quality can become challenging. This is why "tag management systems" have come to prominence (see section 5.8).

Logs

Site logs were the first method of tracking on site behaviour and were originally designed for technical reasons. Logs are records of all requests made to the web server. When a visitor visits a page the request is logged to the server. Unique visitors are typically identified by the IP address and header of the visitor but some more advanced versions like Web Trends also assign a cookie to the visitor for greater accuracy.

Unfortunately the log files also record all other requests to the server which can include search engine crawlers, email harvesters (programs crawling the web designed to gather email addresses) and because many solutions use IP address to identify the visitor the number is consistently inflated. IP addresses are often dynamically changed by Internet service providers (sometimes even in the same session) meaning a repeat visitor to your site could be logged as three visitors (or three, or ten, depending on how often he logs out from his ISP and is assigned a new IP address).

For this reason the standard has shifted from log files to Javascript tagging despite the tagging issues.

Packet sniffers

An alternative solution to tags and logs is the network sniffer. Hardware and software is installed on the web servers that capture all the network traffic. Traffic is identified by cookies and very light Javascript tagging is required to capture information like which links are clicked in the browser or flash events. However the main bulk of the tracking is done at the network level. This is a good alternative for large enterprises as the tagging problem is less of a challenge.

The disadvantage is cost in terms of implementation and price. The network sniffing solution is typically more expensive than tagging both to purchase and implement though savings longer term can be made by having less tag maintenance. You also need to be careful on privacy with packet sniffing. Configuration should mean that no personally identifiable information (PII) is passed or you could be breaking the law.

Crawlers/bots

Social media tools utilise crawler data in addition to data provided directly from the network. So for instance Facebook gives you a lot of data which many tools connect to via an API. However many tools also crawl the websites with a query (such as a keyword) to find out how many times a brand has been mentioned.

You also have ping bots that measure things like site uptime. This is done by pinging a website with a request and sending information packets back to

a system designed to understand whether the website is working or not.

5.8 Tag management systems

As discussed in section 5.7 Javascript tagging of pages is one of the most accurate ways to measure a website but also one of the most laborious because you have to tag every page of the website.

This wasn't a problem in the early days because Javascript tagging wasn't so complex unless you used a high end tool like Adobe Analytics.[10]

However with the increasing number of tools available to the average marketer using page tags the workload exploded. A solution was needed to tackle the ever increasing problem.

When you require pixels to be placed on certain pages for advertising, testing tools on other pages that load synchronously, analytics systems such as Adobe or Google Analytics, click mapping software such as Crazy Egg and want to keep all the data in one repository (i.e. your own data management platform) then the scale and requirements shift dramatically.

Tag management systems (TMS) help with these problems by providing a simple middle layer that allows you to handle all the tagging in the cloud. The idea is you place one Javascript tag on the website and then route all other tags through the tag management system.

This allows you to deploy potentially hundreds of tools on your website all using one Javascript tag. Potentially never having to hard code the website after the initial TMS installation.

In addition by organising all tags in a TMS you automatically create a record of each tag. It means documenting what is happening with your tags is automatically handled. The other great benefit of managing tags from the cloud is quality assurance. You can test and deploy everything prior to launching on the production site as well as control access to different versions.

The solutions didn't come about until around 2009, with early leaders like Ensighten[11] raising funding and buying out competitors like TagMan in 2014. Google however also entered the space on 1 October 2012, giving away their system, Google Tag Manager (GTM[12]), for free.

In a similar way to when Google Analytics was launched in 2005 GTM provided a lot more visibility allowing companies that previously didn't know about tag management to start getting involved.

Tag management is now widespread and if your company is not deploying tags via a TMS then you're wasting valuable time if you still hard code your pages.

5.9 Analytics tool selection

When selecting a web analytics tool business needs should be taken into account and it's why you should define what you want to measure before getting a system implemented.

Tool awareness recommendation

You might want to think about using free tools (such as Google Analytics) if there is the need to prove to your people the benefits of web analytics.

Using free tools means the Analytics hub in the organisation know what is possible and it gives them a benchmark for future selection. A tool selection phase should only happen when the business knows what it wants to measure and knows the free vendor offerings are not up to the task. I have purposely not listed the paid tools and tried to compare them as I don't want to endorse one over the other without knowing your business.

For the record the paid vendors I would recommend looking at in Europe are, at the time of writing, in this order:

- Adobe Analytics (www.adobe.com/marketing-cloud/web-analytics/marketing-analytics-capabilities.html)
- Unica (www-01.ibm.com/software/info/unica)
- CoreMetrics (www-01.ibm.com/software/info/coremetrics-software)
- SAS (www.sas.com)
- WebTrends (www.webtrends.com).

If you do speak to the vendors above then Table 5.1 might help you to find out who you want to work with.

5.10 Analytics implementation

Implementation again depends greatly on your business. The more external agencies involved with the website design and campaign management aspects of your online marketing the more difficult it'll be to control data integrity and quality. Where possible centralise any page tagging with a process that goes through the hub. Whenever the website content is updated a control version of the website should be created with the old tags and old version as a form of quality control. Using a version of CVS control will help this process. Using a TMS is greatly advised.

If you have to use a number of agencies use one agency to project manage the whole exercise as part of your hub team. It should be made clear to them the importance of analytics and the tagging requirements you have.

Implementation planning

The best way to implement is to let the vendor specifically implement the solution basing their implementation around a pre-defined set of measurements important to your business (KPIs).

What's important is that your hub are involved intimately with this process and are taught the technical requirements to make day to day changes themselves or that a vendor/client agreement is in place so that the vendor deals with all future requests within a certain timeframe.

Data quality control

The rest of your implementation planning process should be focused around data quality.

In Table 5.2 the first four entries in part 1 of the data quality checklist show suggested ideas for the hub to implement in order to help verify and control data quality. Again these ideas will depend greatly on your tool of choice but some form of testing along these lines can be carried out with any tool. The last three sections in part 2 of the checklist are important to understand from the enterprise process perspective.

Table 5.1 Tool selection: questions to ask your vendor

Topic		*Questions to ask your vendor or consultant*	*The following considerations should be taken into account about how the product handles data sources*
1. Data collection and storage	Data collection sources	How many methods does the product use to collect clickstream data?	Generally the more ways the product has to handle data collection the better. It means the tool has the ability to be customised with other external data sources allowing you to combine offline data for instance. That means more of your business questions can be answered by the tool.
	Hybrid data collection	Can the product collect and merge data from multiple sources simultaneously?	No ability? Handled as a custom case? This might be important for websites which for instance need to combine site logs with data from tags in order to know when search engine spiders access a website for instance but the primary collection method is page tagging (which doesn't track spiders).
	Rich-client tracking	Can the product track Flash and other rich-client applications? How is this handled?	Important for sites with a lot of rich media applications because collecting and reporting the data is not standard
	Data warehousing	Does the product offer an option to store everything in a data warehouse? How is this option priced (incremental percentage over base CPM rate, flat fee)?	Data warehousing is important for companies that require good segmentation, good historical data and strong filtering/data mining options. It's very important to find out what the vendors charges for this will be so that you don't get any nasty surprises after signing the contract. Many add a percentage to their overall prices or add to their cost per impression basis to have the ability to use a data warehouse. Some charge a one off flat fee. Some don't charge and have it included as part of the base offering.
	Custom variables	Can the product track custom variables?	Usually this is useful for all companies and most vendors handle custom variables. Find out how and what kind of things

			custom variables track. Ask the vendor for an example relating to your business.
	Data storage/ownership: summarised data	How long does the vendor keep detailed and summarised data? Who owns the data?	Less than three months realtime data? 12 months data? Unlimited? The data used to produce reports in the product is called summarised data. Often vendors will keep a certain amount of data instantly accessible and ready to work on but will require a few days to produce historical data. Also does the vendor own the data in their databases or do you as the client? Google Analytics will never let you delete data as they own it.
	Data storage/ownership: raw data	How long does the vendor keep raw/unprocessed data? Who owns the data?	Less than 15 months? Unlimited/term of customer agreement (or licensed software)? Retaining raw data is necessary for some analysis such as data mining.
	Data import	How can data from exitsing systems (e.g., eCommerce, CRM systems) be imported to your tool to augment customer data?	No capability? Is it a manual process requiring the vendor to customise their product? Or is it automatic and inbuilt? Importing data allows you to compare other data sources to the web data. It may not be possible with some vendors or they may need custom set-ups. Some of them can do this automatically.
	Data export	How does the product export data?	How are exports handled? Is there the ability to export directly into preformatted Excel sheets (for KPI processes) via streaming data directly into Excel or is the data exported manually? Can scheduled exports be sent automatically?
2. Data integrity and security	Validation – installation	How is data collection validated at product installation?	No formal validation procedures or tool? Are there Informal testing procedures? Or does the vendor have an extensive procedure? How is the vendor testing the quality of the data? Most free tools only have a limited process, for instance checking if the tag is on your site. Ask the vendor what differences

			they have to the free tools in this respect.
	Content filtering	How does your product handle traffic filtering?	Filtering enables you to for instance filter out internal traffic so you don't get company employees included in the traffic reports. Can the system filter data retroactively to include/eliminate unwanted sources of activity like spiders, robots, or anything else you want to take away?
	Cookie support	What form of cookies does the product support? (if applicable)	No first-party cookie support? Firstparty cookies support at extra cost? First-party support at no extra cost? 1st party cookies are less likely to be deleted by visitors. Paid vendors almost always support this but costs may be incurred.
3. Segmentation and profiles	Retrospective segments	Does the product allow creation of retrospective segments?	Is it possible to set-up segments on data that is already in the database? Can you go back and look at historical data based on a segment you set-up today? Very useful for campaign analysis for instance.
	External segmentation	Custom segmentation based on external data sources	Can you set-up segments based on data imported from external data sources. For instance could you set-up segments based on customer data and start segmenting known visitors from opted in sources?
	Segmentation costs	How many segments can the user create at no extra charge?	Vendors almost always charge extra for the service. Find out how many segments each tool allows as standard, what extra costs are involved and whether for instance you need extra features to support segmentation (like data warehousing)
	Customer profiles	Does the product support the ability to create, update, and share customer profiles via database access or a profile API?	Customer profiles could be set-up for instance from CRM data sources. Does the vendor allow access to an API or database to upload customer profiles to the vendor? This might be useful in later stages for testing recency and frequency models of actual customers or customer profiles.

4. Reporting	Dimensions/correlations	How advanced is the product's support for correlating data dimensions?	For instance is it possible to cross tabulate or drill down on different parameters like language, geography, campaign, keyword or referrer? How many correlations can be made? Is the product limited in this respect to out of the box features or is it unlimited (i.e. can you easily create your own correlations?)
	Scenario analysis	How many scenarios/funnels can the user track simultaneously?	Scenarios might be things like setting up a custom funnel analysis of a shopping cart or lead generation process. Are extra costs associated with scenarios? Are there limits on numbers of scenarios you can run at any one time?
	Excel interface	Can Excel-based reports be refreshed automatically when the workbook is opened?	Does it work by generating a simple XLS file generation (a new file is generated every time you want new numbers?) Or is there an API through third-party tools or an API from within Excel to the Web analytics database? The latter two are better because it means you can get trended data much faster into Excel and generate KPIs based around automatically updated Excel data. The first means cut and pasting the data into your preformatted Excel sheets to get the same result which is time consuming.
	Calendars	Can clients base analyses on custom calendars relative to key events like the launch of products or media campaigns?	Most tools have this on a basic level (i.e. monthly, weekly, daily) so find out if there is support for reporting in variable time periods or advanced calendar reporting.
	Partner integration and APIs	Does the product provide APIs to enable partner integration and access to data?	It might be important to your business for a CRM partner or datamining partner to be able to access your raw data.
	Ad-hoc query capabilities	Can you query a data warehouse for ad-hoc queries or data	If not then what is the process involved? The data warehouse is very valuable source of information

		exploration without vendor intervention?	and if there is the requirement to get permission to access it then you need to know how it's done and how long it will take.
	Path/scenario visualisation	How advanced is the product's support for scenario/clickpath data visualisation?	How good is the out of the box reporting of the funnel paths in a visual graph with incoming and outgoing traffic to/out of the path steps visualised.
	Report visualisation	How advanced is the product's report visualisation?	Important that the system is well visualised and easy to follow for hub analysts but also if any access is to be allowed then it needs to be easy for the average business user.
	Experimental design support	What native support (i.e., not through partnerships) does the product offer for conducting and analysing designed experiments (also called A/B or split-run tests)?	Most products have the ability to compare tests. The question is does the vendor you're working with have more than simply the ability to compare datasets? Does it have any integrated testing technology allowing you to do on the fly A/B or multivariate testing? If not can external solutions be integrated into the vendors product?
5. Onsite Search	Capabilities	How advanced are the product's capabilities for reporting on site-based search results?	If you have an on-site search engine it's a superb source of qualitative data about what your potential customers are asking for. Nearly all vendors have a way of capturing this data. Ask what and how they do it and how it is displayed. The reporting quality varies greatly from vendor to vendor.
	Email Campaign management	Does the product have features for email campaign analysis or interfaces to third-party email campaign management vendors?	Almost all prospects and customers are sent emails, whether it's newsletters, sales letters, thank you for your custom messages, or even simply answers to questions. How does the vendor integrate email and/or CRM. How does it report on email activity out of the box?
	Campaign Management	Does the product have features to automate search engine marketing campaigns, affiliates	Undoubtedly at some stage you will run a campaign, be it a search marketing campaign, affiliate program or banner ad campaign. How does the vendor handle this?

		and/or banner ad placements?	Is the cost of tracking this activity included in the package you're being sold and how well visualised is this?
	Benchmarking	Does the vendor offer a benchmarking service across its client base (e.g. retail conversion rates)?	Competitive data can help you to benchmark how well you're performing against the competition. Is this part of the vendors package. If so ask to see how it's reported.
6. Service and support	Technical support availability	When is live (standard service) phone technical support available? Is backup (phone or email) available for off hours?	Usually this is business hours at least. However some vendors only offer email support. Make sure you know what technical support is available.
	Technical support locations	Does the vendor have more than one support location for in-market support?	Are the vendor in your country? It's possible they may have sales operations in your country but do they have a network of technical support analysts available? If not do they work with partners. If so who? What are their prices? You may need their services.
	Technical support service levels	How do reference clients rate the vendor's technical support responsiveness?	Call the reference clients and ask them for their best and worst experiences. Ask the vendor for names and drop them an email.
	Training curriculum richness	How do the vendor's training courses compare to competitors?	Do the training packages come as part of the cost? Is the training technical or analysis based? How long and how detailed are the courses? You will need to know this so you can allocate time for the hub analysts to train with the tool.
	Consulting/professional services	How robust is the vendor's professional services offering?	It may be that the vendor only has professional services in certain countries. If they don't consult around the tool do they work with agencies that do? Can they recommend any and what are their prices?You may need their services especially in the early stages.

Table 5.2 Data quality control

Data quality checklist		*Quality control test to undertake*	*The following considerations should be taken into account in order to ensure data quality and integrity*
Part 1	Validate data	**Data validation test** – monthly spot check and site upgrade testing after any new content uploaded	Is the data being recorded in exactly the manner it should? Is it being recorded in the right analytics profile or account? Finally and most importantly is there an effective plan in place to upgrade websites with new information while retaining data quality? Is there a good system of version control in place? IE CVS? Validation version should also be recorded in any audits.
	Data types	**Data classification** – certify the data type based on the difficulty of maintenance	1. Clientside data – flash applications, rich media. 2. Count data – tagging required on specific actions and specialist requirements such as campaigns. 3. Automated data – syndicated or dynamic content that is automatically tracked upon going live.
	Define terms	**Terms definition document** – internal terms glossary	Make sure when you're discussing a conversion rate that everyone knows what a conversion rate is.
	Count Accuracy	**The big 3 test** – quarterly quality testing	Do a page views, visits and visitors (the so called big 3) quarterly deviation check. Use standard deviations to measure overall count trends. If they deviate from the normal levels investigate further.

Part 2	Data ownership	**Hub quality manager** – overall responsibility for data quality	Name and contact details of the person responsible for overall data quality and names of hub specialists responsible for each spokes data quality. If this is handled by vendors or consultants list them and share the information across the organisation.
	Audit trail	**Paperwork or data quality record** – any report containing data without an audit trail is classed as high risk data	1. High risk data – non-audited. 2. Standard data – data validation test carried out. 3. Low risk data – data validation test carried out for specific report or analysis.
	Staff training and process	**Quality process** – how will the process be communicated and implemented?	All staff should be made aware of the audit trail and understand that they cannot rely on high risk data being accurate. They should have the hub responsible personnel carry out a test and make sure the data is valid.

Data ownership

It is important that a senior person in the hub is responsible for enterprise wide data ownership and data quality. He or she should be responsible for setting up and ensuring processes are followed and audits carried out. I would suggest that your platforms manager should work closely with senior analytics specialists for data ownership and quality.

Audit trail

The audit trail should be documented and clearly defined.

If reports and analysis are submitted to decision makers with data that is not clearly defined then make sure they have the necessary disclaimers. No

responsible manager is going to rely on disclaimed data to take major decisions but some data is necessarily "a grey area" and indicative and could be included with a disclaimer as to what the actual figure represents.

In reality someone may request data that isn't fully validated due to urgency (early results on hastily set-up campaigns for instance) and that in itself is not an ideal situation. However at least the person knows the data is not validated and decisions he/she takes could be based on erroneous data. Transparency is better than hoping the numbers are right.

With regard to tag management, make sure good taxonomy is in place for tag rules. For example, a good taxonomy might be:

Company name : Tool Type : Tag variable : Tag description
ACME : Adobe Analytics : eVar 34 : Download instances
ACME : Google Analytics : Event : Download clicks

Finally, set-up tool alerts for data fluctuations. If for instance uniques visiting your website on a Monday is 75 per cent lower this Monday than it was last Monday, is there a tagging problem?

Staff training and processes

Best practice here is to make sure everyone understands and follows the audit trail. How the hub sets this up depends entirely again on the business in question but where possible integrate your data quality process with other existing processes. This again helps you to build things that are followed by the rest of the organisation.

5.11 ACME analytics implementation

ACME had issues with data integrity and Bill wasn't happy with what he saw as poor governance, quality control and documentation of what was going on with data collection in the company.

For years ACME had been using Adobe Analytics[13] on their website which is an extremely powerful tool but only if the tagging was correct. Additionally the new case studies his team had produced across the business meant that all sorts of new requests were coming in and his team couldn't

cope when they had technical problems getting good data from other sources than simply Adobe. They were becoming victims of their own success.

He gathered the team to discuss it on the one day of the week they all sat with him in his office area.

“How do we scale our operations so we don’t get bogged down with technical problems?” Bill asked.

“Tag management”, Tommi quickly replied. “We’ve mentioned it before but we didn’t have a business case to purchase the product. Now there are a number of reasons to get a TMS. Adobe have a free product as well called DTM, but there are other options we could use, like Ensighten.”[14]

Bill had heard of tag management but wasn’t fully aware of what the tools could do.

“How will it help?” he asked.

“Well, according to other folks I know in the industry, using a TMS has scaled up the speed to deploy tags by 8–10 times”, said Tommi. “Whereas before we’d get one deployment out every couple of weeks we’d be able to get up to five out per week. It depends of course on what you want to do but to me this is the way to make us much more agile.”

Tommi then went onto explain that the process would be to install a single tag on every webpage where upon everything related to on the page Javascript or beacon tagging could be handled in the cloud. He also said that the tags could run alongside one another until we were ready to move to the final tag management system. So that would mean they wouldn’t lose data until they had fully tested everything.

“What about governance?Who will do the tagging?Who will audit our current tags and train our folks how to use this?” asked Bill. Tommi volunteered to be the data quality manager for ACME and be the TMS “super admin”. He would personally handle who was allowed to view, edit and publish tags.

He also said that auditing the tags would be an ongoing quality control process actually instigated by the tag management tool. To actually create tags in the TMS system he would need to duplicate old ones and thus could verify as he went. Documentation would also be irrelevant as you could log in and see the status of every tag the company had.

Training would only come if they paid a vendor though Tommi was quite confident he could train Jan and the others to get basic tags deployed.

Bill asked Tommi and Jan to talk to Harri the platforms manager in order to come up with recommendations for the business.

What tool?

Tommi had basically volunteered to run the project but was a little unsure about what to do now he'd been given the job. So he asked Jan what tool they should use. Should they go paid or should they use a free version?

"I think that's the wrong question to ask" replied Jan. "What we need to do is determine what we need as a business and then get the best fit for purpose. The first thing we need to do is list our requirements in a TMS. Then I reckon we should pilot the free version of Adobe (DTM) or Google Tag manager (GTM). Do they meet our needs?Then finally we have a look at a few paid vendors to see what they offer above and beyond the free versions. Yes it might take a bit longer but we've at least covered all our bases and we give Bill the most informed choice we can about where to put any investment we make."

Tommi understood the wisdom in this approach and armed with this plan spoke to Harri to determine the next steps. After outlining the discussions he'd had Tommi decided he'd pilot Adobe DTM[15] on one market and GTM on another before then reaching out to the paid vendors. Harri advised on the sites to use to do the tests because he knew which ones would give Tommi enough information and have minimal impact on the business if something went wrong.

Tommi had already gathered all the KPIs across the business as the analytics team had been involved with setting them all up and had them documented.

Technical audit

He then went about doing a technical audit that outlined what kind of tools he'd need to deploy.

1 It had to deal with Adobe Analytics tagging out of the box. The analytics tool wasn't going to change and he needed to be able to transfer all the rules and custom variables across to the TMS easily and quickly. If a new flash video went on the website Tommi needed to be able to tag it with

Adobe Analytics tags easily and quickly, but he also needed to be able to migrate existing rules across.

2 He needed to easily be able to add marketing pixels on various pages of the website. The KPIs James had developed for the SaaS they were selling required a lot of new tags be placed at key points across 30 markets. They were using Google Doubleclick floodlight tags[16] and had many conversion pages that needed to be quick and easy to update.

3 Jan had suggested testing tools be deployed for marketing teams looking to improve conversions. ACME were looking at 2 tools, Adobe Target which was a quite complex on page tag or another testing tool called Optimizely (www.optimizely.com). Adobe Target (www.adobe.com/marketing-cloud/testing-targeting.html) could be picked because it combined so well with other Adobe products, but Optimizely was a lot cheaper and easier to use. Both had different tagging set-ups and impacted the pages in different ways so he needed to test the systems out as well.

4 John's KPIs included a recommendation for using a tool called Click Tale or Tealeaf to measure the impact of form abandonment in hardware lead generation and ecommerce abandonment. Sam in customer support also had similar requirements. Click Tale (www.clicktale.com) and Tealeaf (www-01.ibm.com/software/info/tealeaf) measure how many people fill in forms and allows you to analyse what people click on better than Adobe Analytics could. Again they had on page tags to manage and the tools hadn't yet been decided on, but Tommi needed to take future requirements into consideration.

5 Remarketing[17] also needed to be taken into consideration. Jan had mentioned this in her work with marketing that they wanted to target people that had seen the ACME brand but not yet converted to sale. This could be done with a variety of display and search engine marketing scripts and Tommi needed a simple way to deploy all of them.

6 Page load speed was of paramount importance to the business. Harri had mentioned that if a tag management system was deployed that it should not break the website, stop it loading, or in any way reduce performance. Harri's main responsibility was to keep platforms working at 99.99 per cent uptime and he had no intention of allowing any tag management tool to reduce that efficiency.

7 Scalability was also extremely important. ACME had 30 different websites, one for each market they operated in. In addition there were some 10 campaign sites or specific product sites that had no localised versions but still needed to be tracked and managed. Tommi needed to be able to easily handle and deploy rules for all of them.
8 Finally governance had to be centrally controlled. Tommi was sure that in the future all sorts of folks would want to access the TMS. Agencies, programmers, marketers, IT folks would all have different needs and requirements and Tommi needed a centralised way to take care of all these different stakeholders. He also needed the whole thing to be fully documented so he needed to understand who had changed various things and what each tag did.

DTM deployment

After defining the technical requirements Tommi passed on the DTM code supplied by Adobe to Harri who deployed it on one of the lesser trafficked markets. Actually placing the tag itself was very easy, it took minutes to do and would be the same if they did it on 30 markets or one.

The first thing Tommi and Harri did was test the page load speed.

The scripts loaded in less than 200 milliseconds which once Tommi could replace the Adobe Analytics scripts and others that currently were on the website should be an improvement, though currently Harri noted that there were no rules firing through DTM.

User set-up was good, DTM allowed multiple users with different access rights be controlled by an admin user hierarchy.

Adobe Analytics could easily be handled by the tool. There were lots of ways to make tags work faster, easier and in a what you see is what you get environment, rather than manually coding. Adobe Target also had some specific rules that could easily be applied out of the box.

Then you had a custom code form that would allow you to add basically anything else (covering all tools that Tommi needed). This was OK for Tommi because he could code, but he noted that other users would not have his experience with coding and might easily make mistakes with DTM.

It took Tommi about a week to deploy all the Adobe codes to DTM and then remove the old codes on the site. The 200 millisecond load time

remained about the same for DTM but overall page load speed was radically improved as there was much less code on the page for a browser to load.

After running a number of stress tests on different browsers, devices and operating systems Tommi was satisfied with DTM performance levels.

GTM deployment

Similarly Tommi deployed GTM on another market.

The load time was superfast, at under 100 milliseconds, though again no rules were deployed.

User permissions were also fine in GTM, he had the ability to allow people to only view, view and edit or view edit, delete and publish across all of the different tag containers he would set-up.

GTM had out of the box rules set up for ClickTale, DoubleClick, comScore (www.comscore.com) and Google Analytics, as well as remarking tags for Adwords. He would need the rules for double-click, Adwords and ClickTale but not the others. He also would have to use custom scripts for all of his rules for Adobe. This was a major downside for Tommi.

Again everything else Tommi needed would have to be set-up in a custom manner. It took him 2 weeks on this particular market to get everything deployed as he needed via GTM.

Performance wise it had been less efficient that DTM because it took him a lot longer working with the Adobe Analytics scripts. It was almost as slow for the Adobe Analytics part as hard coding natively on the website which would be an issue with 40 sites to deploy and very little time with everything else Tommi had going on.

Speed wise it was around the same level of performance after a similar amount of scripts loaded via GTM though it took longer to load the Adobe Analytics scripts.

The vendors

Tommi then approached 3 paid Vendors to establish what they could do better than the free tools available to him.

In terms of governance the paid tools were much better. All of the ones tested had great tagging quality control, very good user management and

strong out of the box functionality for controls over specific tag rules and websites.

Quality controls were in built meaning that if you put the tags onto a test environment you could publish to a live production site immediately after testing with the push of a button. In the free versions this functionality also existed but nowhere near as sophisticated and for a large business like ACME the sophistication was desirable.

The other thing they offered was out of the box modules for different tools. Plug and play options had been developed for all the tools Tommi needed. This meant an even faster time to delivery.

Most impressively though the paid tools offered more than simply tag management. Tag management was core to what the vendors did, but there were also add-ons to their software allowing third party tracking of pixels, data segmentation (allowing ACME to own their own data) and triggering of marketing messaging.

If ACME owned their own data they would know who their users were rather than have to pay Google for the privilege. So for instance if ACME had gone to Google to buy display ads from 35–45 year old males in the software industry, ordinarily they would have to keep paying Google to re-market to those users every time they wanted to.

By having their own pixels placed on third-party display websites they could buy the segment once from Google or whomever they were buying from, record the segment themselves and then use the data to re-market to the 35-45 year olds at no extra cost later on.

This could end up being very valuable in the not too distant future.

Tommi asked the vendors to build him a business case for purchasing their products that he could then take forward to Bill and Jan. If the business case out performed the free versions then there would be a possibility that ACME could purchase a paid tag management system for future proofing.

5.12 Checklist and exercises

What follows is a checklist you may use for your own business case with regard to data.

1 Which of the following data sets applies for your business:

(a) Clickstream data?
(b) Experience data?
(c) Ecosystem data?
(d) Social data?
(e) Big data

2 For each data set list the tools you use to collect the data. Are there any tools missing?
3 If there are tools missing identify free or paid tools that can solve your data problem.
4 If you need to talk to a vendor send them a questionnaire (see section 5.9). If you're using a free version how many of the questions in section 5.9 can you answer yourself? For the questions you can't answer consult with a partner whom will in many cases offer free advice (in Google's case there is a certified partner portal[18]).
5 If you are tracking via beacons or tags select an appropriate tag management tool. It's not good enough to simply put tags on your website manually any longer.
6 Validate the data integrity across all devices, operating system and browsers.
(a) Is the data being recorded in exactly the way it should?
(b) Is the data being allocated to the correct profile or account?
(c) Do you have a way to upgrade the website without impacting data quality?
7 Identify an individual whom you can trust to be responsible for data quality and integrity.
8 Make sure there is a method to record all the different variables you're tracking. Either via documentation or if you're using a tag management system via identified tag rules with a clear taxonomy.
9 Do your technical audit. What tools do you need to add to fully track your website?
10 Implement and test the outcomes. Validate the data (see point 6 above) and setup alerts to monitor any fluctuations. If for instance uniques visiting your website on a Monday is 75 per cent lower this Monday than it was last Monday is there a tagging problem?
11 Train your staff about the importance of data integrity and if possible ensure quality control by piggy backing on existing quality control

methods. For instance if updating website content has a process can tagging be added to that process?

Notes

1 See http://en.wikipedia.org/wiki/Big_data.
2 See http://en.wikipedia.org/wiki/Net_Promoter.
3 Neil Mason, "Defining Web Analytics: Market Intelligence", 23 August 2005, www.clickz.com/clickz/column/1699348/defining-web-analytics-market-intelligence.
4 See http://en.wikipedia.org/wiki/Android_(operating_system).
5 See www.pamorama.net/2014/01/04/10-keyword-research-tools.
6 See www.quantcast.com/wp-content/uploads/2013/05/Q-News-One-Pager-FINAL2.pdf.
7 See http://en.wikipedia.org/wiki/Apache_Hadoop.
8 See www.internetlivestats.com/google-search-statistics.
9 See "The Big Data Revolution", *Harvard Business Review*, https://hbr.org/2012/10/big-data-the-management-revolution/ar.
10 See www.adobe.com/marketing-cloud/web-analytics/marketing-analytics-capabilities.html.
11 See www.ensighten.com/products/tag-management/manage.
12 See www.google.com/tagmanager.
13 See www.adobe.com/marketing-cloud/web-analytics.html
14 See www.ensighten.com/products/tag-management/manage/
15 See www.adobe.com/solutions/digital-marketing/dynamic-tag-management.html
16 See www.google.com/doubleclick.
17 See http://en.wikipedia.org/wiki/Behavioral_retargeting.
18 See www.google.com/analytics/partners.

6 Segmentation and personas

> When dealing with people, remember you are not dealing with creatures of logic, but with creatures of emotion, creatures bristling with prejudice, and motivated by pride and vanity.
>
> Dale Carnegie

6.1 An introduction to personas

Since Forrester published the best practices in 2005 I've seen many websites designed with persona-based models at their core. Most user-centred design (UCD) philosophy has been preaching the use of persona models for well over a decade.

I've long been a student of the excellent work done by Bryan Eisenberg[1] as well as Martijn Van Welie[2] around personas. Over the years I have found myself experimenting with personas in order to learn how to measure and improve the websites I've worked with.

I've shamelessly adopted both methodologies because they both work in different ways depending on the situation and they can be easily combined.

One method is based on Bryan's work around the Keirsey personality model.

Because the method is based on how people tend to behave rather than a one fits all primary persona it means you can segment your audience into behavioural groups based on the type of behaviour they demonstrate.

That's a little different to standard UCD persona design. Segmenting the audiences means you can measure which persona type is not being as well catered for on your website and take action accordingly.

The other method is a template based approach. Alone this can be used for heuristic assessments and was developed by Martijn Van Welie. He usually used four or five of these motivationally driven personas in his design work for websites.

I've had a lot of success with this method when using it for e-commerce or lead generation operations when testing a single user motivation or goal. If you've never used personas before it might be a good idea starting with the persona template method and moving onto Bryan's method when you have some wins under your belt.

For full website design I would combine the two techniques by using Martijn's method to "flesh out" the character of behavioural personas developed using Bryan's techniques.

6.2 The Van Welie method

This simple template method is used to design a persona based around the web visitors' motivations. The persona creation template (provided as an Excel file at www.blackbeak.com/cult2/files) is an adapted version of a persona template created by Martijn Van Welie. Really it's just a checklist that allows you to develop the persona.

If for instance you go online with the intention of buying something it's rare that you know exactly where to go to get the best deal. Every decision you make is based on your previous knowledge and experience. If you feel something is risky, you're less likely to do it.

The template helps you define what the risks might be and the questions that need answering in the mind of the visitor before they will take action. When you have defined the questions you can compare how well your website answers those questions.

It can help you learn a lot about rights, wrongs and how your competitors compare with you.

The advantage is you get a lot of insight around a specific motivation.

The disadvantage is that your website might have 20 different reasons for attracting visitors in which case there will be 20 different personas and you can't design a single site for all of them (This is where Bryan's method comes in handy). However for measuring a single motivational process like

lead generation or ecommerce it's very good. This is why I'd advise you use this kind of method to develop quick win, reach, engagement or conversion assessments.

Research stage

When designing your persona you should look at a number of different sources of data to define how you get to your role model. These are:

- **Demographic data** – age, gender, geography (data acquired from customer surveys, CRM data or other sources like Hitwise/Comscore).
- **Customer psychographics** – what the customer does in the pre-purchase phase found by looking at a number of different sources in addition to the demographic data, for instance:
  - web analytics keyword data
  - blog post phrases angled at user motivations
  - top search engine phrases (Google trends)
- **Market data** such as how the branding and market place effects the decisions of the persona.
- Who else sells your product/service (competitor analysis)?
- Which competing websites do your visitors engage with (Google Trends)?
- What other things are your customers looking at (Google Trends/Keyword tool)?

Develop the persona (a checklist)

Goals

Determine your personas key motivation or reason for visiting you website. Note that it's the personas goal not necessarily your business goal. This data about key motivations comes from the research stage.

Scenario

Describe the personas typical scenario and reasoning for visiting your site.

- What would be the typical way that the persona would know about your product or service?
- How would they find out about the product or service?
- What specific methods are involved in reaching you (REAN – Chapter 2)?
- How would the person do his/her research?

Tasks prior to taking action

Determine all the tasks the persona will need to accomplish before being in a position to take the action you want them to take.

- What does he/she need to find out?
- What is his/her first major concern about the product or service?
- What are the next four factors that need to be understood out in order to have a better idea of all the risks involved?

Considerations and questions

From the scenario and the task list determine what questions your persona may have about your product or service. It should be relatively easy at this point to come up with around 10 key questions that the persona would want answered before deciding to purchase.

Pain points

Will be different for each persona and very personal.

- What makes the persona cry out in frustration?
- If he/she is potentially defecting from one product or service to yours, what pains him about the current situation?
- If not what are the needs that really pain him/her the most?

Trigger terms

Based on the pain points and considerations

- Trigger terms are words people look for or use to try and solve their problems.

- Used in search engines, on blog posts, in emails.
- Determine five terms that will be used to search on Search engines that relate to the pain points and problem the persona has. It's good to do prior research around these keywords on Google trends/Google keyword tool.

Scoring

After creating your persona and defining your personas needs, his pain points and his trigger terms, you're now ready to score how you're doing. The point then is to do a heuristic analysis of your website and answer the questions as best you can. The needs are answered on a scale of 1–5.

The scoring method is really designed to help simplify things, which helps build your analytics culture. Scorecards can level the playing field in large organisations and help build understanding.

6.3 ACME persona using the Van Welie method

In order to build his business case (see Chapter 4) John developed a persona called Frank Butler to do the heuristic analysis.

Definition stage

Looking at his customer demographics from the CRM system, John determined that Frank Butler would be European, with the largest market being the UK (43 per cent). He'd be fairly well off, in the $60–80K per year income bracket, and most likely have a spouse, maybe children. He'd be around mid-thirties (80 per cent of ACME customers being in the 25–45 age bracket). He asked the support team what the customers were concerned about and was told that installation and technology were the biggest worries for most consumers.

John also knew that the market, the competitive landscape would influence Frank's research procedure. He'd know or easily be able to find out about all the major brands by simply Googling.

To make it a bit more interesting John decided that one of Frank's barriers would be his wife, who didn't want Frank spending a lot of money on a variety of equipment.

Building Frank's persona

In order to put a picture in the mind of anyone looking at his work, John got a stock picture of Frank Butler to use as his persona (Figure 6.1).



Figure 6.1 Image of Frank

Source: © Markus Spiske (www.flickr.com/photos/markusspiske/14648242613)

He then fleshed out his description based on the demographics data he looked at:

> Frank is a 43-year-old English entrepreneur working for himself. He wants a no hassle upgrade his in house entertainment system that could also potentially be used for presentation purposes. Frank has a variety of different needs from his in house entertainment system.

He wants a big screen (a projection system), better sound quality than his standard TV, the ability to stream movies and websites via the internet to the big screen and connect his MAC book direct to the screen.

His wife is putting up with Frank's ideas but doesn't want any difficulties with installing all of this stuff in their home. She doesn't see the point of spending a lot of money when a flat screen TV would do the same job.

John then went on to fill out his persona template:

Frank's goal

Purchase an in-house entertainment system.

Scenario

Frank has to upgrade his in home entertainment system. He has just moved apartment and needs a new service. His wife isn't really interested and won't like him spending a lot of money so he's using his freelance work as an excuse, saying he could run his presentations to clients at home instead of hiring office space. Frank knows all the major brands he is looking for but doesn't really know the best way to set it all up. He has a budget of £3000, which he is telling his wife is £1500 (the price of a good flat-screen TV).

Tasks prior to purchase

Frank needs to research what he'd need for a full entertainment system. He knows he needs an amp and projector but already has Apple TV, wifi and speakers to connect his PC to this internet system but he doesn't know how the entertainment system would all work together. Ideally he'd like one system or service that comes preconfigured and will search for such solutions. His plan is to:

- Check brand websites directly.
- Compare brand prices (Google).
- Compare full service offers (Google).
- Learn how to install the system.

Considerations and questions

Frank also had a number of considerations, many related to pricing but also installation and technology:

1 What is the cost per year?
2 Are there any options like hire purchase or leasing?
3 Does the cost vary based on technology? If so how? If not is the price fixed or could there be hidden surprises?
4 Do any free accessories come with the service that might help his wife come around to the idea?
5 Does the service work with HDMI sound and picture quality?
6 Can he connect his computer to the entertainment system?
7 Are the brands used for the various items well known and trusted?
8 Does the system come home fitted? Is it manual installation? If so is it easy?
9 Are the installation instructions clear on the website?
10 Are there warranties and guarantees? Any associated costs?

Pain points

Frank had three main pain points:

1 Frank is not that technical so is concerned about installation.
2 Wife is very cost-conscious and isn't sold on the idea.
3 Frank also doesn't know about the different jargon in the industry (HDMI?) he has just heard he needs HDMI to be future proof.

Trigger terms

Finally, based on his pain points and considerations, John hypothesised that Frank might be influenced by the following trigger terms:

- home entertainment system
- pricing/leasing/hire purchase
- amp, speakers, projectors, Apple TV
- installation, support

- HDMI, screen quality, sound quality.

Scoring

Most of what Frank wanted was very well covered by ACME's website. See the scoring sheet in the Excel spreadsheet provided at www.blackbeak.com/cult2/files.

However once John started scoring Frank's experience when he got to their website he identified 5 poor areas that needed addressing.

He knew that Frank's wife would block him if he wanted to pay £349 per month as that was way over his budget. He also found that it was very difficult to find the custom leasing that Frank would need (he already had an Apple TV and speakers so he just needed to add a projector and Amp to the 99 a month package that would cost him 159 per month.) When John then looked at the analytics he noticed a similar trend that almost no-one bought the premium package. They were buying standard packages or custom deals.

When John then tried to sign up for a custom deal he knew he'd stumbled onto a massive problem with their website.

The items added up the price as a purchase price but didn't have the leasing option till after you got to step 2. He checked the analytics. Custom options were by far the most popular choice because unless you didn't have any home entertainment system you would simply be adding hardware to a current collection.

This would probably mean that Frank would leave the website and look for something else. Result!

6.4 The persuasion architecture method

The persuasion architecture (PA) method of creating personas was developed by Bryan Eisenberg and his team when he worked at FutureNow Inc. (www.futurenowinc.com), and is based on the Myers/Briggs behavioural studies adapted by David Keirsey in the 1950s.

Where the prior template approach is a good starting point that will help you to start using personas scientifically, the PA method could be the foundation of your website copy and content strategy. It goes into much

more depth and closely follows Keirsey's observations about how people act.

Keirsey spent decades studying how people behave. He observed personality traits, such as habits of communication, patterns of action, and sets of characteristic attitudes, values, and talents. He also examined personal needs, the kinds of contributions that individuals make in the workplace, and the roles they play in society.

He discovered that in essence people have 4 personality types that dictate how they are likely to act. FutureNow adapted this approach to how people behave on the Internet and started developing personas based around behaviour. Because we're dealing with measuring behaviour using web analytics tools it's a natural progression to start measuring behavioural personas.

You start to deal with the reality of how people act and not try to determine who they are.

It's *not who comes to your website* that is important it is *how they behave* when they get there that is key to your success. I know personas work because I can measure how effective they are by segmenting them out with web analytics tools. Your visitors *behave in a more predictable way* based upon their temperament.

There were four primary temperaments defined by Keirsey:

- sensing/judging (SJ)
- sensing/perceiving (SP)
- intuitive/feeling (NF)
- intuitive/thinking (NT).

Keirsey argued that the four primary temperaments are basically hard wired into human beings. We all act like one of these temperaments primarily and depending on the situation will change to be more like one of the other temperaments when our attention is engaged on something else.

FutureNow pioneered using Keirsey's methodologies in order to design personas. The Keirsey temperaments were renamed by FutureNow to be more intuitive and relevant for today's audience:

- methodical (SJ)

- spontaneous (SP)
- humanistic (NF)
- competitive (NT).

The competitive persona

- **Attitude:** Business-like, power oriented.
- **Time:** Disciplined, strategic/fast paced.
- **Typical question:** What can your solution do for me?
- **Approach:** Provide options, probabilities and challenges.

Competitives need you to clearly define what the benefits of your product are. They want to know why they should choose you over their other options. They are looking to keep a competitive edge and have products/services that will make them feel and look better than others. There should be an introductory paragraph that explains who you are, what you do and what benefits you offer the visitor. You have guarantees in the active window listed as bullet points.

The humanistic persona

- **Attitude:** Personal, relationship oriented.
- **Time:** Undisciplined, slow paced.
- **Typical question:** Who has used your solution to solve my problem? Who are you and why do you care about me?
- **Approach:** Offer testimonials and incentives.

Humanistics need more information about who the founder of the company is and who the people in the organisation are. They want to see pictures of the people who work there. They want to see testimonials on how purchasing your products has worked out for other customers, as well as testimonials that indicate you offer incredible customer service.

The methodical persona

- **Attitude:** Businesslike, detail oriented.

- **Time:** Disciplined, methodically/slow paced.
- **Typical question:** How can your solution solve my problem?
- **Approach:** Provide hard evidence and superior service.

Methodicals want to make sure they know everything before they move forward. They do not mind reading through to the bottom of a page to find out what they need. They are concerned with how you operate and need to know that you're competent. You should provide them with all the details so they don't go elsewhere to find what they're looking for.

The spontaneous persona

- **Attitude:** Personal, activity oriented.
- **Time:** Undisciplined, fast paced.
- **Typical question:** Why is your solution best for my needs?
- **Approach:** Quickly address values and provide assurances.

Spontaneous personas want credible opinions rather than options. Offer immediate gratification where appropriate. Your homepage should speak to this personality type, begin to build rapport, raise confidence and propel them toward immediate gratification. They will not spend a lot of time reading content; rather, they are skimming for interesting and captivating information within the most obvious sections of each page.

My own temperament falls into the intuitive/thinking temperament. I am the kind of person who'll use the best possible way to get information. I'll use a search engine on an individual site, or examine web pages scanning for keywords and links before flicking to the next page if I don't find what I'm looking for. I spend most of my day online and haven't got much time on my hands to browse around.

On the other hand if I am browsing around a subject that I'm emotionally tied to like Football (specifically SAFC.com or ReadyToGo.net message boards) I can easily spend hours looking at everything and absorbing all the information. Behaviour that is closer to the Sensing/Judging temperament. I can also get very emotional and spontaneous (SP). This doesn't mean I'm some sort of Jekyll and Hyde, it just indicates that my normal behaviour

changes when I am dealing with something I'm more personally invested in. But the point is that I fall into one of the "types" regardless. Either I am an NT or in the football case I'm an SP or SJ.

This is why designing for the 4 kinds of temperaments is more effective from a design perspective than trying to figure out the primary persona visiting your site. Design for all 4 types and then measure which type you succeed with the most. This gives you a starting point to develop a roadmap for improvement.

I think it's a better description to describe me as competitive, or when I'm looking through SAFC.com and ReadyToGo.net as methodical or spontaneous. It's easier to understand.

Future now have translated Keirsey's work into modern day buying modes adding fast, slow, logic and emotion to the mix which simplistically shows emerging behaviour patterns.

Behavioural patterns

Competitive people (like me) take logical decisions quickly and strategically based on what they see. Methodical people are also logical but slower paced and more deliberate. Spontaneous people are quick, fast paced people that do things on the spur of the moment based on how they feel. Humanistic folks want to feel like they are doing the right thing and want evidence that other people have also enjoyed doing the same thing.

It's possible that in any given situation you can see yourself represented in the above behavioural patterns.

I know my primary type is a competitive personality because I have taken the tests and saw the same result – NT. Actually my exact personality type was what Keirsey calls "The Field Marshall – ENTJ". I can recommend two very good books if you want to learn more about temperaments: Keirsey's *Please Understand Me II*, and for lighter, much easier reading, Stephen Montgomery's *People Patterns*. If you want to take the test yourself it's on the Keirsey website at www.keirsey.com/sorter/register.aspx.

The point here is to show that while my primary personality is competitive, fast paced and logical, it doesn't mean that I will act that way all of the time. This means that knowing your audiences type won't tell you which mode they will be in when they buy from you.

If SAFC.com relied on my primary personality type of competitive, fast paced and logical they would only succeed in irritating me immensely. I am emotionally tied to them, I want to connect with club, I want to read in depth articles, watch video highlights and can easily spend hours with the subject of football and Sunderland. I act irrationally when normally I'm rational. It's the curse of being a Sunderland football fan.

Luckily for SAFC.com they have Sunderland fans in charge of writing and developing the website, so most of the time they get it right.

This is one reason why traditional UCD (user centric design) based on traditional research is not always relevant when building websites.

Doing the research and team selection for a PA workshop

To select the best team and gather as much information as you can in order to design good personas. The team should be from the business with diverse backgrounds that are responsible for the product or service. You should always include someone who is dealing with customers on a daily basis such as an account manager or a face to face sales person. The people in your team should all have a strong understanding of the benefits of the product or service to the customer.

Upon selecting your team you should collect all the data you can gather about your customers. This information can be gathered from the following sources:

- **Demographic data:** age, gender, geography (data acquired from customer surveys, CRM data or other sources like Hitwise/Comscore).
- **Customer psychographics:** what the customer does in the pre-purchase phase found by looking at a number of different sources in addition to the demographic data, for instance:
  - web analytics keyword data
  - blog post phrases angled at user motivations (BlogScope)
  - top search engine phrases (Google trends).
- **Market data** such as how the branding and market place effects the decisions of the persona:
  - Who else sells your product/service (competitor analysis)?

– Which competing websites do your visitors engage with (GoogleTrends)?
– What other things are your customers looking at (Google Trends/Keyword tool)?

The workshop facilitator should present this information to the team to kick off the workshop and then move into the brainstorming.

The brainstorm should focus on listing the benefits and unique selling points of the product or service. Ask the team:

- What is the best benefit of this product/service from the customer's point of view?
- How is this product/service completely unique?
- Why would a customer buy from us and not the competition?
- When will the customer be better off buying from someone else?

I've found that these four questions are very good to get people focused. The facilitator of the workshop should end up with a list of key benefits on a whiteboard.

Categorising the attributes

Once you have all the product/service attributes that everyone agrees on and is clear about you should determine whether each benefit appeals to logical or emotional decision making on the part of the customer and what decisions about the service or product are fast decisions or slow ones.

I personally ask everyone to write this on a Post-it note and suggest why they feel it is one or the other. Then stick the post it notes next to each attribute on the white board and discuss until the attribute is agreed upon as either one or the other.

Emotional connection happens when a feeling is caused usually without thinking. Branding, for instance, has an emotional effect.

Logical connection is simply a decision you make based on a logical thought process. For instance a home fitted product doesn't appeal to your deeper emotions and make you feel good inside but it might make a difference to someone who doesn't know how to fit it. Similarly pricing,

billing and service agreements are purely logical decisions, not emotional ones.

Next you need to decide what is fast and quick to explain and what takes more time. How hard is it to understand each attribute? If it's immediately obvious it goes into the Fast section. If it's going to take some careful reading it goes into the slow section. Again, in your team use the post it notes.

Table 6.1 Decision type (logical or emotional, fast or slow)

Logical	*Fast*
Pricing or billing options	Easy to make a decision
Clear options with little complexity	Very little input or action required
Simple benefits	All the information required is present
Slow	*Emotional*
More difficult to take a decision (more than 3 variables to take into account)	Brand is clearly a factor
	Association with a feeling or concept
More information required to take action	Benefits beyond simple options
Information is drawn out and might be better explained in person	Being part of something bigger

For instance – it doesn't take five minutes to say "It comes home fitted", it is clear as soon as you read the words. The branding is either known or it isn't, there is no middle ground.

It does take longer however to explain things like customer service levels. It also takes some explanation to cover complex services and may need documentation and in depth explanation to answer all the questions.

Mapping attributes back to the whiteboard

Now you can map the attributes back to a persona template.

- Competitive personalities are strategic/fast paced and look for logical attributes.
- Spontaneous personalities are fast paced but undisciplined and emotional.
- Methodical personalities are slow, deliberate people also focused on the logical attributes.
- Humanistic personalities are slow paced, deliberate and focus heavily on emotional attributes.

Depending on the case you'll have more attributes in two or perhaps three types of persona than others. This isn't to say that you only design two or three personas catering for the types people. The point is you now have a clear starting point to design your website based on the strengths of your service. You would also know what your weaknesses are in terms of how you're communicating.

By knowing how I'm likely to behave ACME could improve how their website satisfies customer needs. What follows is an example of how ACME have designed one of their services based on the PA method.

6.5 ACME persona-based strategy development

Mari had been asked by Jan to help build a business case around software sales that had come from education articles posted on the company blogs in Belgium.

She was particularly excited because the studies Jan had done uncovered a business case but it also enabled her to get internal backing to further improve conversion rates.

Table 6.2 Persona attributes mapping

Competitive	*Spontaneous*
Logical Fast	Emotional Fast
Methodical	*Humanistic*

Logical	Emotional
Slow	Slow

She'd read about the behavioural persona development online from a number of books and resources and decided with her team running B2B software sales to tackle the first persona based website design the company had undertaken.

The first stage was all about the preparation and planning of how the persona would be developed so the changes to the websites or marketing materials could happen later.

She gathered her team. She had Nathan who was an account executive from B2B Software sales, Jean who was a copywriter and blogger for the software team and of course Artur the legendary director of software whom had given Mari the go ahead for the development after seeing Jan's presentation (see Chapter 4).

Artur was legendary for a couple of reasons.

First, he had developed the first versions of the software the company now sold and worked his way up through the company spanning a 25-year career. He was a major shareholder, one of the biggest individual owners the company had. He owned more shares than the CEO.

Second, he didn't suffer fools. He'd been known to throw cups at walls, ranting and raging at people who wasted his time. Some people thought he was an asshole. Some people loved him. But there was no grey area with Artur. You either liked him or you didn't.

Artur respected Mari for being what he called a "doer". She'd been assigned to his team to help out analytics and marketing for their department and Artur was impressed that the analytics team had already found some opportunities. He was intrigued when Mari had approached him with the idea of doing a persona based strategy for social media and education of prospects.

Research

Mari had already done the research before she went to the meeting of her team. She knew the average buyer profile was typically a CIO, CTO,

software director, platforms purchasing manager or CEO (in smaller companies).

She'd also done her research into the people in the CRM system across the top companies and noticed that the buyers were different to the influencers. Typically below the management you would have senior software specialists, analysts and coders. She had a hunch these were the people who went out and learned how to use the software in their respective companies and told the management what they needed.

She looked into the web analytics of the website and noticed that people doing internal searches on their software sites were searching for technical documentation and specific motivations that weren't necessarily aligned with what the buyers profile suggested. This was something she needed to confirm with sales (Nathan):

- Average age: 30–40.
- Male.
- 40 per cent US, 60 per cent European (UK, Germany, France and Italy were the biggest markets).

She'd also looked at the competition. There were a lot of related searches that competition were higher ranked for in Google searches. There was an opportunity here to improve search rankings and thus increase traffic based on searches made on competing (but related) products.

Kicking off the workshop

Mari introduced the group to the purpose of the workshop. She explained what her colleague Jan had uncovered with the social media campaign successes in Belgium and said while she wanted to re-produce the success of that campaign she also felt the conversion rates of the website was very low. She believed that by building in better persona based methodologies to their web and social media presences they could improve the business even further.

She then opened up the philosophy behind Persuasion architecture, explaining that it was all about uncovering the common behavioural traits that exist when people browse websites of certain kinds.

She opened up the market research she'd done that showed the typical example of the buyers and their influencers. Nathan immediately agreed that the people doing the buying weren't necessarily the same people who downloaded white papers and read articles online about the software they were selling.

So then Mari started the asking the questions of the team:

- What is the best benefit of their software from the customer's point of view?
- How is this software completely unique?
- Why would a customer buy from us and not the competition?
- When will the customer be better off buying from someone else?

The open discussion between the team brought out a lot of consensus. Nathan in particular knew all about the benefits and features of the software they sold, but Artur threw in a couple of gems about what made their security software unique.

Artur:	Software is used by 25,000 organisations and 55 million users worldwide.
Nathan:	Centralised user management system.
Nathan:	Integrates with 400 other partner solutions.
Artur:	Risk algorithms. The software had its own risk prediction capabilities about users that might be trying to "break the code".
Jean:	The ability to mix and match hardware authentication with software authentication was unusual in the security field.
Nathan:	World class uptime. The system had near 100 per cent uptime.
Nathan:	Pricing has many options – one reason why it was a difficult sell. Software alone could be less than $1000 per year but configurations and options meant the price could go as high as $150,000. This was one reason why people went to competing products.

Once Mari had gathered this she drew up the whiteboard into four quadrants and discussed with the team where they should place the attributes. In each

case she explained why she thought it fit into one of the quadrants and got feedback from the team until they reached a consensus. The breakdown was as shown in Table 6.3.

Mari then counted the attributes and mapped them back to the persona template:

- logical fast = 2
- logical slow = 4
- emotional slow = 0
- emotional fast = 2.

Table 6.3 ACME workshop outcome – decision types

Logical	*Fast*
Integrates with 400 other partner solutions (LF)	Software is used by 25,000 organisations and 55 million users worldwide (EF)
Centralised user management system (LS)	ACME well known in the field as the number 1 in the market (EF)
World class uptime: the system had near 100% uptime (LF)	Integrates with 400 other partner solutions (LF)
Pricing varies from $1K to $150K → requires a lot of explanation as to why the differences exist (LS)	World class uptime: the system had near 100% uptime (LF)
Risk algorithms: “breaking the code” (LS)	
The ability to mix and match hardware authentication with software authentication was unusual in the security field (LS)	
Slow	*Emotional*
Centralised user management system (LS)	Software is used by 25,000 organisations and 55 million users worldwide (EF)

Pricing varies from $1K to $150K → requires a lot of explanation as to why the differences exist (LS) Risk algorithms: “breaking the code” (LS) The ability to mix and match hardware authentication with software authentication was unusual in the security field (LS)	ACME well known in the field as the number 1 in the market (EF)

Table 6.4 ACME persona attributes mapping

Competitive	*Spontaneous*
Logical = 6 Fast = 4	Emotional = 2 Fast = 2
Methodical	*Humanistic*
Logical = 6 Slow = 4	Emotional = 2 Slow = 0

From this Mari and her team concluded that the two main types of people they needed to cater for were competitive or methodical personas. Both had logical decision making at their core but she also recognised that a fast paced spontaneous approach might also work for the competitive behaviour types as well.

In the final part of the workshop before Mari went onto segmentation they had an open discussion about pain points and questions each persona might have (see sections 6.2 and 6.3).

Mari fleshed out her personas in template form (see Hans, Steve and Bruce in the associated Excel document at www.blackbeak.com/cult2/files). This allowed her to understand the issues each persona might have before summarising them as such.

The competitive persona

- **Name/demographics:** Hans Schmidt, German, 35, platforms purchasing manager.
- **Attitude:** Businesslike, power oriented.
- **Time:** Disciplined, strategic/fast paced.
- **Typical question:** What can your solution do for me?
- **Approach:** Provide options, probabilities and challenges.

Hans would want to know whether the solutions worked with what his business already used (integration), understand how easy the user management was for his company, pricing options and what made the software special. If pricing wasn't going to be clear then Hans wanted to know what he had to do immediately to get a quote for his own situation. He wasn't interested in talking to sales people, he wanted to be able to serve himself the same day. He'd be interested probably because of ACME's reputation in the field, he will have definitely heard of the company.

Mari and Jean discussed developing an introductory paragraph that explains who ACME are, what they do and how visitors might justify expenses to superiors.

They also suggested developing a pricing calculator that Hans could use instead of contacting sales. They also thought of developing a table that compared their product to competing systems. Finally, something they hadn't considered previously that Mari wanted to test was localising the website in German. Hans might understand English well, but everyone prefers their mother tongue.

At this stage because very few of the attributes matched a humanistic persona Mari elected not to focus on it with her testing and segmentation strategies. It would be enough to see if the results worked primarily for the two main personas (competitive and methodical) and then they could build upon those.

The methodical persona

- **Name/demographics:** SteveWhitfield, American, 38, senior software engineer.
- **Attitude:** Business-like, detail oriented.

- **Time:** Disciplined, methodically/slow paced.
- **Typical question:** How can your solution solve my problem?
- **Approach:** Provide hard evidence and superior service.

As a senior developer Steve is responsible for writing software similar to the type produced by ACME. He worked for one of ACME's competitors in the past. Now as an influencer in a key role helping with purchasing decisions, Steve needs to understand the intimate details of how the software works. What is the programming language? Do ACME provide samples? Ideas? Methodologies? White papers? Steve wants to completely understand what he's recommending and for what purpose. He doesn't mind doing the research or being in touch with Sales/Support because he knows all the questions to ask that will prove that ACME know what they're talking about or not. He's not responsible for buying so his name being in ACME's database is not a concern to him. It's a benefit as he might get information he wouldn't otherwise get by giving his email address. He's an experienced professional so no sales gimmicks will work. The guy just wants to know how ACME operate and need to know that they're competent. Steve also knows what the budget is, so understanding pricing is useful to him.

The team came to the conclusion that they should provide them with all the details so they can, plus a way to get more. If Steve would be happy to give his email address, white papers, software demo's and sales support should all be provided to him.

The spontaneous persona

- **Name/demographics:** Bruce Cooper, 43, CEO.
- **Attitude:** Personal, Activity oriented.
- **Time:** Undisciplined, fast paced.
- **Typical question:** Why is your solution best for my needs?
- **Approach:** Quickly address values and provide assurances.

Bruce doesn't really understand or want to understand how the software works, but he knows he needs a solution and is looking around himself at a few options his team have provided to him. He'd need to know if ACME are credible, who else uses the solutions, who the team running ACME are – are

they approachable? How quickly can he find what he needs to put ACME in the "OK we'll try this" box? Is there something free ACME can offer Bruce and his team that starts the conversation?

Once Mari had gotten to this point she then needed to figure out how they could target these kinds of personas one their website. This would require careful segmentation. She ended the workshop. Artur and the others were happy because they had defined 2 major things that they could work on to improve the website a calculator and a free software trial. However Mari knew that the segmentation would provide a lot of the answers they needed going forward.

6.6 ACME persona segmentation

Once you have mapped the personas likely behavioural patterns you can now begin to measure them. This is again where scoring comes into the equation. First, though, you should plan the behaviours of the personas into the REAN model.

Scenario planning based on the REAN model (Chapter 2) should allow you to map all the points of acquisition, the points of engagement and points of conversion. This framework is a good planning procedure for your persona measurement. Once the scenarios are mapped you can then define behavioural segments based on the actions the visitors in each typographic scenario take.

For instance looking at the REAN map dimensions, which of its component parts are more likely to be used by competitive people? Which by more methodical? Humanistic? Spontaneous?

- **Reach:** By identifying trigger words used by different personas to find your website you can segment your audience by keyword groups.
- **Engage:** By identifying pages of content, images, links, linear processes or offline touch points in your engagement model you can segment by visitor or session behaviour.
- **Activate:** By identifying conversion points you can segment by actions taken.

- **Nurture:** By identifying how people repeat their actions (or don't) you can segment based on likelihood to re-convert.

Identifying segments that represent a persona type is very useful. If you can identify by way of a persona segment which part of your audience is taking the actions you want them to take and which ones are the most likely to re-convert you can design your online sales pitch more scientifically.

Competitive personas REAN segmentation

Reach

Mari knew that reaching their competitive persona "Hans" was done primarily through search engines. Usually Hans would be searching for the direct solution to the problem, or he'd be looking to see if one of the tools he was using in his business worked with the ACME solutions. Mari went to Google Webmaster tools and listed all the potential keywords he might use. The other ways Hans might find ACME was via partner websites, LinkedIn, Facebook and Twitter which ACME posted all their blog posts, white papers and articles to.

Mari developed her segment starting with traffic sources that came from direct entry, partner websites, LinkedIn, Twitter, Facebook and the three major search engines.

Engage

If Hans arrived at ACME he'd either land directly to the home page if he knew the brand or typically a blog post. Depending on how well the search optimisation on the website was, he may also drop directly to a product page.

Mari developed added landing on the home page, a blog post or dropping directly to a product/solutions page to her segment.

Activate

The ideal conversion for Hans would be contacting sales, using a sales calculator that hadn't yet been developed or starting a trial that might help

him make a decision. Mari knew that segmenting based on conversions was likely to limit the sample size of her persona so what she did was check out the most likely conversion scenario for the segment she'd already started developing. It was clear that the type of persona she was looking at only converted after being at the website three or four times.

Mari segmented using three visits to the site as a minimum to drop into her segment.

At this point she actually saved her segment as "Hans: Pre-Conversion Competitive".

This segment equated to about 6 per cent of all the traffic ACME got, so was a good starting point. One of Mari's goals would be to aim at increasing this segments audience size to double even treble that percentage. Very hard to achieve, but if she did manage this then she knew her strategy to increase conversions would work.

Nurture

Once Hans had converted in some way she would then devise strategies to persuade him to buy either via email, phone call or whatever. However, she wanted to see what worked in various stages of the journey before and after the sale.

She copied the previous segment she'd created and added conversions to any goal as her final criteria in her segment. She saved her new segment as "Hans: Post-Conversion Competitive".

This segment equated to about 0.1 per cent of all the traffic ACME got. She knew this was because the segment criteria was quite tough but the workshop they'd already done had brought another 2–3 test ideas to increase this conversion rate and doubtless there would be many more ideas as a result of upcoming testing.

Methodical persona REAN segmentation

Reach

Similar to Hans, Steve primarily did his research through search engines. Like Han's he would be searching for the direct solution to the problem, or

he'd be looking to see if one of the tools he was using in his business worked with the ACME solutions. However he'd also be searching on more complex programmatic terms and around specific solutions to problems.

Similar to Hans, Steve might also find ACME was via partner websites, LinkedIn, Facebook and Twitter which ACME posted all their blog posts, white papers and articles to.

Mari re-used her segment starting with traffic sources that came from direct entry, partner websites, technical forum websites, LinkedIn, Twitter, Facebook and the three major search engines.

Engage

If Steve arrived at ACME he'd either land directly to the home page if he knew the brand a blog post but more likely he'd drop to a technical article or support forum discussing programmatic issues rather than sales issues.

Mari developed added landing on the home page, a blog post or dropping directly to a forums page to her segment.

Another thing Mari did was add a page view filter. Typically Steve would be looking at more than 3 pages after he landed. If he didn't view three pages, he was unlikely to be very interested. However if he did view more than three pages he was much more likely to convert.

Activate

The ideal conversion for Steve would be contacting sales, downloading a white paper, starting a software trial or using the calculator. However, she figured that Steve was more likely to convert on the first visit so she didn't filter on visits or on actual conversions from the same reason as previously.

At this point she actually saved her segment as "Steve: Pre-Conversion Methodical".

This segment equated to about 7.5 per cent of all the traffic ACME got so again a good starting point. She knew there would be some crossover between Steve and Hans but it didn't matter because she knew that both personas had logic as their core thinking. One was merely faster paced and had slightly different needs than the other. Hence the difference in segments.

Nurture

Post conversion Steve had very similar needs to Hans.

She copied the previous segment she'd created and added conversions to any goal as her final criteria in her segment. She saved it as "Steve: Post-Conversion Methodical".

This segment equated to about 0.5 per cent of all the traffic ACME got. Again this was because the segment criteria was quite tough but she noted that this segment converted 5 times better than the competitives. The competitive segment needed a faster/softer solution at the moment because they were the buyers and they didn't want to be pitched. But the methodical people didn't mind.

Spontaneous persona REAN segmentation

Reach

Bruce would have been told about ACME if he didn't already know them. He might also use a brand keyword on a search engine.

Mari considered the social channels (Facebook, LinkedIn and Twitter) were less likely to be used by Bruce. She segmented based on Direct traffic or search engines only. There were also some new paid campaigns they were going to try aiming at Bruce's type on LinkedIn so she added paid LinkedIn media traffic as part of her segment.

Engage

Once Bruce found the site he's most likely going to enter via the home page. If he's searching he's using a brand name so it's likely that he'd click the top link. ACME were top of the search engines for their own brand name so Mari segmented on landing at the home page or (if it came from LinkedIn) directly to the landing page around a software trial.

Activate

Mari knew spontaneous folks were less likely to buy the kind of software ACME were selling. It was a big purchase so unlikely to be something you'd

buy quickly. She did however want the user to see the software free trial landing page, understand that ACME were credible, had big customers and were run by professionals.

In order to understand all that you had to read the software trial landing page. That meant scrolling to the bottom of the landing page. Or it meant clicking on the about us link of the company as well as reading the top of the landing page. She added either scrolling down or reading clicking the about us page as part of her segment for activations. This segment equated to about 50 per cent of all the traffic ACME got. She knew it was likely that it was because generally people were less engaged with the website and that the criteria was wider than the others that this would be a larger segment. She saved it as "Bruce: Pre-Conversion Spontaneous".

Nurture

Post conversion Bruce had very similar needs to Hans (the buyer).

She copied the activate segment she'd created and added conversions to any goal as her final criteria in her segment. She saved it as "Bruce: Post-Conversion Spontaneous".

This segment equated to about 0.5 per cent of all the traffic ACME got. It was comparable to the methodical segments conversion rate and Mari believed that a lot of crossover would exist.

Humanistic persona REAN segmentation

Reach

So far not much with ACME's solution signalled any reason for the humanistic persona to be particularly catered to. Branding might be a reason if the visitor has an emotional tie to the brand, but in the case of software (a commodity) Mari deemed it unlikely to influence the buyer. However, you may find that this persona has come from forums discussing software and heard from other people that your service is good, something which you could use as a filter for a segment. However Mari had already filtered for this kind of user in the methodical segment.

Engage

Again this scenario doesn't have enough unique selling points to appeal to an emotional type of person. It's therefore worth testing by adding humanistic traits to the website, like great support, online live chat (where you can connect with a person in real time), testimonials, case studies, blogs and anything that the visitor can interact with. ACME could help make the difference between selling their service as a solution and offering friendly, helpful advice and support from real people. By offering this kind of service ACME show that they care about their customers (live chat). That they care about customer opinion (blogs), that the user is not the guinea pig for their service (others have tried and love it). All these little triggers will help the humanistic persona to feel like they are in the right place.

Activate

Humanistic people buy when they have enough confidence in ACME as service provider. In ACME's case adding a lot of humanistic traits to the website would cost a lot and require a business case developed to see if things like LiveChat and user communities would work.

Nurture

Re-conversion scenarios that could be created for these types include emailed blog response (a perfect way to get the visitor to re-connect and build up a community feel). Following brand evangelists on Twitter, LinkedIn and Facebook could also be something that ACME could go about rewarding (and measuring inbound clicks from).

Persona segments

Once the segments were in place Mari could now start her testing program to see if she could finally move to improving conversion rates.

6.7 Checklist and exercises

1 Define all the data sources you can get to help develop your personas. Include:

A Demographic (age, gender, geography, work titles, CRM data, external data like Hitwise or comScore).

B Psychographics (see section 6.2)

C Market data.

2 Develop a persona template (the Van Welie method) as shown in section 6.2. List the following:
   - Goals
   - Scenario
   - Tasks prior to taking action
   - Considerations and questions
   - Pain points
   - Trigger terms.

3 Score your website based on the scoring model provided (see section 6.3).

4 Read and understand section 6.4.

5 Do your background research (as with point 1 above).

6 Introduce the workshop concept to your team. Ask the following questions about your service to get the conversation started:
   (a) What is the best benefit of the product/service from the customers' point of view?
   (b) How is this service/product completely unique?
   (c) Why would a customer buy from us and not the competition?
   (d) When will the customer be better off buying from someone else?

7 List benefits and attributes of the product and service and categorise them as logical decisions, emotional decisions and whether the attributes are a fast or a slow decision.

8 Determine whether the attributes are logical/fast (LF), logical/slow (LS), emotional/fast (EF), emotional/slow (ES).

9 Conclude which type of personas you need to create.

10 Flesh out your persona characters by matching either your Van Welie method with the persona type and/or using the PA method. See also the template of examples at www.blackbeak.com/cult2/files.

11 Map your segments to your REAN points and integrate your persona profiles (section 6.6).

12 You're now at an advanced level of analytics segmentation that few are using well. Practice and tweak your segments until you're ready to start testing. Only by testing the segments in live environments can you really start saying whether your segments are correct.

Notes

1 See "Persona's the Magic Behind the Mirror" at www.bryaneisenberg.com/personas-magic-behind-mirror.
2 See www.welie.com.

7 Testing personas to improve conversion

> Most people use statistics the way a drunk uses a lamp post, more for support than enlightenment.
>
> Mark Twain

7.1 Define your testing approach

In Chapter 1 you saw a situation described that affects most businesses trying to do analytics today. The number one problem faced by most businesses today is not the lack of information, it's data abundance! The abundance of data and the way most analytics systems measure things mean generally we're looking at the wrong information when we look at out of the box metrics.

That's why it's so important to get persona or profiled segments into the analytics tools so you may start to score your audience in order to make more informed decisions with your digital marketing.

This was true in 2008 when I wrote the first edition, and it's even more of a problem in 2015 as I write this second edition. I don't see it getting any easier in the coming decade, I see it becoming a bigger problem.

Too much information and not enough action! The problem is that it can be overwhelming for those who don't have an analytics background how to decipher the data.

Many vendors are earning their living by supplying tracking tools that deliver the thousands of reports. The trick is to know what to look at which

is why in Chapter 2 we already discussed REAN and in Chapter 3 we already discussed KPIs. These should be considered starting points.

Take a KPI like bounce rate. The following passage is lifted directly from a guy I highly rate called Thomas Baekdal:

> The bounce-rate metric is another example of a single-metric counter gone wrong. The concept of the bounce-rate is that it counts the number of times people leave after only viewing a single page, compared to the number of times they stick around for more.
>
> In theory, this sounds like an incredibly useful metric to have. Surely, getting people to stick around longer is better. Right?
>
> Well, it depends. Let me illustrate this with five different user behaviours.

- The first person came to your site, and left without doing anything. This is obviously not the kind of user behaviour you want, but it's also not that unusual. He is still counted as a view (even though he shouldn't be), and has a bounce rate of 100 per cent.
- The second person came to your site, read the article, and then he left. This is obviously a lot better. The difference between a visitor who is doing nothing and a visitor that is taking in the content is huge. For a site like this one (where most of my articles are 15–30 pages long), whether someone is a reader or a non-reader makes all the difference in the world. But, in the traditional analytics sense, he is counted the same as before. One view and 100 per cent bounce rate.
- The third person came to your site, and immediately clicked on a link to another page, and then left. This person is also rather useless, but is counted as if he is not. He is counted as generating two views and has a 0 per cent bounce rate, because of that click.
- The fourth person came to your site, read your article (or looked at your product), then went on to another article, but left without reading that. Again, we have a huge difference here between readers and non-readers, but he didn't actually do anything useful afterwards, and he is counted the same as the one before: 2 views/0 per cent bounce rate.
- The fifth person came to your site, left almost immediately, but then came back again two days later, where he read the article. This person is hugely

valuable, because he is illustrating both interest and intent, but he is measured as the least useful of all. He generated 2 views with 100 per cent bounce rate on each.

You see the problem?[1]

On its own, bounce rate can be very misleading. However, if you have developed a persona based segmentation or profile as Thomas suggests you can use bounce rate in a much more effective and impactful manner.

If for instance people who come back twice or three times were scored higher (and added to different segments) despite bouncing on their first visit like "Hans" was scored in the previous chapter then you can start defining the tests you need to create. It does depend on your business model.

Testing persona gaps

Persona gaps are the types of things you find missing from your website or user experience that have been scored from your persona template. By defining your most valuable type of persona (i.e. most profitable) and understanding how they behave you can identify what kinds of things are missing from your user interfaces. A/B testing these things to see if you can improve the experience would help significantly.

1 First, look at your analytics persona segments and find out which type has driven the most value. You should correlate information from profit margins to your most influential segments.

For instance if you understand the type of business that hits your website (a vertical like financial companies) drives the most profit then testing for them clearly makes sense financially.

It works equally as well for consumer sites. If your customers fall into a certain type of persona tend to buy 5 or 6 products what can you test there?

2 Define the tests for the segment. What are they looking at? What are they missing?Where do they abandon from (see point 3 in testing sequences below) Can you test showing them something and see how the persona

responds? Can you up-sell, cross sell, do email marketing, or re-marketing, create funnel drop-out tests, etc.?

Testing sequences or calls to action

You could also take an approach that says "let's develop internal benchmarks which we try to improve" which is not a bad approach for specific things that can be A/B tested. Areas of "low hanging fruit" which you deem quite obvious test points. For instance one customer we worked with had a link called "documents". People simply weren't using the link in the menu so we surmised it wasn't clear that actually they were free. So we tested renaming it "Free documents" and the click through rates and downloads increased.

However, there are certain things you can look at (you don't need to do all of these at once, but you could try each method and see what sticks);

3 Where do most of your visitors leave your site from? Is your home page working? Are your calls to action effective? Are your menu items clearly viewable?

In their book *Call to Action*, Jeff and Bryan Eisenberg write:

> If you want to optimise and increase your online sales you have to begin at the beginning. Your home page is literally 50 per cent of the battle. Virtually every visitor will come back to your home page to re-orientate themselves.[2]

Another massive waste is shopping cart abandonment. If your visitor wants to buy something then that's great but if the process of buying is painful they're a click of a button away to buying from your competition.

4 Optimising trigger terms: use a spreadsheet and split into 6 columns.
   A For each of your major landing pages list the top 20 search terms or phrases (both paid and organic) that people have used to visit your site. These are their actual trigger terms, it's what they're looking for so this should match your persona's trigger terms as well.

B Assign a score between 1 and 5 to indicate how relevant the term is to the landing page being clicked upon. See again the Van Welie template[3] we used earlier for scoring trigger terms.
C List the bounce rate for each page.
D List the top CTR for the next three best pages. So if they click from the landing page to the home page, blog and product pages what are the CTRs?
E Apply your persona segment to the same trigger terms as your general traffic. Are the percentages better or worse?

Once you have determined where your persona segments are leaving the process (either via higher bounce rates or poor CTRs as compared to your general traffic benchmark) you can then define your testing ideas.

5 Use the worst scores across your persona segment to define ideas to test. Come up with a hypothesis and run your test.

Tools to test with

Once you've found the areas of interest that you test there are a number of tools I'd recommend looking at to actually do the testing. There are many tools but at the time of writing a good comparison is provided by Conversion Rate Experts from the UK.[4] The ones I can vouch for having personally used them with a simple list of pros and cons are below.

Google Content Experiments

Website: https://support.google.com/analytics/answer/1745152?hl=en
Pros: It's free and integrates with Google Analytics.
Cons: Unless you have a tag management system in place you have a lot of coding to do as the code is generated per test.

Optimizely

Website: www.optimizely.com

Pros: Free for the first 50,000 monthly unique visitors tested. One time code install. Simple to use. Tests can be running in 5 minutes.
Cons: Limited functionality in the free version compared to their enterprise version which they price on volume.

VWO

Website: www.vwo.com
Pros: One-time code install, simple to use. Relatively low pricing for a single company.
Cons: As compared with optimizely at the time of writing it's expensive for agencies to set up. Pricing depends on volumes.

Monetate

Website: www.monetate.com
Pros: Enterprise level, one of the best in class, combines multiple approaches to testing.
Cons: Price. Enterprise pricing starting at five figures.

Adobe Target

Website: www.adobe.com/marketing-cloud/testing-targeting.html
Pros: If you're using the Adobe Marketing cloud this has everything you'll need.
Cons: Without a tag management system again it's difficult to tag. Four-figure monthly fee means using it is restricted to those who can afford the Adobe licences.

7.2 ACME's Frank Butler persona

In section 4.9 we saw that John had developed a persona called Frank Butler (see also section 6.3).

Just to refresh, Frank is a 43-year-old English entrepreneur working for himself. He wants a no hassle upgrade his in house entertainment system

that could also potentially be used for presentation purposes. Frank has a variety of different needs from his in house entertainment system.

He wants a big screen (a projection system), better sound quality than his standard TV, the ability to stream movies and websites via the internet to the big screen and connect his MAC book direct to the screen.

His wife is putting up with Frank's ideas but doesn't want any difficulties with installing all of this stuff in their home. She doesn't see the point of spending a lot of money when a flat-screen TV would (in her mind at least) do the same job.

John needed to take a walk in his customers shoes when he visited ACME's website, so he literally took the tasks they had already defined into mind as he did that walk.

What is Frank's scenario?

Frank has to upgrade his in home entertainment system. He has just moved apartment and needs a new service. His wife isn't really interested and won't like him spending a lot of money so he's using his freelance work as an excuse, saying he could run his presentations to clients at home instead of hiring office space.

Frank knows all the major brands he is looking for but doesn't really know the best way to set it all up. He has a budget of £3000, which he is telling his wife is £1500.

What will Frank do? How will he find ACME?

Frank needs to research what he'd need for a full entertainment system. He knows he needs an amp and projector but already has Apple TV, wifi and speakers to connect his PC to this internet system.

What he doesn't know is how the entertainment system would all work together.

His plan is to:

- Check brand websites directly (including ACME).
- Compare brand prices (Google).
- Compare full service offers (Google).

- Learn how to install the system.

What are Frank's questions?

- What is the cost per year?
- Are there any options like hire purchase or leasing
- Does the cost vary based on technology? If so how? If not is the price fixed or could there be hidden surprises?
- Do any free accessories come with the service that might help his wife come around to the idea?
- Does the service work with HDMI sound and picture quality?
- Can he connect his computer to the entertainment system?
- Are the brands used for the various items well known and trusted?
- Does the system come home fitted? Is it manual installation? If so is it easy?
- Are the installation instructions clear on the website
- Are there warranties and guarantees? Any associated costs?

What might stop Frank from buying? What are his pain points?

- Frank is not that technical so is concerned about installation.
- Wife is very cost conscious and isn't bought on the idea.
- Frank also doesn't know about the different jargon in the industry he has just heard he needs HDMI to be future proof (HDMI?).
- Top cost per year is £3000.

What kind of Trigger terms is Frank Using?

Finally, Frank had a number of key terms based on his motivations. He was looking for words that would tell him he was in the right place and guide him to keep on clicking.

These terms were:

- home entertainment system
- pricing/leasing/hire purchase
- amp, speakers, projectors, Apple TV

- installation, support
- HDMI, screen quality, sound quality.

Once John knew all these points of reference he could now see what persona gaps existed by scoring the persona based on the terms defined.

7.3 ACME scoring Frank Butler

For the first 10 questions John used the following criteria to score how Frank proceeded through the ACME website.

5 = Frank found answer within 3 clicks of the mouse and under one minute.
4 = more than 3 clicks of the mouse but under 2 minutes.
3 = more than 3 clicks of the mouse between 2 and 5 minutes.
2 = more than 3 clicks of the mouse between 5–7 minutes.
1 = more than 3 clicks of the mouse between 7–10 minutes.
0 = Anything more than 10 minutes.

The time spent were based on persona segment averages they had already bench-marked via their analytics tools as were the clicks of the mouse (measured in page views).

John figured that Frank would know ACME's brand and would have checked out the competition by the time he reached ACME's landing page. Because Frank was a methodical type of guy he'd need to find out everything was good before he did anything. John now went through every question Frank would have and scored based on the criteria above.

The first place Frank would orientate himself was the home page. His first question:

1 What is the cost per year?

On the home page of ACME's website was a call to action:*Buy or Lease a Full Home Entertainment system with ACME from as little as €99 per month, click here to find out more*.

John scored this a 5 for the first concern Frank had. The cost was clearly listed as an example, within one click of the mouse he could find out more

and he found the link in seconds, right from the home page. An excellent start!

2 Are there any options like hire purchase or leasing?

This second question also scored 5. The home page on ACME showed clearly that leasing was possible.

3 Does the cost vary based on technology? If so how? If not is the price fixed or could there be hidden surprises?

Again John scored 5. Upon clicking through from the home page call to action you could see that the cost varied based on the plan purchased. So far so good, John surmised.

4 Do any free accessories come with the service that might help his wife come around to the idea?

This was not answered anywhere on ACME's site. John scored a "0". This could be the first "persona gap" test that John needed, but he wasn't convinced this would be a deal breaker for Frank so he kept working through the questions.

5 Does the service work with HDMI sound and picture quality?

Again John scored a 5. The technology was clearly listed in the platinum plan 1 click from the home page (and also listed on all other plans). HDMI was standard to all products.

6 Can he connect his computer to the entertainment system?

John scored "0". This was simply not answered anywhere on ACME's site and might contribute to a frustrated visitor experience. John surmised they should add a test where they might show examples of "typical home set-ups" with perhaps a printable schema or download of how to do various common set-ups.

7 Are the brands used for the various items well known and trusted?

Again John scored a 5 for this. The various brand logos were on the website and mentioned in the product descriptions.

8 Does the system come home fitted? Is it manual installation? If so is it easy?

5 again, John figured. All the packages come with home fitting and this was clear within two mouse clicks. From the home page.

9 Are the installation instructions clear on the website?

From Frank's perspective John figured he wouldn't need instructions unless he was buying parts (a different persona scenario). Again this scored a 5, though John made a note that times/schedules of fitting were not mentioned which might be a follow on question for Frank. Again something they may need to test out as part of their sequence testing.

10 Are there warranties and guarantees? Any associated costs?

For this final consideration John again had to score a 5. There were full warranty for the full lease period, clearly mentioned alongside the package descriptions.

At this point John was a little concerned. He'd found a couple of test points but nothing major that stopped sales from happening.

The pain points

He made his notes and moved onto Frank's main pain points. The pain points were covered off by scoring the amount information that addressed the pain like so:

5 = 5+ sources of information on the website about the pain point.
4 = 3-4 sources of information.
3 = 2 sources of information.
2 = only 1 source of information which is quite easy to see.

1 = 1 source of information which is difficult find.
0 = No sources of information to help you.

There were two that were very well catered for on the site, the installation was answered very well from 5 or 6 places on ACME's site and the technology issues around HDMI should not be a concern any longer for Frank.

The issue then had to be something to do with the major pain point, the price.

This was when John stumbled over a big problem.

Comparing the prices of the two plans ACME was pushing hard both of them included things that Frank didn't need.

The platinum plan was a full service that covered off all the things a new home entertainment system might need for £349 per month for 3 years (whereby the equipment was either upgraded or became owned).

This however would be a deal breaker for Frank's wife. Cost was a huge issue when it became £9000+ over 3 years. Also from competitive searches Frank had done he knew he could get it way cheaper than that elsewhere.

The £99 per month option was more attractive but again for Frank a lot of the hardware was un-necessary. He only needed an Amp and projector to connect everything together. He didn't need a PC/Wifi, Apple TV or speakers, some of which were included in the £99 per month package.

Frank could take a custom plan that would be much cheaper but that was the issue. You only saw the custom plan if you signed up for either platinum plan or the standard plan and then customised your items.

Frank could get the items he was looking for leased, home fitted and guaranteed for £49 per month, cheaper than the standard plan and perfect for his price range but it wasn't advertised until you were half way through the shopping process!

He scored both the pain points around cost a 1 because even though you could find the information on the site it took way more than 10 minutes if you didn't know what you were looking for.

John knew he'd found the persona gap they were looking for.

The trigger terms

Trigger terms had been defined already from a number of different sources, like internal search and phrases used to reach the website. They were scored like so:

Score one point for each item below (max five)

1 Keyword used is visible on the landing page.
2 Keyword subject is discussed at length on the landing page (more than 250 words).
3 Links to more direct information regarding the keyword subject exist.
4 Links to related topics exist.
5 Graphics/images used relate directly to the keyword.
6 Customer testimonials discussing the subject exist.

All the trigger terms used to tell Frank he was in the right place were there, so this also shouldn't be an issue.

Most terms were also linked to on various parts of the site (installation had its own area, support had its own area). Words related to technology even had little mouse-over pop-ups explaining what each acronym meant.

So for example if you moved your mouse over HDMI a little pop-up would come up saying:

> High-Definition Multimedia Interface, used for transferring high definition digital audio and video to devices like your projectors, amps, screens or PlayStations, basically any HDMI compatible device in our range. Long story short, we install the best-quality sound and video throughout your system.

Anything technical was explained in this way and John believed this was something a non-techie like Frank would appreciate.

Customer testimonials were also spread through the website discussing great service and saying things like "I didn't know what system to choose what with all the terms and technology, but ACME sorted it all out as part of the monthly fee. Very cool."

For these reasons John scored 5 across all the trigger terms. There wasn't anything not mentioned or covered off by ACME's website so Frank should know he was in the right place.

John however was excited, they'd found what he considered a huge problem. His next step was to verify and test the persona gap he'd found.

7.4 ACME testing the persona gaps

After uncovering what he thought might be a major problem John excitedly went to his analytics tools. He looked for the shopping cart funnel report.

He applied his methodical segment and ran the numbers. The methodical segment showed better performance percentage wise overall than the general visitor segment but abandonment on step 2 was huge on both segments.

He also noted that on the time range he was looking at not one person on either segment had bought a "custom package".

This made absolutely no sense to John. Custom packages were not advertised by ACME because they could be cheaper than the standard and way cheaper than the platinum packages. But after running through how to order a custom package he totally understood.

It was *because* ACME didn't advertise the custom package anywhere until it was probably too late. No-one knew about it.

Testing

He opened up his testing tool and selected the first page in the shopping cart funnel. His "control" page (version A) would be the page as it stood now and his test page (version B) would have a small call to action button he created in a different colour.

The image was placed alongside the "add to cart" option and said "customise your leasing options".

He also used the same technology as ACME had on some of the trigger terms, to have a pop up appear upon mousing over the button that said: "Click here to customise this package to suit your own needs. Don't need a flatscreen? Cool, just remove it and see the new price."

A versus B

He ran the test for a week and came back to check the results. He was astonished. His test version "B" outperformed the version "A" by 1500 times. 99.99 per cent statistical significance. The success metric was sales (not just click through).

That meant that those that had seen his control version had done the same as they always did and simply abandoned the shopping cart, probably leaving for a competitor.

Those that had seen his test version (a 50/50 split of all the traffic) had signed up with new leasing options at a huge rate.

Actions from analytics

His analytics data was telling him there was a massive potential if they placed this button into the shopping cart as an actual function rather than simply running a test on 50 per cent of the visits.

He contacted Harry from IT and excitedly told him the case. He asked him to keep it to himself until it was proven as this could be a great case study for their upcoming KPI workshop.

Harry agreed. History was made (see section 4.9).

7.5 Testing best practices

Things to look out for when testing are generally about understanding when you have a result or you have something you should keep testing and iterating till you get it right. The following guidelines may help.

1 Your hypothesis. Always define your idea in terms of improving what you currently have with something you think your persona will need. You're either going to prove a case for, prove a case against, or fail to prove anything. If you prove the case for, keep the change or make it permanent. If you prove the case against or fail to prove your argument, refine your hypothesis.
2 90–95 per cent statistical significance. Statistical significance means the likelihood that a result or relationship is caused by something other than mere random chance. As a general rule we use a 90–95 per cent chance that our results are more than random chance to declare a hypothesis

proven. Anything less than that is unproven. There is a good article about statistical significance from Avinash Kaushik[5] which also includes an excel calculator you may use to do this.

3 Calculate the time it should take. Also make sure that you calculate time you should test for based on conversions (or success actions like a click through) and traffic. This again helps you understand statistical significance is real or is hindered by the primacy effect (a change to your site being positively responded to in the early stages of your test).
4 Test one thing at a time. Many disagree with this and say multivariate testing (MVT) is the way forward. I don't disagree with MVT as long as each variable is the same variable. You can test 10,000 different graphics with MVT as long as the results are about 1 variable. So the same call to action. The minute you start testing two calls to action on the same page is the minute testing gets difficult and hard to prove conclusively. See also point 3 with regard to time.
5 Clear success metrics. From a testing point of view make this very simple. Improve click through or improve sales are the two simplest I use. Use only one metric (clicks or sales, not both).
6 Split size. This is dependent on the risk of your test. If you're not sure start small (95 per cent control and 5 per cent test versions) and work up until you gain statistical significance.
7 Do an A–A test. Run a control versus control to make sure there isn't some kind of weird tagging problem before doing your test. The results should not be statistically significant.
8 Randomise email testing. If you are planning a test on email, make sure your audience is a random sample of the whole list. Don't pick a geographical or demographical sample as the result may be skewed.

7.6 Checklist and exercises

1 Look at your analytics persona segments and find out which type has driven the most value
2 Based on your persona (see Chapter 6) heuristically walk through the website and score the results:
   – Score on considerations

   - Score on pain points
   - Score on trigger terms
3. Define persona gaps in the user experience from the scores registered in the heuristic walk through you just did.
4. Define a test hypothesis for your control experience and your test experience.
5. Check your analytics to see if there is any evidence your hypothesis is correct (do your segments tell you there is a potential problem?)
6. Do an A–A test on your control (see section 7.5, point 7)
7. Set up a test with a simple control (the same site you're trying to improve) versus your test (the item or items you think will improve things).
8. Leave the test running for a full 7 days minimum. Depending on your site running the test for a full month may be wise (see section 7.5, point 3).
9. Check statistical significance. This should be 95 per cent or better (see section 7.5, point 2).
10. Define and implement actions you can take that permanently improve your website or define a new test hypothesis (if the previous one has failed).
11. Go back to step 1. Rinse and repeat (this is a never-ending process improvement).

Notes

1 Thomas Baekdal, "Introduction To The Future of Learning Analytics", www.baekdal.com/analysis/introductionnbspto-the-future-ofnbsplearning-analytics.
2 Bryan and Jeffrey Eisenberg, *Call to Action: Secret Formulas to Improve Online Results* (Thomas Nelson, 2006).
3 Available at www.blackbeak.com/cult2/files
4 Conversion Rate Experts, "Which Split-Testing Software Should You Use?", www.conversion-rate-experts.com/split-testing-software/
5 Avinash Kaushik, "Compute Statistical Significance", www.kaushik.net/avinash/excellent-analytics-tip1-statistical-significance.

8 Using data and telling stories

> Please be good enough to put your conclusions and recommendations on one sheet of paper in the very beginning of your report, so I can even consider reading it.
>
> Sir Winston Churchill

8.1 Numbers, dashboards, insights, stories

I did some work for a company once where a marketing manager showed me the reports they were looking at.

Their agency had compiled the numbers from a web analytics tool into a PowerPoint show that was 50 slides long.

The report had lots of numbers explaining the effect of the campaigns the agency was running, showing numbers like banner impressions, clicks, costs and CTRs.

The person I was talking to was very proud of the report suggesting that he measured everything down to the last click and could account for every cent of his marketing spend.

Then I started making him feel uncomfortable.

I started asking questions like:

- How did you use the numbers?
  - He kept a list of campaign costs using cost per impression as a gauge of where to spend his marketing money.
- What actions did you take in the middle of the campaign because of the report you got showing progress?
  - He only got a report at the end of each campaign.

- How did these numbers compare to other campaigns?
  - He had a list, he could check, he knew the cost information immediately.
- Were these visitors engaging more than all the rest of the traffic to the site or were the campaigns engaging less?
  - He didn't know.
- I asked him what the story was about with the numbers he was showing? How would he summarise?
  - He asked me what was wrong with using the PowerPoint and summarising key points from within it?
- I asked the business value of the campaigns?
  - He knew the cost but not the value.
- If he didn't know the business value how could he truly know which campaigns he was running worked?
  - He looked like he was going to cry.

The things he didn't know rocked him a little and he challenged me to be able to produce those kinds of insights. I told him it could be done but not with the data the agency was producing.

Laziness is the key to reporting

I pointed out that every part of the data tells a story. You just need to uncover what the stories are and tell the most interesting ones. This has been my approach to data delivery since as long as I can remember working in the field, not because I was clever, but *because I was lazy*. I have always wanted to get to maximum impact with the least amount of work, and developing reporting PowerPoint is a tedious waste of time.

There should only be three types of reports, the numbers, the business insights with associated evidence and the analysts KPI dashboards. These are the sources from which the good analyst derives insights and is able to tell a story.

Reporting the numbers

The numbers are important for any business, even the basic ad impressions, visitors, page views are important when you're looking at the very high level of how your business is doing. The numbers form part of your dashboards and are the foundation upon what you build your insights upon. They are the data points for your KPIs. Alone and without context they are useless.

Reporting and using KPI dashboards

Ideally your KPIs per business unit are a measure of how that unit is performing. If you have a KPI like bounce rate lower than 10 per cent being exceptional, between 10 and 40 per cent normal and above 40 per cent critical then what are you going to do about it if an exceptional or critical situation arises? The dashboards are formed by your numbers, your segments and your KPIs but they are indicators to keep going as normal or "do something" depending on the colour of the traffic light.

Reporting the business insights

The idea of reporting business insights is to condense all the findings and actions required into one easily readable form that can be understood by anyone responsible for the business.

This is the hardest part of any analysts job because uncovering insights requires digging into data, working out what's happened before finally forming and testing a hypothesis. Only when you have tested a hypothesis can you claim to know something you didn't know before.

The dashboard might tell an analyst there is a problem. The numbers might tell him what has happened. The hypothesis is a potential explanation as to why, which when tested will be proven, disproven, or point to further investigation or actions that mean you get better data.

Telling a story

Once you have a business insight you come to the story telling part. This is when you need to develop what you're doing into an art form. Putting the insights into the context of your business is the main point. Making the

work understandable by all is the other. The whole point of dashboards, numbers, KPIs and analytics is to be able to tell a story.

An example story

The following story (on next page) was written in 2008 for the then Web Analytics Association (WAA)[1] competition about their data in Google Analytics. The purpose of the competition was for the competitors to tell the WAA what they should do to develop their website based on the analytics data they had.

Analysis of the story

Note this doesn't have any form of bar chart, Excel or PowerPoint but does use data points, metrics and KPIs to tell a simple powerful story to the WAA. Also note that they use sales data (site revenue), search data (Google trends), web analytics data and qualitative data (their own heuristic analysis and hypothesis). They haven't relied upon one source of data (like web analytics) to tell their story.

The report could be printed in a newspaper and understood by everyone who read it. You could also have that data sat in various silo's and never put it into PowerPoint or Excel. Therefore simply by using the data sources you could compile your story without having to do a lot of work (hence my point about laziness!).

The three golden rules about reporting and telling your story

1 *Always* start with a business question.

section 8.2 covers starting with a business question, followed by some simple ways to monetise your business questions in section 8.3.

Click Thinking Business Insight Report

The site does not provide for users outside North America,[a] mostly relies on people that already know about or have heard of the WAA,[b] is not optimized to drive conversion for any industry related keywords that don't include "web analytics"[c] and while it provides for formal analytics education, it fails to provide analytics help for common analytics problems.[d] The site itself needs some updating to with the most optimal use of Web 2.0 features and best practices navigation (probably should get rid of the Tendenci) as the page reloads when clicking navigational items need to go![e]

Our recommendation is a concerted effort to include the rest of the world through copy, keyword optimisation, events, help/FAQ and seamless navigation to allow the WAA to truly become the worldwide provider of web analytics education, support, help and jobs in all fields related to web analytics.

Localization will be key.

Notes:

a Singapore and India does three times more searches for the term "web analytics" than the US and more than double that of the US and Canada combined (Google Trends data), yet the WAA site receives 62 per cent of its traffic from North America while India and Singapore only provides around 4 per cent!

b Almost 19 per cent of direct site visitors' account for 46 per cent of the site revenue with the majority of corporate memberships coming from direct traffic. The search term *web analytics* converts 70 per cent better for professional memberships, while *web analytics association* converts 138 per cent better for professional and 63 per cent better for corporate memberships when compared to the overall site.

c "Web data" is searched 6.65 times more than "web analytics" and "web stats" 1.74 more than "web analytics", but these keywords don't bring any visitors to the WAA site. (Google trends)

d Education is not only a formal thing. Google launching Google Analytics on November 14, 2005 brought web analytics into the mainstream. This resulted in more people confronting data anomalies, issues with terminology, etc. The WAA site is not addressing these issues. The only place where this can be addressed is in the Forum and this isn't even hosted within the WAA framework.

e There is an inconsistency with the cookie trail (aka breadcrumb), the navigation requires page reloads, the blog is stale with old postings and very little comments, too many things link outside of the WAA site like the Social wiki, the forum, etc and does not seem to contain analytics data.

2 *Always* write a 'newspaper ready' summary.
3 *Always* conclude with what to do next.

section 8.4 covers the final two rules before we wrap up with an example from Jan at ACME and the checklist for Chapter 8.

8.2 Starting with the business question

There is only one rule about numbers and KPIs for that matter.

Never start with the numbers. Start with a business question. You can't find a number (and thus develop a KPI) without having a business question.

If someone asks you for data ask them what the business question is they're trying to answer.

The business question is the foundation of your story. It's your plot. You're the detective looking for the answer to that question. Your quest as an analyst has a beginning, middle and end.

Using numbers

In the beginning you identify the 'cast of characters' or query parameters

In Chapter 3 we discussed KPIs that are derived from business questions.

The first question asked was "What is the best source of traffic in terms of volume and sales?"

So to answer this question you would need the following numbers:

1 Volumes of traffic broken down by traffic source.
2 Volumes of sales broken down by traffic source.

Get your characters in place (your parameters)

You now have your query parameters, your stories characters to get your question answered and develop your business insight. You have traffic, traffic sources, and sales at the moment. Do you need other characters?

The question is a little vague. It doesn't give any dimension of time or type of product so it's an open, general question. Answering what the "best" source is requires a dimension.

Dimensions are things like time of day, date range, type of device, type of product, service, download, conversion, persona segment, whatever you need to answer your question.

I typically study a full year of data to determine words like "best" because I then have a full data set including all the seasonal results and I can account for any outliers. Any outliers need to go into your story as well as either an unusual occurrence that can't be repeated or something that could be tested again.

Then we need to look at external factors. These could be the villains of your piece. If you get no traffic from search engines but all your competitors do your best source of traffic might be email, but that doesn't mean its "the" best source of traffic you could be getting. You then have a villain in the shape of a competitor that can go into your story.

Finally you should also look for correlating factors best viewed through the lens of a segment. Is your persona segment coming from a specific traffic source and converting much higher than other traffic sources? To answer this question well you want to focus especially on your post activation persona segments.

So the process for using numbers is quite simple. First you put them into context and then you gather your data points.

1 Never start by finding numbers. Start with a business question.
2 Determine your query parameters to get your question answered and develop your business insight.
3 Determine the dimension you're looking at.
4 Study the data.
5 Integrate external information if possible.
6 Look at segmented data (personas for instance) to determine if your preferred segments convert higher than the general data.

Using KPI dashboards

KPIs developed should be used as indicators of an issue. Usually they have already been developed based around business questions and the resulting dashboard is there to give an alert when something good or bad happens.

Let's take the same example question "What is the best source of traffic in terms of volume and sales?" and assume the KPI was a 25–50–25 variation around volumes for each source across traffic and sales. Meaning the KPI alerts based on something good or bad (the top or bottom 25 per cent) but not the middle 50 per cent.

Let's assume for arguments sake the KPI showed a green light across 3 different traffic sources that week for volumes but not much of a noticeable impact on sales.

Your KPI by default is already answering the business question and has its parameters set. But you still need to determine the dimensions, study the general data, integrate external information and then look at segments to generate insights from your dashboards.

KPI dashboards therefore save a little time in your analysis because they are usually set-up to inform if something "abnormal" is happening across your most common business questions meaning your analyst should then go and find out.

That's why this book dedicates a whole chapter to them. (See Chapter 3 for running your own KPI workshop).

Finally, business question pretty much always require monetisation, so start looking for ways to monetise the data before writing your summary and telling the story.

Using KPIs and numbers to explain your story is one thing that you need to do, but also monetising the value points of each website you work with is key.

8.3 Monetisation

Determine points of value

List anything that engages, activates or nurtures the visitor in some way. There are three kinds of value: direct value, indirect value and potential value.

Direct value

Direct value is perhaps easiest to monetise. If you have direct online sales, leads, customer services or advertising revenue then all of these activities can be directly accounted for. A conversion can be assigned a value in the terms of sales or leads. Customer service can have an associated cost saving and ad revenue can be directly attributed to the amount of pages you can serve to an audience.

Indirect value

The vast majority of websites don't sell anything directly. However they all have points of value to the visitor otherwise there would be no point for them to visit. Many websites are educational in nature about a product or service.

If that is the case then referrals from your site to a partner could be one point of value if such sites are then referring traffic elsewhere.

Downloads could be a point of value. Watching a video could be a point of value. Reading a product page or sequence of product pages could be indirect value.

Buying a product offline could be a point of value, though tracking it would require survey data or a voucher printed from an online page that would then be a point of value in itself.

Potential value

This is either direct, indirect or a combination of both in relation to the market size. The potential value if you could reach more of the target audience you need to reach and persuade more of them to take action in the way you need them to.

Monetise the value points

Direct value monetisation

If for instance 500 leads resulted in 50 customers with a profit of €100 each then the value point of a lead is: (€100 × 50)/500 = €10. This means in

effect that the lead value is €10 because on average it takes 10 leads to yield one sale.

Profit per sale (PPS) of the same sale above in an e-commerce operation would be (€100 × 50)/50=€100 showing that the formula is consistent. (IE the value of a customer is €100 and it takes 10 times as many leads to get a customer so the lead value is €10).

If the cost per support query was €5 and you saved 1000 support requests by website actions then you have just saved €5000.

If you manage to increase the websites capacity for page views and can increase advertising revenue at the same level then 10 per cent more page views means 10 per cent more revenue.

Direct business objectives like these can be easily measured. If however the content had a different purpose than lead generation, sales, service or advertising you would need to use a different calculation to monetise indirect value of content. This calculation would be based on propensity to buy, or customer satisfaction.

Indirect value points

You can design a calculation in order to work out the value of anything on a website based on propensity to buy.

Propensity to buy is measured by taking a survey of a sample of the audience who looked at the content in question. You might ask them to fill in a survey after for instance reading a particular article.

The survey would ask something like, "After reading this article are you more likely to buy product *x*?" You then compare the people that answered "yes" to a control group who have not seen the article, they might be asked "are you interested in buying product *x*?" and then the amount of conversions measured after purchase from those that said yes.

You can then use the following calculation to work out the value of the content:

$$PtB = [Interest(s) - Interest(c)] \times [Real(c)/Interest(c)]$$

Interest(s) = Interest to purchase on the measured content (%)
Interest(c) = Interest to purchase in the control group (%)

Real(c) = Real purchasers in the control group (%)

This would then give a value point percentage of the profit of 1 customer.

For instance let's say the PtB interest (s) to buy after reading the content was 10 per cent and the control group (c) showed 4 per cent while the real purchases from the same control group was 2 per cent. The content value point percentage is [10 – 4] × [2/4] = 3. Therefore the PtB = 3 per cent.

We use a control group because without it we would not be able to determine the value of the lift from the content viewed as our subject.

What we're trying to do is show what value a specific piece of content is worth therefore we have to subtract the normal level of interest from the new level of interest raised by the content. This makes the valuation of the content fairer.

The second part of the calculation works out the number of people who actually became customers from the control group as a percentage of those that said they were interested. This second figure will never be more than 1 (100 per cent), unless of course your survey visitors mislead you about their intentions saying they wouldn't buy when actually they did.

Once you have calculated the lift in propensity to buy you can then look at the euro value.

In the example shown the content has lifted the propensity to buy by 3 per cent (on average). You would then put this into a final calculation to get the Euro Value of the content. This would be:

€value of content = ((change in PtB × amount of visitors) × avg profit) × market share

So if there were 100,000 visitors to the site, the average profit per sale was €100 with the company only having a 10 per cent market share then in the above case this calculation would be:

((3% × 100,000) × 100) × 10% = €30,000.

Following this logic you can comfortably say that this will put a value on the content read.

If the site has bad content then PtB will be negative. This is very useful for sites that are educational in nature but don't sell anything or direct anyone anywhere.

Other indirect measures are less complex. Matching offline sales to website visits can be done with vouchers. Matching retail sales with web based store finder conversions is possible to do in the same way as the lead generation calculations.

Visitors should determine the brand value based on their level of engagement. So if you have a brand engagement segment that is 10 per cent of the total visits to the site and you can track how many became customers you have a value on the level of your branding activities.

Indirect values are more difficult to monetise but that doesn't mean they shouldn't be calculated. If based on regular surveys you can determine the value of an educational website, you can determine the level of investment you need in the same way as you would look at an e-commerce website.

There is *always* a customer somewhere and in order to evaluate the value of the website you need to get that average profit of a customer.

Add up the various values to get a website valuation

Once you have the individual value points it's then just a case of adding them up to get the total value of the web property. If you determine PtB lift for indirect content you can take a sum of those numbers per year.

If a referral to a partner is potentially worth €300 and there are 1000 clicks then potential value to that partner is €300,000 (then tracking the partner conversion rate is needed to give you a real value). Working this out can lead to advertising revenue from the partner that then becomes a direct addition to the overall value of the website.

Work out the potential value

Finally put this back into the market size by working out the potential value based on the size of the market you can grow into.

If you have figured that your current Website value is €1 million a year based on adding up current direct and indirect value points you can firstly work out what your conversion improvements would be on your own reach

factors. If 1 per cent conversion overall is equal to a value of €1 million then 2 per cent overall is worth €2 million and 10 per cent overall is €10 million. This is a simplistic example but the point is valid. By working out the PtB with an increase in market share (20 per cent instead of 10 per cent for instance) you will be able to calculate potential value if you reach your targets. Your KPIs can then be adjusted to match doubling your traffic to the content.

You then could also factor in hitting your KPI target of doubling your market share that would mean also doubling your potential revenues. This can really make a difference when asking for investments.

In short as an analyst you should attempt to monetise wherever possible in order to talk the language of the people reading the business dashboard. That language is all about the money.

It's more than possible that when you show real business numbers that the executives in charge of the spokes will ask to see proof of your findings. This is where the analyst data and KPIs come into play and is what the analyst uses to determine where the opportunities are.

It's where he/she gets to show off.

8.4 Writing and telling the story

Developing insights

Developing business insights is a skill that can take years to develop. Hopefully by using KPIs, and some of the techniques in Chapter 4, you have some starting points, but asking the question "why does this happen?" and being hyper critical about the outcomes of the results you see is how you become good at developing insights. Ask yourself "can I prove this hypothesis wrong?" and if you can't you either aren't thinking hard enough or you have your insight.

It's a good process for your business to follow because it makes you think outside of your own comfort zone. If you're running through the checklist in section 4.11 and doing this with every case you're looking at then you're half way there. The rest is about communication.

If you can now write a lead explaining what the numbers mean instead of just graphs and trends, you're already part way there to developing your insights and telling a story.

Once you have examined the data in section 8.2 across the numbers and KPIs that are answering your business question try writing the whole answer as if you were writing an email to your CEO.

Your CEO is not going to accept any long drawn out email they want something fast, bite sized and executive summary style. But they will also need the data.

Writing the story

Writing the story is the most difficult part of the job for many analysts. People who are used to analysing data are often not that good at getting a powerful message across.

Of course they do the analysis first which could give them plenty of ideas but the key is to then condense the findings into something that nails the business question the stakeholder is asking.

This book is not about teaching you to write. There are lots of free and paid resources that will help you with this. However you don't need to be a Shakespeare or a Hemingway, you just need to clear in your writing. All stories have a beginning, a middle and an end.

The beginning – the headline

This is a summary of why the whole thing is important. Your CEO gets a lot of communication every day, you need to point out why your communication is important, even if he has asked for it. The recipient may not be the "CEO", it may be his secretary, it may be an executive that the CEO passes your email onto, or even another analyst for verification purposes. Remember that your communication doesn't live in a vacuum, it gets shared, can be quoted and used in ways you didn't expect.

I usually leave writing the beginning till the end, it's the most important part of your story. However a good headline achieves one thing: it persuades the reader its important enough to read the first paragraph.

The first paragraph then persuades the reader to continue reading while giving the reader the context about why this is so important.

The headline should transmit urgency, if it does nothing else. Why does the reader need to pay attention at this very moment?

The middle – the meat, what you know and what actions you want to take

Remember that not everything that you can measure should be included. You need to prioritise based on what you know about the business and what makes the most sense.

For instance, if bounce rate from search engines is 88 per cent, say:

> Nearly 9/10 of our visitors don't find our website relevant.

If conversion is only 1 per cent from paid search:

> Only 1–100 people sign up at the moment and we feel there is huge room for improvement.

Monetise where possible (using external data sources):

> Our competitors do 3 times as much business as we do right now and if we performed as well as they do it would be worth £100K per month in extra sales. Speak in easy language and get the key points across in your first paragraph.

Table 8.1 The story-building tabel

What we know	*Actions*	*Priority*
Bounce at 88%	Start A/B testing on high priority landing pages and keyword campaigns	2
1% conversion from process	Start A/B testing on our lead generation system	1

Competitors out-perform us by 3 times on sales	Verify this is true	4
We could be losing 100K per month in extra sales	Verify this is the case and rectify	3

If you have evidence that your competition is doing a better job than your business you might want to start with that. Put each number or KPI into a sentence that makes sense.

I often start with a table and three columns to help me build my story though once you get used to doing this it's not always necessary. The key thing is determining outcomes and the priorities for your business.

Write like Churchill

Once you have figured out what you know, how to act and prioritised then it's time to put this into sentences.

Winston Churchill was a great writer and speaker. There is a common theme to his best speeches (and he wrote most of them himself). They were incredibly simple English. He wrote where possible with a one-syllable rule, meaning he would choose words of one syllable if it conveyed the meaning clearly enough. Consider the amount of syllables in the following sentences (syllables in brackets):

> We (1) will (1) fight (1) you (1) on (1) the (1) beaches (2). We (1) will (1) fight (1) with (1) growing (2) confidence (3) and (1) growing (2) strength (1) in (1) the (1) air (1), we (1) will (1) defend (2) our (1) island (2) whatever (3) the (1) cost (1) may (1) be (1).

Only two words in the entire passage had three syllables and the words are very simple and well known English ("confidence" and "whatever").

One good guideline for any text is to write like this. Churchill could have easily said the following and made much less impact:

> His Majesty's forces will strategically place defences around coastal locations as well as marshal the already significant and ever developing Royal Air Force to prevent our enemies from gaining a foothold in Great Britain.

So when your write for the CEO try to make the words fewer than 3 syllables. You may now have starting paragraph that looks something like this:

> Only 1/100 people sign up at the moment and we feel we can grow that to 3 from 100. Also 9/10 of our users don't find what they are looking for at the moment. Our competitors may be selling three times as much business as we do right now and if we sold as much as they do on average it would be worth £100K per month in extra sales.
>
> This would get the CEO's attention.

The end – the call to action (what does your reader do next?)

Adding recommendations depends on what's practical and this is where the judgement about what's possible comes in. The priorities listed in the table above show what would drive the most value in the shortest time frame.

Priority 1 is fixing the lead generation process with A/B testing. It's an easy fix that requires no massive investment and if the point of speculation is correct (the part where the potential value has been monetised) then there is a big gain to be made. Priority 2 is the bounce rates on key landing pages, again low cost but not as important as the lead generation process so slightly lower priority.

Priority 3 and 4 still need to be verified as true or not, the testing will go a long way to validating if 100K a month could be made. More competitive intelligence would need to be carried out to verify if competitors are indeed converting at a higher rate or not.

I often use three categories for prioritisation: short-term, mid-term and longterm:

- Short-term changes are something that can be changed in house without having to seek approval or large budgets.

- Mid-term changes are something the business can do but usually with higher approval processes or increased budget.
- Long-term changes require corporate approval or steering groups decisions and usually effect the entire company. You should focus on the smaller issues first and the larger issues later.

Your call to action might now look like this:

> We recommend the following corrective actions:
>
> 1 Start A/B testing on the lead generation process.
> 2 Start A/B Testing on landing pages and keyword campaigns.
> 3 We should also carry out other qualitative studies, why aren't people converting as well as the industry average? (mid-term).
>
> Note that A/B testing is very cost effective in terms of time and would require no budget as we have already got the tools to do the job.
> Give us the go ahead and we'll get started immediately.

Next add the supporting evidence

Let's review the above paragraph with added notes where data needed to be used to back up the claims I was making:

> Only 1/100 people[a] sign up at the moment and we feel we can grow that to 3 from 100. Also 9/10 of our users don't find what they are looking for[b] at the moment. Our competitors may be selling three times as much business as we do right now[c] and if we sold as much as they do on average it would be worth £100K per month in extra sales.[d]

The notes giving the supporting evidence could be as follows:

> a 1 per cent conversion rate steady over the last 3 months. (Source: web analytics data.)
> b 88 per cent bounce rate for the last 3 months. (Source: web analytics data.)

c Average conversion rate industry wide is 3 per cent. (Source: Google Analytics benchmarking.)

d 1 sale = £1000 and 3 per cent conversion would result in 100 more sales per month at euro value £100K. (Source: product value.)

Keep it very simple, you can add trend graphs to prove the points if required, but often the text is good enough alone. Always include the data sources and back up findings with a time scale you've looked at so that there is some business context.

How the whole story looks

We may be leaving £1.2M per annum on the table

To: CEO/Executive board

We did a study which showed how we could make over £1M if all goes to plan. A quick summary as follows:

> Only 1/100 people[a] sign up at the moment and we feel we can grow that to 3 from 100. Also 9/10 of our users don't find what they are looking for[b] at the moment. Our competitors may be selling three times as much business as we do right now[c] and if we sold as much as they do on average it would be worth £100K per month in extra sales.[d]

We recommend the following corrective actions.

1 Start A/B testing on the lead generation process. (Short term)
2 Start A/BTesting on landing pages and keyword campaigns. (Short term)
3 We should also carry out other qualitative studies, why aren't people converting as well as the industry average? (Mid-term).

Note that A/B testing is very cost effective in terms of time and would require no budget as we have already got the tools to do the job.

Attached are the files and supporting data.

How do you want us to proceed?

Supporting data

a 1 per cent conversion rate steady over the last 3 months. (Source: web analytics data.)

b 88 per cent bounce rate for the last 3 months. (Source: web analytics data.)

c Average conversion rate industry wide is 3 per cent (Source: Google Analytics benchmarking.)

d 1 sale = £1000 and 3 per cent conversion would result in 100 more sales per month at euro value £100K

Source: product value.

You now have a story backed up by data.

Getting back to ACME we'll go through the first question Jan had prior to the KPI workshop as an example.

8.5 ACME storytelling

When Jan built the quick win in section 4.7 she started off by using 2 questions as her starting point and then took it from there:

- What is the best source of traffic in terms of volume and sales?
- Which channels are the most effective?

In order to answer the questions Jan knew she had to break down the numbers she would need:

1 Volumes of traffic broken down by traffic source.
2 Volumes of sales broken down by traffic source.

She decided that in order to give the numbers context she should look over the entire year, and so went back 13 months for a fair comparison of each month.

After she'd selected her date range she immediately noticed a peak in February page views (by default her analytics tool showed page views and she had yet to select traffic sources).

Before going any further she wanted to verify that this peak in traffic was unique to ACME.

She ran a query in Google Trends (www.google.com/trends) across ACME and the major competitors brand names. This verified that there was no special lift in traffic volumes from search over that given time period for the industry. She also ran a comparison across brands on Amazon's Alexa service (www.alexa.com) just to see if there was anything she should note around that period across a couple of her major competitors.

Jan figured if everyone had a significant lift around February that the peak in February might possibly be attributed to general industry lift or peak in interest.

She knew the data she had from this would not be 100 per cent reliable but it would give her an indication of any external trend she should take into account.

While some competitors were outranking ACME in terms of global visibility she knew it was simply because the companies were bigger with more branding budgets. What she noted was that the data around February was fairly stable and therefore any lift ACME got was unlikely to be due to a wider interest in the industry.

This was good news. It meant ACME had somehow achieved something.

Jan went back to her analytics data. When she had logged into her analytics tool it had told her about the peak in February page views, but she hadn't drilled down to find out the market she should zero in on.

She selected a tight period around the lift and ran a what's changed report (see Chapter 4) looking at the countries. That's when she noticed Belgium had had a 500 per cent rise in traffic, whereas most of the other countries hadn't seen any lift.

She filtered to only look at the lift in Belgium and could clearly see the peak in traffic (see Figure 4.1, page XXX).

Now she was ready to find out what traffic sources had contributed to this lift.

She ran more segments against the date range she was looking at.

- Organic – any free visit from a search engine.
- Paid search – any visit from a paid search engine traffic source.
- Display – any visit from a paid display/banner campaign.
- Social Media – Any visit from Facebook, LinkedIn and Twitter.
- Email – Any visit from an email campaign.
- Referrals – Any visit from a free referral excluding the social media websites listed.
- Direct – Any bookmark or typed URL entry.

The metrics she looked at were for completeness, Unique visitors, sessions, page views, bounce rates and conversions for each traffic source.

She saw a big proportion of the lift had come from Social media and that it was contributing over 70 per cent of all conversions in February. The lift in conversions in February had all come from either social or direct traffic.

The lift in direct traffic correlated to the lift in social traffic, an indication that people who had seen the social media had also directly come to the site to sign up.

This really got her interested. She went to see where the traffic had landed and noticed the vast majority of people had been interested in a specific article that was posted across the social channels ACME used (Twitter, LinkedIn and Facebook).

The article was about a new way to use existing ACME software that generated excellent cost savings for those that bought the software. A white paper download explaining the whole thing had already been prepared by these clever people from Belgium.

Jan took a look in ACME's internal directory and called the Belgian marketing manager (Maarten) responsible for the campaign. He informed that the article was a success from Belgium's point of view. They had gotten 375 new leads over the period of the campaign. She asked what that equated to in business but he could only guess.

Monetising the case

Jan called her colleague Mari who was attached to the software team to find out how much the software was worth and anything else she could find out about the impact of this campaign.

Mari had told her that the software the customer might buy from the articles was approximately 50K per year.

The website closed 375 new leads over this period when the peak happened from about 30,000 visitors (1.25 per cent conversion rate). The lead close rate was 10 per cent being booked to meetings within 3 months (again Jan found this out from Mari) and from there approximately 10 per cent would become customers.

So it meant that it was fair to say that the lead conversion to sale was about 1 per cent. That meant that the lead value was €500 (1 per cent of 50,000).

Jan quickly worked out that this period of successful marketing was worth somewhere in the region of €187,500 in new business to the company in Belgium alone. (She reported this onto Maarten who was delighted).

This was one market. This method of social marketing had only been tried in Belgium according to the analytics she was looking at and so by doing the same thing across 30 markets (literally translating the method and marketing it across social channels in the same way) the company could earn potentially €187,500 × 30, or €5.6 million in new business.

Jan wasn't finished though.

She reckoned 375 leads from 30,000 visitors was poor and by running a testing program to increase conversion and reducing bounce rate she could get an extra 1 per cent. That would mean 675 leads instead of 375 from the same 30,000 visitors. By applying her previous maths to the equation her single market value would have been €337,500. If that was ramped up across all markets you'd be looking at over a €10 million opportunity.

She also worked out the cost. Again she got Mari to help by asking how long it had taken to develop the content in the first place as well as calling her translation agency and getting a quote. The cost per market worked out at about €2000 to localise and a day or so per market in time to publish – this was time her colleagues could be doing something else so she factored it in.

Maximum cost therefore for doing the campaign globally was €90K. That meant a return on spend of €62–112 per euro spent. *Huge*.

Now Jan was ready to share her findings. At this stage she didn't need anything else from Persona segments. She could look later. She had found low hanging fruit that needed to be shared as fast as possible.

Telling the story

In Chapter 4 you already heard how Jan used this case to show the KPI workshop stakeholders how to implement and use KPIs. Jan also developed a PDF with all of the supporting evidence mentioned above. She then wrote the following email to Carol at her request (copying in the CEO), her direct report.

Subject line: €5.6M business opportunity – we need your help to implement it.

TO: Carol & Executive team.
FROM: Jan (Marketing & Analytics Senior specialist)

Hi all,

As you know we've been looking at all the ways our marketing can be more effective based on the data we have at our disposal. I am pleased to inform we found an opportunity worth €5.6M.[a]

This opportunity stems from research we've done around social media articles posted in February 2014 that resulted in Social reach channels attributing to 70 per cent of all conversions for that month.[b]

This lift in conversions was no accident, we uncovered concrete evidence that Belgium the market in question made new sales of €187K as a direct result of the social activity.[c]

Our Belgian team headed up by Maarten (Ccd) wrote articles that explained clearly how our software worked for a specific niche we haven't looked at before, they also had an excellent white paper to

capture the inbound lead and follow up on the customer. This can be repeated at scale.[d]

Even better than this we believe we could do better next time by implementing on site testing and conversion programs to convert more traffic than we did before thus potentially doubling the €5.6 million estimate.[e]

We estimate the return on investment of such activity to be €60–120 for every euro we spend[f] depending on the market size, conversion improvement from baseline.

We would now like to use our scale and ramp up conversions across all applicable markets by utilising the same model where possible. Please inform your teams we'll be getting in touch to run feasibility studies for marketing in different markets in the same way.

Please find attached a PDF documenting the study. Please direct all questions to me or Carol and we'll follow up as soon as possible.

Supporting data

a Based on 375 leads converted at €500 per lead value.

b Total 535 leads in February 14, 375 from social, 50 direct, 25 from paid search, 50 from organic search and 25 from display.

c Previous conversions run over the previous 13 months approximately 125-150 per month with very little business impact.

d The sales angle they developed can be used across markets with minimal localisation expenses.

e Conversion rate in Belgium is only 1.25 per cent. We believe we can improve it on other markets with testing to 2.5 per cent effectively doubling our incremental value.

f Our average cost per market is approximately €2K plus a day of internal work per market. Doing the campaign globally would cost 90K maximum. Returns on investment would be €5-10M

The key was that Jan had told a story that was really easy to understand, she backed up her points with data that were possible to read in less than 1 minute and give the executives an easy call to action.

8.6 Checklist and exercises

In the first edition of this book, this chapter was not received well because I didn't show any graphs, charts or dashboards. I refuse to apologise for that.

If you want to design a good dashboard, read Stephen Few.[2] He's dedicated an entire career to developing good dashboards and his books are excellent.

It's not that I am saying that dashboards or reporting is not useful. Far from it. I use dashboards and numbers every day in my job.

I just believe that you're better off learning how to analyse dashboards, numbers and KPIs to deliver insights, monetise your data and best of all tell a great story. If you want to develop a cult of analytics your communication has to be better than showing people an Excel file or pretty dashboard.

The following checklist and exercises can help you solidify your story. Try it out with your business or colleagues. I guarantee if you get it right it is better than sending an Excel or PowerPoint to anyone.

1 Never start reporting by finding numbers. Start with a business question.
2 Determine the "characters" or query parameters you need to get your question answered and develop your business insight.
3 Determine the dimension you're looking at.
4 Study the data.
5 Integrate external information if possible.
6 Look at segmented data (personas for instance) to determine if your preferred segments convert higher than the general data.
7 Look at dashboards you may have to see if any of your KPIs are in the red light zone (see also section 4.5 in Chapter 4 on the 20–50–25 rule).
8 Run "what's changed" and disengagement reports over trended periods of time (see Chapter 4 again).
9 Monetise your opportunities (see section 8.3).
   – Look for direct monetisation possibilities
   – Look for indirect monetisation possibilities
   – Look for potential monetisation possibilities
   – Total these up
10 Use a story-building table (as shown in Table 8.1).
   – What do you know?
   – What action can you take?
   – What priority to the business is this?

11 Try writing the story as if to your CEO developing footnotes for data points you'll need to summarise as you go. This is your first draft.
12 Edit your story for the 2–3 syllable rule (section 8.4).
13 Add the call to action (what do you want your reader to do).
14 Add the beginning, the headline of your piece.

Notes

1 Now the Digital Analytics Association (www.digitalanalyticsassociation.com).
2 For example, Stephen Few, *Information Dashboard Design: The Effective Visual Communication of Data* (O'Reilly Media, 2006).

9 The starting point

Do or do not. There is no try.

Yoda

9.1 From ground zero to the cult of analytics

So at the end of the book in the last chapter I start talking about where to start. This is because Culture-building never ends, it's a cycle. You have to plan things before you can change them and in order to keep moving forward you need to keep encouraging behaviour you want to see. Like Yoda says: "Do, or do not. There is no try." Consistent "doing" of the right things is what you need to encourage.

In order to do that with analytics you need to know where you are as a business or organisation right now.

A lot has been written since I wrote the first edition of this book about, for instance, web analytics maturity.[1] I don't really believe that it needs to be too complicated though I do understand why a large organisation might need a sanity check and an excel sheet to guide them.

You won't build a culture by talking about it, or working out where you are in an excel spreadsheet, you and your executive team have to make a plan to make it work. However where you start depends on where you are in the analytics adoption model.

The following simple levels of maturity still stands in 2015.

Level 1 Analytics ground zero
Level 2 Analytics test and play
Level 3 Analytics insight

Level 4 Analytics systemisation
Level 5 The cult of analytics.

Use common sense and see where you fit in the 5 levels I mentioned. Just be sure to be honest. It's fine to say "somewhere between level 3 and 4" if it gives you the starting points to move forward.

Level 1 Analytics ground zero

Ground zero means that your organisation doesn't have a web analytics system. It may be that some IT staff know that they can get log files but no-one understands the data they produce correctly. The need for analytics may stem from some random campaigns that someone is asking to measure.

It could be that the CEO has decided it's time to move into the twenty-first century and start measuring performance, a good catalyst for change, but currently your organisation has not yet started any activity. You have a black hole where your data should be.

What to do next

Your first milestone should be to generate internal awareness about what web analytics is, how your competition is doing it and the tools they use.

You may at this stage consider outsourcing to a consultancy to run a workshop. At this stage you might not have executive backing.

You need to get backing from the executive level at some stage. It may be that your executive level don't know the importance of analytics.

In this case you need to convince them by showing them the outcomes of the awareness workshops and where your business opportunities lie.

If at least some of the executive level buy into the idea and start to create an internal vision about how to use analytics for your business then you're ready to take things further.

If you don't have backing you can still move to the second level without them and try to prove business benefits by producing some quick wins (see Chapter 4) but it will be harder to achieve anything without executive backing.

Level 2 Analytics test and play

Level 2 means that at least a few of the people in the organisation realise the potential that using analytics can bring them and they're starting to make a noise about it.

You might find a tool like Google Analytics in place and some enterprising individuals that are trying their best to make sense out of their campaigns.

The reports your enterprise might use tend to be scattered and sporadic, perhaps campaign specific.

Little actual action happens because the information is in the hands of so few people and usually it's the wrong kind of people.

Basic level implementation of tools means potential analysis and reporting capabilities but so far you haven't got down that road.

What to do next

If you have buy in at this stage from the top management you might now be ready to determine roles in the business using the hub and spoke model (Chapter 1), design a REAN model (Chapter 2), do some KPI workshops (Chapter 3) for your spokes prior to finding out what tool specification you need (Chapter 5).

At this stage you need to build awareness that these models exist and try to encourage management to take the next step.

If you don't have buy in at this stage you need to continue the awareness push either with previously mentioned awareness workshops or, because you have a tool, you might want to plan some quick wins (Chapter 4) to help convince your management.

If you don't have the analysis know how then ask for help from a consultant who is competent with the tool that you're using and learn from them.

You're still not ready at this stage to start building incentives into daily working practices, especially without executive buy in.

Level 3 Analytics insight

You've gone beyond simple experiments with tools and are now using the tools and fully understand the potential of analytics to your organisation.

You're using web analytics tools to deliver basic reporting and random analysis happens across the business as key people start to learn how to use the tools.

These "analytics stars" are starting to really get to grips with things but the majority of people still haven't enough time or training to understand analytics intricacies.

Information reaches a lot of people through company communication channels such as intranets, emails and newsletters.

Because of analytics you're beginning to compare campaign performance, have seen the benefit of SEO to your organisation and may even be developing content to be search engine friendly because of this.

Wins are slowly forming. There is basic measurement in place across the entire organisation and in certain key areas you may even have advanced implementation with good KPI dashboards measuring performance.

You may even be segmenting and using data already like suggested in Chapters 6 and 7.

Management is almost certainly backing the initiatives but if they're not you should now have ample evidence that they need to continue the process building. Make them well aware of all the gains to the business.

What to do next

A hub should be formed (see Chapter 1). It's vital to get the right people in this key area, the "analytics stars" should be part of the hub and you need to utilise their knowledge and build processes around them that allow knowledge to be transferred throughout your business.

KPIs may already exist in some units and others they may not.

REAN should be clearly explained to all concerned and used as an underlying framework going forward.

The strategy should ideally come from within the business.

You can bring consultants in to help with the practical things but you should own business and your strategy.

These models should be applied to your business, if REAN doesn't work as an acronym call it something else but use the concept.

Nokia, where we first introduced REAN in 2006, have developed variations around Acquire, Engage, Convert, Retain (AECR) because their business uses these terms. Hewlett Packard have their own variation.

It's better if you have words that your business is already using so that they see the point faster.

KPIs should be developed in units that aren't already measuring them. The ones that are measuring them should have their KPIs placed into the context of the REAN model.

Level 4 Analytics systemisation

You have developed a hub and spoke like model and have an underlying framework like REAN in place.

You've developed advanced KPIs for each business unit and have a scorecard or dashboard system that monitors business performance.

You have a REAN scorecard reported to the executive management and tactical business insights regularly being sent around the business. Actions are being taken based on the information in most cases rather than reports being filed and forgotten.

You plan quick wins to share with the business units that still need the most help to understand analytics. The hub is really becoming a functioning measurement organisation helping to provide intelligence for the entire company and you have advanced analytics measurement coming from your tools.

All the data sources are utilised and data abundance doesn't frighten you.

What to do next

You're almost there and you should keep up the momentum. This is the perfect time to start rewarding good analytics practices internally.

You might also be at the stage where you want to automate more of the data going out to the business units. It won't be long before the business units understand quite a lot about analytics and you need to gear up for that.

They will start asking more and more difficult questions that need more of the hubs time so look at automation wherever its possible.

Data integration is also a big question for level 4 analytics.

Streamlining CRM and BI data to truly combine it with web analytics data is the next thing to tackle and start predicting what people will do.

Level 5 The cult of analytics

Web analytics data is fully integrated with all marketing systems and measures on and offline marketing/sales activity to a reasonable level of accuracy.

You're doing everything you did at level 4 but you're doing it every day. You can predict how your customers will act in a given situation to a certain level of accuracy.

Your business is now basing the vast majority of their actions on what the data tells them they should be doing, while the hub quickly finds out the answers to key business questions.

Investment decisions are made at the executive level based on analytics data in most cases.

Your customers can tell you something through their online behaviour via segmentation and you can react on the fly to help them and nurture them.

Your lead generation systems drop hot leads into the email inboxes of your sales people.

Your e-commerce systems are integrated to react to who your visitor is based on behavioural or customer segments and you can react in real time through marketing automation.

You can really start to measure the ROI on your analytics expenditure and you're finding out that while the cost of implementation and systemisation was high, it was most definitely worth it.

What to do next

It would be easy to say, do nothing different because you're already at "Analytics utopia".

However you've only just begun. Amazon probably hit the analytics cultural level around year 2000 and they are still at it, constantly improving their services, increasing their revenues and proving the doubters of the "dot bomb" completely wrong.

You need to consistently do the things that work and consistently stop doing the things that don't work. Because you're measuring everything that matters that shouldn't be a problem.

You now know where you stand in the adoption model. You now know what you have to do next.

One of the things you may start doing is help your employees to embrace the culture and to do the right things by rewarding the right kind of behaviour.

This is sometimes called empowering employees.

Incentives, rewards, competitions, prestige, all of these things will help people to start taking the courses of action you need them to take in order to get your analytics to the next level.

Your business is full of people that have been taught in university and then had years of working experience. I doubt even today in 2015 as I write this they had more than a summary smattering of analytics training.

The courses are becoming more available but they are not everywhere and still not widely part of university training.

Finding good people is the first hurdle. Finding ways of keeping the people you find in an intensely competitive market is the second hurdle. Keeping them happy longer term (5 years is a long-term job in this field) is the mark of a company with a good culture.

In order to keep good people you need to keep building your culture so that the money they're offered doesn't become the only issue.

9.2 Culture-building best practices

Culture-building tip 1: reward passion

Anyone with good people and business skills who demonstrates passion for the job should be promoted. In an analytics culture the key is to get people in charge who understand why analytics is truly important and have a passion to follow that. If a guy in the business starts talking about what the data is telling him, about how he dug deeper, that he got curious, that he found something strange and went further, with real passion ask him/her if they would like a job leading their business unit.

If you hear people groaning about a high bounce rate or cheering about a 5 per cent improvement in abandonment rate you have a potential candidate as an analyst.

Only people who have really got a passion for the work should be promoted to positions of authority analytics hub or the business units.

It's important for the business that everyone have people leading the staff that truly understand why they are doing all this extra work.

Without real passion in this field and a lot of patience you will fail to do a very good job.

Culture-building tip 2: promote the business case

You should promote the people providing business cases and quick wins as people of analytical excellence.

This does two things. First, it raises internal awareness that the management are looking for this kind of behaviour. Building a business case, demonstrating potential and achieving results for the business.

Second, it singles out the person leading the initiative as someone to look up to, congratulate and respect. It helps them feel rewarded for a job well done and helps them to feel its worth doing it again.

Culture-building tip 3: incentives

Pay people more when they win for the company. Pay them more when they really win big. Individuals can be paid bonuses that contribute to direct cost savings or improvements.

The team as a whole should also be rewarded so that when one of the staff do well the whole unit who helped all get a piece of the action. This encourages team building and team thinking.

For example if an analyst spots that a funnel process has high abandonment and suggests that the business unit runs a test. If the funnel conversion rate increases the analyst attached to the team should be rewarded as well as the rest of the team.

Single out the leaders and the analysts but reward everyone (financially or otherwise) if there is a success.

Culture-building tip 4: promote the business result

When cases happen, collect them and present them at the Annual staff get togethers or the quarterly business reviews. When people see things start to add up they will try to repeat the successes.

Most folk in an enterprise have high pressure roles and the people in those teams should understand their value to the business over the longer term as well as for the short term cases.

Culture-building tip 5: prestige

When someone does something that is particularly insightful, out of the box, innovative or just plain clever reward him or her with a prize of some nature.

Put the idea on a pedestal for all to see and make sure that the awareness is spread.

Use communication tools at your disposal to sing the praises of the idea or innovation. Try to find small wins everywhere to reward.

One example of this is the Stinky award covered in the book *Always Be Testing* by Bryan Eisenberg and John Quarto-von Tivadar.[2]

Intuit's Stinky

When it comes to companies that embrace a culture of testing, we have to take our hats off to Intuit. You see, Intuit has this small, cute, squishy toy skunk – the Stinky. It's an award, but it isn't given to the testing winners who achieve positive returns on investment through their testing successes.

Testing losers earn the right to display the Stinky with pride.

Thomas Edison once quipped, "Results! Why, man, I have gotten a lot of results. I know several thousand things that won't work." When you test, you always learn something: You learn from what went right; you learn from what went wrong.

Usually you learn more from the failures, but whatever the outcome, you've got a lesson. What you choose to do with it defines your TCR –

your Testing Culture Rating.

The Always Be Testing culture at Intuit recasts every negative as a positive. Intuit groks testing. Intuit has an extremely high TCR.

Culture-building tip 6: remove barriers

Make it easy to do things inside the hub and spoke system.

- Paperwork and red tape should be kept to a minimum.
- Access to analytics systems should be quick and controlled by an effective administration system from the analytics hub.
- Holding regular meetings with open communication should be encouraged.
- Discuss any problems that might exist in terms what the business needs to know from their data.
- Take action on problems raised and fix issues.
- If people aren't immediately sharing the vision discuss their ideas, show them how Analytics can help them back up their arguments, earn the business more money and bring them a lot of internal prestige.
- Show people how they fit into the hub and spoke model and ask them to ask questions, don't simply expect them to "get it" immediately.
- People are complex creatures and they may "toe the line" for a while, trying out what the management are telling them but without seeing "wins" (see Chapter 4) they will lose heart and start grumbling about the good old days when they weren't measured by a scorecard. Show them why this isn't the point. Make sure enough "wins" happen. Make sure they understand why performance measures are there.
- Make sure that business units are independently measured for success. If one unit is failing they should try to copy what the other units are doing not hold back everyone.

Culture-building tip 7: HR should replace or move problem people

With any vision there are some people that simply won't share the idea. Offer these people an easy way out of the data guided business either by

moving them to somewhere that they will be better suited, offering them early retirement or worst case scenario let them go.

This may sound controversial but if you're trying to change to a data guided company and someone in a particular area doesn't like the idea then they will breed contempt of the new ideology.

This kind of person can cause a lot of damage, especially if the person is well liked. Well liked people that can't manage change are possibly the worst problem you might find.

Politics in big enterprises often get in the way. This is one of the reasons why it's vital that the management vision is strong.

I've seen individual people jump on reports that are 50 slides long with little value and use the same arguments that I have shown in Chapter 8 as a weapon against analytics. This by the way is another reason to have good reporting practices in place, don't give the politicians an excuse.

They might argue, "All these numbers and for what? I get sent these reports every week, I am supposed to be using them to help meet my quotas yet they take weeks to read and understand. This will never work!"

It could be that they aren't educated enough in which case you need to lower the barrier and find ways to get them on board. But if they persist that this new analytics drive is just one big waste of time then they have to go or you'll never get anywhere fast.

Culture-building tip 8: HR should hire good people

Just as you let people go that don't fit with the new cultural approach you have to bring in people that do. They don't all have to understand the intricacies of web analytics but they should understand the data you're using is very important for the success of the company.

They should be doers. People prepared to get their hands dirty and do the work not just sell the result of other peoples work.

Also make sure that HR (Human Resources) is in line with the new policy. It's no good allowing new hires to come through that are being screened (formerly or informally) in old ways that don't match the new hire requirements.

In interviews people will tell you what you want to hear in order to get a job so new employment practices should be considered that screen people

in an appropriate manner.

9.3 Amazon case study

One company who get that analytics culture is vital for their success is Amazon, and a public case study is worth sharing even today, many years after it originally became public.

In their 2008 SEC filing,[3] Amazon describe the vision of their business as to "Relentlessly focus on customer experience by offering our customers low prices, convenience, and a wide selection of merchandise." The vision is to offer Earth's biggest selection and to be Earth's most customer-centric company (notice this is bigger than analytics, but it requires world-class data management at its core to achieve that vision). The case study in the box below is reproduced from an online article by Dave Chaffey.[4] Dave also uses a marketing framework very similar to REAN called RACE (reach, act, convert and engage). Dave edited the first copy of this book so it's an honour to have the work built upon as it has been by many people.

How "the culture of metrics" started in Amazon

A common theme in Amazon's development is the drive to use a measured approach to all aspects of the business, beyond the finance.

James Marcus in his book *Amazonia*[5] describes an occasion at a corporate "boot-camp" in January 1997 when Amazon CEO Jeff Bezos "saw the light":

"At Amazon, we will have a Culture of Metrics", he said while addressing his senior staff.

He went on to explain how web-based business gave Amazon an "amazing window into human behaviour".

Marcus says:

> Gone were the fuzzy approximations of focus groups, the anecdotal fudging and smoke blowing from the marketing department.

A company like Amazon could (and did) record every move a visitor made, every last click and twitch of the mouse.

As the data piled up into virtual heaps, hummocks and mountain ranges, you could draw all sorts of conclusions about their chimerical nature, the consumer.

In this sense, Amazon was not merely a store, but an immense repository of facts. All we needed were the right equations to plug into them.

Marcus then goes on to give a fascinating insight into a breakout group discussion of how Amazon could better use measures to improve its performance.

Marcus was in the Bezos group, brainstorming customer-centric metrics. He summarises the dialogue, led by Bezos:

"First, we figure out which things we'd like to measure on the site", he said. "For example, let's say we want a metric for customer enjoyment. How could we calculate that?"

"There was silence. Then somebody ventured: "How much time each customer spends on the site?"

"Not specific enough", Jeff said.

"How about the average number of minutes each customer spends on the site per session" someone else suggested. "If that goes up, they're having a blast".

"But how do we factor in purchase?" I [Marcus] said feeling proud of myself.

"Is that a measure of enjoyment"?

"I think we need to consider frequency of visits, too", said a dark-haired woman I didn't recognise.

"Lot of folks are still accessing the web with those creepy-crawly modems. Four short visits from them might be just as good as one visit from a guy with a T-1. Maybe better."

"Good point", Jeff said. "And anyway, enjoyment is just the start. In the end, we should be measuring customer ecstasy."

It is interesting that Amazon was having this debate in about the elements of RFM analysis in 1997, after already having achieved $16 million of revenue in the previous year.

Of course, this is a tiny amount compared with today's billions of dollar turnover. The important point was that this was the start of a focus on metrics.

9.4 How ACME got started

John Bull the CEO of ACME sat in his fifth annual general meeting talking to his board.

They were relatively happy with his performance. After all he'd managed to turn things around, ACME was profitable again for the third year running.

But John knew it was more to do with a general uplift in the market and cutting staff that had led to the turnaround rather than anything that ACME were doing particularly well.

He knew he needed to instigate change. He'd hired a big change management firm to come in and assess what he needed to do. That in itself riled him. His last firm would've done things that made sense quickly and efficiently but ACME was a different kind of animal.

His consultants had told him three things:

1 The company was still largely driven by personalities rather than data driven.
2 Politics internally was rife.
3 Silos existed.

One of the recommendations from his consultants that John was keen to instigate was to start taking advantage of data and turning it into good information.

He believed if he could do this it would lead to more cross business cooperation and efficiency which would in turn reduce the silos that existed.

Politicians he would deal with brutally to simply cut it out from the business.

John outlined his vision. He wanted to be the best and the measure of this was becoming number 1 or 2 in each market they competed in. They would do this by providing the best hardware and software in those markets. If they couldn't become market leaders or close to it they would sell off or cut the business from their portfolio.

After explaining the vision to his board they agreed they needed a much better analytics capability.

Collectively they decided to hire an analytics director reporting directly to John and so the new ACME vision was born.

Getting the executive buy-in

John needed the executive team to buy into his vision so he knew he'd have to make this whole approach acceptable.

He talked privately to three of his key executive team – Carol, Artur and Tim – and asked them how they would run analytics within their business areas.

He asked them to come up with a few ideas for the upcoming management strategy meeting where he would outline the board's new approach to making ACME fly again.

After calling the strategy meeting together he then outlined the vision and explained the board's decision about needing a better analytics capability. About using that as a catalyst for increasing efficiency and cooperation between business units.

Carol, Artur and Tim had great suggestions that kicked off a very positive discussion around what they would need to do that.

The outcome was that they would probably need to centralise the reporting function of the business. Only Carol or Greg might be good candidates to implement that approach and they both had plenty to do already.

In the end it was decided unanimously that a new director of analytics be appointed to ACME's management team who'd be responsible for improving all their respective business functions and reporting to John, the CEO.

John grinned inwardly while saying "Great idea guys and girls, I'll make an appointment by the end of the coming quarter." He had his buy in.

Bill Brannon

The reason John picked Bill were the questions Bill asked alongside from his impressive track record. A lot of candidates were good but none of them had asked him;

Do you understand what you need to do now in terms of people, processes and tools to succeed at making this change?

John had to admit when it came to the tactics, he didn't know. He only knew he needed to make the changes.

In effect then the interview turned into Bill Brannon interviewing John Bull about what tools, processes and people they already had at ACME. Once half an hour or so had passed, with John answering as best as he could, Bill outlined what his approach would be if he took the job.

1 Determine the level of internal analytics maturity in the business. Where were they on a scale of 1–5? And what would the next steps be from a tools/people/process point of view?
2 Develop a hub and spoke model where the hub would report to him and the spokes would be different divisions of business in ACME. Bill excitedly got up and drew it on a whiteboard explaining how it would work to John.
3 Then develop an internal framework that no matter what business you were in you could be measured in the same way and start talking the same language. This was a model called REAN he'd used successfully in his past life at a huge company. There was no reason to believe it couldn't work in ACME.
4 As the hub and spoke model required analysts to work as part of different units and those analysts would all use the same framework, KPIs could be compared and contrasted to each other.
5 He'd then ask his teams to create business cases and quick wins to get the flywheel turning.
6 Once they had all these basics (as Bill referred to them) in place they could look at whether they needed new tools, new approaches, new segments, personas, reporting practices and other requirements in place as the flywheel kept turning.

Bill sat down after 20 minutes of animatedly drawing diagrams and asked "What do you think?" John replied, "I think we should talk about salary and benefits now."

Determining the analytics maturity in ACME

Bill's first task after he'd been hired was to assess where they were in terms of maturity. What tools, processes and people were available? What would need adding?

First he looked at people. He had 10 people that were asked to compile reports with no real context as to why. They were asked "send me this report" and the standard response was simply to send the data requested from the variety of tools they used.

He would need to alter the process so that the 10 folks were assigned to the key parts of the business ACME had identified as important business functions or units.

Second, he looked at Process. Reporting was pretty much the only function this team provided and it came from legacy requests (largely outdated). There was no process for providing business insights to the units that could use the information, nor was any of the reporting that automated.

Third, he looked at tools. There were dozens of tools in place designed to serve different functions. The potential to save money and improve efficiency was considerable. The tools were often in a silo of their own and maybe only one person would use some of them. This would be a streamlining process and he figured a tag management system would work well to save some of their technical analysts a lot of time.

Upon spending a few weeks interviewing people and assessing where they were Bill was ready to say that ACME were somewhere between 2 and 3 in the maturity of their business.

He needed to form his hub, develop REAN and KPIs and then get some business cases formed.

ACME had got its starting point. The good thing was that Bill knew the CEO and board had his back which was the most important catalyst to make the changes he needed to make.

9.5 Checklist and exercises

Ask yourself what best describes your situation:

1 You have a web analytics system.

If yes, move on to question 2. If not you're at level one: see section 9.1, level 1: what to do next.

2 You have a web analytics tool at least, some sporadic reporting, often campaign specific is going on with no official process in place.

If this is you you're at level 2: see section 9.1, level 2: what to do next.

3 You've gone beyond simple experiments with tools and are now using the tools and fully understand the potential of analytics to your organisation.

If this is you you're at level 3: see section 9.1, level 3: what to do next.

4 You have developed a hub and spoke like model and have an underlying framework like REAN in place. You have dashboards, KPIs and actions come from the analytics you use.

If this is you you're at level 4: see section 9.1, level 4: what to do next.

5 Analytics data is fully integrated with all marketing systems and measures on and offline marketing/sales activity to a reasonable level of accuracy. Investment decisions are made at the executive level based on analytics data in most cases.

If this is you you're at level 5: see section 9.1, level 5: what to do next.

6 How do you reward passion?

See section 9.2, culture-building tip 1.

7 Do you have communication methods in place to promote business cases?

See section 9.2, culture-building tip 2.

8 Can you build in incentives to help employees?

See section 9.2, culture-building tip 3.

9 Do you have communication methods in place to promote business unit results?

See section 9.2, culture-building tip 4.

10 Do you celebrate prestige?

See section 9.2, culture-building tip 5.

11 Have you removed all barriers to change?

See section 9.2, culture-building tip 6.

12 Are all your problem people in a place where they can't hurt the culture?

See section 9.2, culture-building tip 7.

13 Are good HR policies in place to hire people who get the point about analytics?

See section 9.2, culture-building tip 8.

14 Have you identified how your vision aligns with your analytics strategy? See sections 9.3, 9.4 and 9.6.

9.6 The one big thing that drives change

The one big thing that all successful cultures have at their heart is the vision. Not a grand vision statement or some difficult to grasp concept but something simple and easy to understand.

A great vision has the power to inspire. But at the very least the vision should be believable and conceivable. Like Amazon: the vision is to offer Earth's biggest selection and to be Earth's most customer-centric company.

Analytics is never likely to be the vision itself. It simply isn't interesting enough to inspire everyone to change daily routines.

The reasons for using analytics should be the core of the vision. In order to build a culture of analytics or a data guided company you need to have a reason to be guided by data at all.

The analytics strategy should align with the vision statement rather than be the other way around.

This isn't to say I don't believe that analytics is important enough on its own to be the reason for a company to change.

In many ways I think analytics should be the foundation of all business activities, however I don't think you can inspire the people on the front lines of the business with a statement like "We will become data driven." People would ask the question "Why?"

In ending this book I'd like to share if I may what I consider a simple believable vision communicated with urgency by a true leader: Winston Churchill.

This was a vision that when implemented used more data, intelligence and analysis than any other campaign that had gone before it.

He utilised competitive intelligence, analysed his own situation and drew upon human endeavour in a time when computers were the size of houses.

He never mentioned data or analysis in his "vision statement" but before he released the vision he had already started on gathering data and finding new data sources.

It was 4 June 1940. It was his fourth speech as Prime Minister. His accession to the position had been controversial and was in fact a default ruling. Members of both the major political parties in Britain openly criticised him.

His speech opened[6] with a factual account of the French collapse, the evacuation at Dunkirk, and preparations for home defence. It was a brutal

wake up call for the British Empire. We'd been hit and hit hard.

But he then said his government was determined to "ride out the storm of war, and to outlive the menace of tyranny, if necessary for years, if necessary alone".

This was the vision.

This single sentence hushed the entire House of Commons.

He went on:

> Even though large tracts of Europe and many old and famous States have fallen or may fall into the grip of the Gestapo and all the odious apparatus of Nazi rule, we shall not flag or fail.
>
> We shall go on to the end, we shall fight in France, we shall fight on the seas and oceans, we shall fight with growing confidence and growing strength in the air, we shall defend our island, whatever the cost may be, we shall fight on the beaches, we shall fight on the landing grounds, we shall fight in the fields, and in the streets; we shall never surrender!

His immediate audience was stunned, and then erupted into a prolonged standing ovation.

The entire nation heard it, understood it and more importantly believed it. We were to fight, defend and never surrender.

A vision that had analytics at its core turned Britain into one of the finest counter-intelligence agencies in the world, bred exceptionally talented analysts and helped defeat one of the greatest tyrants to walk the Earth, was started when Churchill said "We shall fight on the beaches."

That in my view was his finest hour. Now it's your turn.

Notes

1 For example, see Stéphane Hamel, "The Web Analytics Maturity Model", www.cardinalpath.com/wp-content/uploads/WAMM_ShortPaper_091017.pdf.

2 Bryan Eisenberg and John Quarto-vonTivadar, *Always Be Testing: The Complete Guide to Google Website Optimizer*, available at www.johnquarto.com/Always_Be_Testing.pdf.

3 See http://phx.corporate-ir.net/phoenix.zhtml?c=97664&p=irol-sec.

4 Amazon Case Study - Matt Pounds & Dave Chaffey www.smartinsights.com/digital-marketing-strategy/online-business-revenue-models/amazon-case-study/

5 James Marcus, *Amazonia: Five Years at the Epicenter of the Dot.Com Juggernaut* (New Press, 2005).
6 Winston Churchill, "We Shall Fight on the Beaches", www.winstonchurchill.org/resources/speeches/1940-the-finest-hour/we-shall-fight-on-the-beaches.

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